



VIBRATING MACHINERY FOUNDATIONS ON PILES

PURPOSE

This document establishes guidelines, recommended procedures, and sample calculations for the design of pile supported foundations for large reciprocating compressors, centrifugal compressors, and other similar vibrating equipment. The vibration analysis is based on frequency dependent soil stiffness and damping in a procedure originally described by Gazetas and Novak.

SCOPE

This document includes information about the following major topics:

- Definitions of several terms related to dynamic design.
- A detailed list of required design data.
- A description of the formulae and sequence to perform a vibration analysis.
- A listing of acceptable design results.
- Additional design conditions including situations not normally encountered.

A sample design is included as an aid in producing actual designs. A foundation for a centrifugal compressor is analyzed and designed using a computer program.

Attachment 10 is included for detailed information on the vibration analysis procedure. Basic theory, dynamic equations, and a sequence of analysis are covered. Tables are included to aid in evaluating complex pile stiffness and damping criteria. A detailed reference list identifies sources of information for further examination of practices and procedures. A hand calculation is provided to illustrate how the vibration analysis formulae are applied.

APPLICATION

This document applies to reciprocating machines greater than 200 horsepower and centrifugal machines greater than 500 horsepower supported on block type foundations. The design of foundations for small pumps is described in Structural Engineering Guideline 000.215.1227: Pump Foundations.

DEFINITIONS

This section presents a discussion of a few topics relating to dynamics that need elaboration.

Reciprocating Machine Forces

Two types of forces act within a reciprocating compressor: gas forces and inertia forces.

Gas forces result from the action of piston motion and valve action which generate time varying head and crank end pressures. The magnitude of gas forces depend on the



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differential between suction and discharge pressure, the area of the cylinder bore, rod area, pulsations, and external resistance. The near uniform pressure within each compressed volume make the force on the cylinder a direct reaction to the force on the piston. Thus, gas forces act on the crankshaft with an equal and opposite reaction on the cylinder.

Inertia forces result from varying accelerations of rotating and reciprocating machine parts. The magnitude of inertia forces depend on speed, rotating inertia, and on reciprocating inertias of crosshead, piston rod, and piston. Equations for the forces resulting from a single cylinder are indicated in Attachment 01. Inertia forces act at the crankshaft bearings without any opposing reaction on the frame.

For the dynamic analysis of a pile cap, gas forces tend to cancel each other within the machine frame and within the foundation pier. Thus only inertia forces need be considered. However, for machine anchorage and pier design purposes, the magnitude of gas forces transmitted into the pier is very important and is dependent on machine rigidity. To obtain reasonably accurate anchorage design forces, the rigidity of the machine must be realistically considered. For detailed pier and anchorage criteria, refer to the appropriate sections in this document.

Centrifugal Machine Forces

These inertia forces are caused by imperfect balancing of a rotor. Centrifugal machine forces can be determined by the supplier; however, this information is usually not voluntarily furnished. Either of the equations shown in Attachment 02 may be used in calculating centrifugal machine forces with the following criteria for eccentricity:

- A value of $Q_g = 0.615$ inch/second produces the commonly used equation from Arya:

$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})}{6,000} \quad (\text{Equation 1})$$

- For large steam turbine compressors, ISO 1940, mentions a value of $Q_g = 0.2$ inch/second for use without supplier data, or

$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})}{18,450} \quad (\text{Equation 2})$$

- For machines built in accordance with API 617, Arya recommends using twice the initial test eccentricity. This results in the following equation:

$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})^{1.5}}{321,673} \quad (\text{Equation 3})$$

If the above criteria is not adequate to produce a reasonable design, vibration sensors may be examined to give an indication about an upper limit of shaft misalignment. If such



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sensors are used, then an eccentricity may be computed based on the shaft vibration level that will cause the machine to be shut down. Judgment must be applied on an individual case basis with the assistance of the Mechanical Engineer.

Whichever formula is selected in lieu of supplier information, the design criteria should be reviewed with the machine supplier for acceptance.

Shear Modulus

The shear modulus is the ratio of shear stress to shear strain and is the most significant factor in computing soil impedance. This value should be obtained by a soil consultant and is usually supplied as a range of values or as a value for each soil layer. Because soil properties can easily vary, a range of shear modulus values (plus or minus 20 percent) should be checked in the foundation design. If a range of values is supplied, that range may be used instead.

The reliability and accuracy of obtaining the shear modulus must be carefully evaluated. Laboratory triaxial compression tests or field plate bearing tests should not be used. There are at least 3 ways a soil consultant may use to determine the shear modulus.

- A crosshole test is the preferred method for determining the shear modulus. This is basically a field test which can be performed at various depths to obtain a shear modulus profile.
- The Resonant Column Test, ASTM D4015, is a laboratory test performed on undisturbed samples of soil. The shear modulus is determined indirectly through simple elastic equations.
- Published correlations relate shear modulus to other more easily measured soil properties. Though simple to use, correlations are better used to examine the results of other methods.

For further detailed information on soil testing methods, refer to R.D. Woods, Measurement of Dynamic Soil Properties.

Pile Impedance

Basically, this is the pile's reaction to dynamic loads and is a combination of stiffness and damping. One set of impedance values will be determined for the piles at the pile cap, and another set derived for pile cap embedment. Various methods are available to evaluate impedance.

Current state of the art procedures take into account the fact that pile impedance varies with frequency. Numerous published articles use various methods to determine pile impedance under a variety of conditions including elastic half space, viscoelastic half space (soil damping), multiple soil layers, elastic layer over a rigid layer, rectangular foundations, and flexible foundations. This document does not preclude the use of any recognized derivation; however, alternate methods should be carefully studied. The



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results in some reports may be limited in scope, consist of small hard to read graphs, or applicable to a narrow frequency range.

The pile impedance and embedment impedance criteria presented in Attachment 10 are from published papers by Novak that identify equations valid for an elastic half space over a wide range of frequencies. For different soil conditions, other criteria will be needed. Details and references are provided in Attachment 10.

INFORMATION NECESSARY FOR DESIGN

This section identifies all data required for analysis and design. The paragraphs are grouped by the source of the data.

Machine data should be requested from the machine supplier. Refer to Structural Engineering Specification 000.215.00920: Structural Data For Mechanical Equipment.

Soil information should be requested from the soil consultant. Refer to Civil Engineering Specification 000.210.02010: Geotechnical Investigation

Client and project data will be in the form of project specifications, client specifications, or meeting notes.

Basic Machine Data

Operating Speed: This is the frequency or range of frequencies of the dynamic machine forces. Individual rotor speeds should be provided if different from overall machine speed.

Outline Drawing: A layout of the machine provides dimensions to the cylinders, shaft, and other components.

Anchor Bolts and Layout: Bolt materials and sizes should be provided in order to verify bolt forces and pier dimensions. Whenever possible, anchor bolts for large compressors should be 12 inches minimum from bolt centerline to face of concrete.

Pier Layout: This layout should provide all dimensions and elevations for machine support. If not provided, pier dimensions may be determined using a minimum of 4 inches from the compressor base to the face of the pier.

Jacking Post Locations and Details: This provides the installation methods to properly align machine components.

Machine Forces

Primary and Secondary Dynamic Forces: These loads are normally provided for reciprocating machines with a brief explanation of their application.

Rotor Weight and Eccentricity: This determines loads for centrifugal machines. If



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eccentricities are not provided, the supplier should be informed of the criteria to be used.

Location of Dynamic Forces: This should indicate the location at which reciprocating forces or rotor eccentricities are applied. Normally, this is somewhere along the shaft.

Machine Component Data

Component Weights: Usually, only the motor and compressor are broken out as separate parts.

Component CG: A center of gravity for each machine component should be indicated, usually on the machine outline drawing. Sometimes, one center of gravity is provided for the entire machine.

Component Mass Moments of Inertia: If provided, mass moments about each axis should be used in the vibration analysis. If not provided, mass moments are normally judged to be minor enough to be neglected.

Additional Machine Criteria

Grout Requirements: The machine supplier is the most important source for grout requirements. Supplier criteria must be carefully examined versus client or construction preferences.

Bolt Post Tensioning Criteria: Inappropriate bolt post tensioning can easily cause excessive movement between the machine and base.

Allowable Amplitudes: If provided, use supplier criteria instead of the criteria provided in this document. If not provided, the supplier should be informed of the criteria to be used.

Measurement Point Locations: Amplitude measurement locations should be requested if allowable amplitudes are specified.

Geotechnical Conditions

Soil Properties: Unit Weight, Shear Modulus, and Poisson's Ratio values should be provided for each layer. Separate values may be needed for backfill if embedment is to be included. Comparison values are as follows:

Description:	Unit Weight: (lb/ft ³)	Shear Modulus: (K/in ²)	Poisson's Ratio:
Granite	150 to 160	4000 to 6000	0.15 to 0.2
Limestone	145 to 155	2000 to 5000	0.16 to 0.22
Sandstone	145 to 155	1000 to 4000	0.17 to 0.24
Dense Sand	115 to 140	10 to 19	0.28 to 0.34
Medium Sand	110 to 130	8 to 15	0.30 to 0.36
Loose Sand	95 to 125	5 to 11	0.32 to 0.38



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Hard Clay	125 to 145	11 to 15	0.38 to 0.41
Medium Clay	115 to 135	7 to 11	0.41 to 0.44
Soft Clay	100 to 125	3 to 7	0.44 to 0.47

Allowable Pile Capacity: This is a standard component of every soil investigation.

Material Damping: Also called internal or hysteretic damping, this is the energy loss within the soil due to interparticle friction.

Construction Recommendations: Also a standard component; recommendations concerning excavation, backfill compaction, and vibration isolation from other foundations may be provided.

Embedment Recommendations: These should be provided to confirm characteristics of backfill materials.

Optional Geotechnical Assistance

Soil Layer Evaluation: The evaluation of soil layers to provide soil stiffness and damping could be computed by the soil consultant. This would be recommended especially if unusual soil conditions are encountered.

Foundation Vibration Analysis: Some soil consultants have the expertise to provide a partial or complete amplitude calculation. This service could become effective if highly unusual soil conditions need to be evaluated or if computer software is not readily available.

Client Specifications

Allowable Amplitudes: If provided, these requirements should be used if more conservative than supplier requirements.

Measurement Point Locations: Location of amplitude measurement points may be specified if allowable amplitudes are provided.

Anchor Bolt Materials and Details: Supplier criteria should still be carefully considered.

Project Requirements

Concrete Strength and Reinforcing Grade: Standard material strengths for foundations are normally used (grade 60 reinforcing and 3,000 or 4,000 psi concrete). A higher strength of concrete may be advisable since tensile strength is important in preventing cracking in the pier.

Equipment Plot Location: This determines how much area the foundation mat can cover without interfering with other structures and equipment. Reciprocating machines with an odd number of cylinders require larger mats because the combined cylinder unbalanced



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forces are larger. Separation of compressor foundations from vibration sensitive equipment or structures should be considered. Problems should be worked out with Piping as early as possible.

VIBRATION ANALYSIS PROCEDURE

This section gives a description of how a vibration analysis is used to design a foundation mat.

Pile Material Properties

The pile material determines stiffness properties for soil-pile interaction. Typical material properties are as follows:

Description	Unit Weight (lb/ft ³)	Modulus of Elasticity (K/in ²)	Shear Modulus (K/in ²)
Wood (Douglas Fir)	36	1500	94
Wood (Southern Pine)	40	1500	94
Wood (Red Oak)	44	1250	78
Wood (Red Pine)	40	1280	80
Concrete (3,000 psi)	150	3122	1951
Concrete (4,000 psi)	150	3605	2253
Steel (A36)	490	29,000	20,714

Pile Layout and Pile Cap Dimensions

For the first trial, this is largely an educated guess based on past experience. As the trial layout and analysis cycle is repeated, a basis for better layout dimensions will become apparent from analysis results.

The pile cap is sized to accommodate the pile pattern, with normal minimum center of pier to edge of pile cap distances. The location of the piles and pile cap relative to the pier may have to be adjusted in order to reduce foundation eccentricities. Refer to acceptance criteria for alignment offsets.

Pile Cap Thickness

The pile cap thickness must be checked for rigidity because the impedance determination and analysis procedure described in Attachment 10 uses the common assumption that the foundation and machine are rigid relative to the soil. If the pile cap is not rigid, then the impedance should be determined by another, applicable, method and a dynamic finite element analysis should be used. The pile cap thickness must be verified by hand because the vibration analysis results will not confirm this assumption.

The pile cap thickness should not be less than 2 feet., in addition one of the following alternative equations should be applied:



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1. The following formula is derived from beam on elastic foundation theory using a flexural deflection pattern that approximates essentially rigid pile cap behavior:

$$L_z \geq 0.02 \left[\frac{G L^4}{b} \right]^{1/3} \quad (\text{Equation 4})$$

where,

L_z = thickness of foundation pile cap (ft)
 G = soil shear modulus (psi)
 L = pile cap cantilever dimension in rocking direction (ft)
 b = width of mat perpendicular to rocking direction (ft)

2. The following equation is commonly specified in many client specifications:

$$L_z \geq 2 + \frac{L}{30} \quad (\text{Equation 5})$$

where,

L = longest pile cap dimension (ft)

3. The following formula from Gazetas is based on a published study of the subject:

$$\left(\frac{E_c}{E_s} \right) \left(\frac{L_z}{d} \right)^3 \geq 1.0 \quad (\text{Equation 6})$$

where

E_c = Modulus of elasticity for concrete (K/in²)
 E_s = Modulus of elasticity for soil (K/in²)
 d = Mat cantilever beyond face of pier, in either direction (ft)

Pile Cap Embedment

Increased stiffness and damping result from soil along the vertical sides of a pile cap embedded into the ground. Embedment must be carefully designed, not just selected as an option in a computer program. Compacted cohesionless soils (sand) should be used to ensure full embedment. Cohesive soils (clay) can easily shrink away from the sides of a foundation mat. If isolation details are not provided, foundations in cohesive soils should be analyzed for the extremes of full embedment and no embedment. Attachment 03 shows typical details used to embed a pile cap, including the use of thickened edges to increase embedment depth.

A conservative value of the embedded depth should be used in the analysis. Normally, 2/3 of the total embedment should be used in the vibration analysis.



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Amplitude Locations

Locations should be established separately for machine tolerance and for human tolerance. If supplier or client criteria is not provided, locations will be determined from the following:

- Machine Tolerance - The shaft bearing locations at each end of the shaft can be selected. If bearing locations are not known, points at each end of the shaft should be selected.
- Human Tolerance - Locations should be selected at grade elevation on top of the mat corners.

Analysis Calculations

After all input data has been obtained, a vibration analysis is performed to compute the dynamic response using the selected foundation dimensions. A computer program is normally used to obtain the following results:

MR	= Mass ratio
SB _{net}	= Net soil bearing (ksf)
OS _x , OS _y	= Alignment offset, in each horizontal direction (percent)
NF	= Natural frequency, for each mode of vibration (rpm)
FR	= Frequency ratio, for each frequency and operating speed
DA _x , DA _y , DA _z	= Double, or peak to peak, amplitudes, in each axis (mils)

The basic steps of the vibration analysis procedure are as follows:

1. Calculate the machine foundation soil center of gravity.
2. Calculate the mass and mass moments of inertia.
3. Compute mat stiffness and damping.
4. Compute embedment stiffness and damping.
5. Resolve all forces at the center of gravity.
6. Perform a 1 dof analysis for vertical translation and for torsional rotation.
7. Perform a 2 dof analysis for horizontal and rocking vibration in each direction.
8. Determine resulting double amplitudes at the desired location.

Attachment 10 provides the technical basis, detailed equations, and a list of references used in a standard calculation procedure. The procedures employed by specific computer programs may vary somewhat. Program documentation should be examined for significant calculation procedure variations.

Coastdown Effects

Centrifugal machines with high operating speeds are usually started and shut down slowly enough to be subject to temporary periods of resonance. If any of the 6 vibration modes have natural frequencies less than the machine speed, coastdown amplitudes should be investigated. Judgment may be used to skip modes which contribute little to total amplitudes. Computed resonant frequencies are used as coastdown machine speeds



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and reduced dynamic forces are calculated using the following formula:

$$\text{coastdown force, lb} = (\text{force at operating speed, lb})(\text{CS} / \text{MS})^2 \quad (\text{Equation 7})$$

where

CS = coastdown speed (rad/sec)

MS = machine speed (rad/sec)

The vibration analysis is repeated for each applicable coastdown frequency. Higher allowable amplitudes are provided for temporary coastdown resonance conditions as indicated in Attachment 04.

Combined Foundations

For multiple machines on a common mat, the vibration amplitude calculations are to be based upon the simultaneous operation of the maximum number of machines representing the design condition. Spare and standby machines are assumed stopped. For those machines operating, all unbalanced forces and moments are to be assumed acting in whichever combination will produce the highest amplitudes.

For relatively small machines spaced close together, the mat may be thick enough to be considered rigid. This evaluation will need to be carefully studied and confirmed. For most cases of multiple machine foundations, the mat is flexible and the rigid mat basis of this practice does not apply. Recommended analysis procedures involve a computer finite element analysis using appropriate soil stiffness and damping computed in observance of a flexible mat.

Because a finite element analysis is cumbersome, a simpler alternate is sometimes used. The following procedure is approximate at best and has a limited technical basis compared to the finite element solution. One can approximate effects of mat flexibility in the following way:

1. Consider a single machine supported on its tributary area of mat as an isolated foundation. That is, disregard the other machines and the remainder of the mat. Evaluate the vibration response of this system and calculate the ratio, R1, of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the upper bound of rocking contribution.
2. Evaluate the vibration response of the common mat system using the procedures outlined herein. Calculate the ratio, R2, of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the lower bound of rocking contribution because the half space solutions are based upon a rigid mat assumption.
3. An estimate of coupled translation rocking displacement at the compressor shaft for a flexible mat is calculated as the translation displacement at the common mat times the average value of R1 and R2 computed above.



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4. The displacement from the torsional mode of vibration must be added to the above translation plus rocking displacement to obtain the total vibration amplitude.

Multiple Rotors

Multiple rotors can be analyzed separately with the resulting amplitudes added together. If the rotor speeds match, unbalanced forces can be combined prior to the vibration analysis. Combinations should include in-phase and out-of-phase rotors as necessary to compute conservative amplitude values.

VIBRATION ACCEPTANCE CRITERIA

This section describes the criteria used to accept the analysis results. Analysis by hand or by computer will be evaluated in the same manner.

Mass Ratio

Mass ratios are used periodically as a traditional gage of how much foundation mass is being provided relative to the machine. The following guideline may be used as an indicator of foundation performance:

$$\begin{array}{ll} \text{for reciprocating machines, } MR \geq 5 & \text{(Equation 8)} \\ \text{for centrifugal machines, } MR \geq 3 \end{array}$$

where,

$$MR = \text{Weight of machine \& foundation divided by weight of machine}$$

Foundations with a low mass ratio should not necessarily be resized. Instead, the dynamic analysis input and results should be carefully examined and used (refer to API 686).

Pile Load

Pile loads should be kept low to ensure elastic soil behavior. The following is generally considered acceptable:

$$PL \leq 0.5 \text{ (allowable pile load)} \quad \text{(Equation 9)}$$

where,

$$PL = \text{pile load}$$

If this criteria is not met, try additional piles or a larger pattern.



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Alignment Offsets

These should be kept to a minimum to prevent unintended coupled modes and to reduce differential settlement.

$$\begin{aligned} OS_x &\leq 5 \text{ percent} \\ OS_y &\leq 5 \text{ percent} \end{aligned} \quad (\text{Equation 10})$$

where,

OS_x, OS_y = distance between CG and center of pile cap divided by pile cap length

If these criteria are not met, relocate the piles and pile cap as appropriate.

Frequency Ratios

Foundations should be outside the resonant range to avoid high amplitudes. The following criteria for frequency ratios apply when soil properties are checked for plus or minus 20 percent variation.

$$FR \leq 0.8 \text{ or } FR \geq 1.2 \quad (\text{Equation 11})$$

where,

FR = machine speed divided by natural frequency

Note that client or manufacturers criteria can vary. Some specifications may use different range values (such as 0.7 and 1.4). Other specifications use a resonant frequency ratio (machine speed divided by resonant frequency). For low to moderate amounts of damping, the difference should be negligible. If needed, resonant frequency can be computed from the following:

$$RF = NF / \sqrt{1 - 2(DR)^2} \quad (\text{Equation 12})$$

where,

DR = damping ratio
NF = natural frequency
RF = resonant frequency

Sometimes, the frequency ratio criteria will be difficult to achieve. For reciprocating machines, 6 primary and 6 secondary frequency ratios will need to fall outside the resonant range. If one or more modes are within the resonant range, check the corresponding unbalanced forces and damping. The offending modes may be ignored if there are no applied forces or if the mode is overdamped, (i.e. damping is greater than twice the square root of stiffness times mass). If dynamic forces are small, an amplitude calculation using half the computed damping may be necessary. Otherwise, try a



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different pile pattern or embedment characteristics to alter the stiffness and damping.

Trying to live within the resonant range carries with it a certain level of risk. On one hand, computed damping levels are usually high enough to preclude significant resonance. On the other hand, machines have failed before due to resonance, especially under unusual soil conditions. Centrifugal machines are especially vulnerable because the magnitude of applied forces is only an educated guess.

Amplitudes

Vibration amplitudes must be kept low in order to prevent machine damage. Computed double amplitudes shall be less than supplier criteria, client requirements, or the limiting values from Attachment 04. Sometimes supplier or client requirements will include velocity limits. If needed, velocity can be computed from the following:

$$\text{velocity, in/sec} = (\text{MS})(\text{DA}) / 2,000 \quad (\text{Equation 13})$$

where,

MS = machine speed (rad/sec)

DA = double amplitude (mils)

If amplitude criteria is not met, the mat size and embedment criteria must be revised. The vibration analysis is performed again until amplitude criteria is acceptable. In unusual cases, acceptable amplitudes may not be achievable with a reasonable foundation size. Amplitude criteria may then need to be evaluated again with the client and supplier until a mutually agreeable solution is obtained.

ADDITIONAL DESIGN CONSIDERATIONS

The vibration analysis is usually of primary concern in designing foundations for vibrating machines. However, several other considerations need to be addressed in completing the design.

Reinforcing Design

Reinforcing is used to eliminate or reduce concrete cracking. As a minimum, mats should be provided with temperature reinforcing (top and bottom bars > 0.0018 times gross area), and piers should be provided with a cage of #5 at 12 inches each way. Refer to Attachment 05 for typical reinforcing configurations.

Piers for Reciprocating Machines

Gas pressure forces must be considered in the design of reciprocating machine piers. The magnitude of these forces depend on the rigidity of the machine frame; the more rigid the frame, the lower the force transmitted to the pier. Frames are rarely rigid enough to eliminate these forces. In lieu of a detailed finite element analysis of the machine frame, The following equation from Smalley shall be used to compute pier forces:

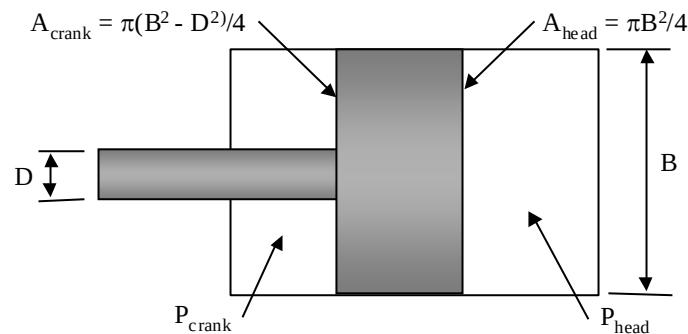


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$$F_{fdn} = [(P_{head})(A_{head}) - (P_{crank})(A_{crank})] F_{cr} / F_{red} \quad (\text{Equation 14})$$

where,

- A_{head} = Area of piston head (in^2)
- A_{crank} = Area of piston head on crank side (in^2)
- B = Cylinder bore diameter (in)
- D = Rod diameter (in)
- P_{head} = Instantaneous head pressure (ksi)
- P_{crank} = Instantaneous crank pressure (ksi)
- F_{cr} = Correction factor (1.15 to 1.2)
- F_{fdn} = Lateral force on foundation, tributary to cylinder (kips)
- F_{red} = Reduction factor, use 2.0 unless better data is available



The instantaneous head and crank pressures can be taken from the maximum and minimum suction and discharge pressures obtained from the mechanical engineer. Refer to guideline 000.215.1234, attachment 06 for a sample calculation implementing this criteria.

The anchor bolts tributary to the cylinder shall be used to resist F_{fdn} .

Grouting Methods

Grout requirements are determined in accordance with the project grout specifications, the compressor supplier's requirements, and the grout supplier's criteria. Conflicts with supplier, client, construction, or design requirements will need to be resolved and a mutual agreement obtained. Normally, epoxy grout is used for compressor foundations. Steel chocks or epoxy chocks may also be used to thermally isolate the machine from the supporting foundation. Anchor bolt sleeves should normally not be filled with grout; machine supplier drawings should be consulted on this matter.

Anchor Bolt Design

Bolt requirements are provided by the machine supplier. This always includes the number and diameter of anchor bolts. Bolt material and post-tensioning requirements may or may not be provided. If not, then these requirements are selected based on the applied forces. In addition, because slippage cannot be tolerated, lateral resistance must be provided through a sufficient clamping force between machine and foundation.



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$$T_{\min} = F / \mu - W_r \quad (\text{Equation 15})$$

where,

- $T_{\min}$ = minimum tensile clamping force per bolt
- F = applied lateral force per bolt
- μ = friction coefficient (use 0.15 based on oily steel on cast iron)
- W_r = applicable machine weight (could conservatively use zero)

For centrifugal machines, the applied force per bolt can be determined from the overall unbalanced dynamic force. For reciprocating machines, the applied force per bolt shall be determined from the gas pressure forces on each cylinder.

If post tensioning is required, bolt tightening instructions to the field should be provided with due consideration for tension measurement procedures and for retensioning at a later time to minimize relaxation.

Concrete embedment should be provided in accordance with Structural Engineering Guideline 000.215.1207: Anchor Bolt Design Criteria. Because the level of actual post tensioning in the field is difficult to control, and because cracking the pier due to overtightened bolts is to be avoided, bolts should be provided with a ductile embedment. The ductile criteria for embedment should be followed for tension and lateral bursting embedment without the use of additional reinforcing. The need for additional embedment reinforcing should be avoided because it requires cracking to become effective. Shear embedment criteria should be skipped because lateral forces are resisted through post tensioning of bolts and friction between machine base and foundation. If not dictated by the machine supplier, anchor bolts for large compressors should be 12 inches minimum from bolt centerline to face of concrete.

Water Table

In a vibration analysis, mass and not weight is what is important. A high water table may affect the apparent foundation weight, but the mass remains unaffected. The water mass is not rigidly connected to any foundations elements and should not be included.

The effect of a high water table on impedance should be verified with the soil consultant.

Elevated Pipe Anchors

Dynamic forces from connected piping are rarely significant enough to be used for the mat design.

Truly rigid anchor points to resist mechanical or pulsation loads are nearly impossible to provide by tall, slender piers without the addition of congested, unsightly bracing. Thus, it is inadvisable to provide elevated pipe anchors. However, if required, such pipe supports attached to the foundation pier or mat should be designed so that their natural frequencies of horizontal vibration are either less than 0.5 times the compressor primary frequency or greater than 1.5 times the compressor secondary frequency.



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Structural Engineering
Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering
Guideline 000.215.1227 Pump Foundations

Structural Engineering
Guideline 000.215.1234 Vibrating Machinery Foundations on Soil

Civil Engineering
Specification 000.210.02010: Geotechnical Investigation

Structural Engineering
Specification 000.215.00920: Structural Data For Mechanical Equipment



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ATTACHMENTS

Attachment 01:

Reciprocating Machine Forces

Attachment 02:

Centrifugal Machine Forces

Attachment 03:

Embedment Details

Attachment 04:

Vibration Limits

Attachment 05:

Reinforcing Detail

Attachment 06:

Sample Design of Centrifugal Machine Foundation

Attachment 07:

PVAP Analysis, Run #1 (Soil Properties As Given)

Attachment 08:

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

Attachment 09:

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

Attachment 10:

Analysis Procedure

Attachment 11:

Directional Nomenclature

Attachment 12:

Mass Moments Of Inertia

Attachment 13:

Typical Pile Impedance Coefficients

Attachment 14:

Embedment Impedance Coefficients

Attachment 15:

Vertical Group Factors



VIBRATING MACHINERY FOUNDATIONS ON PILES

Attachment 16:

Horizontal Group Factors

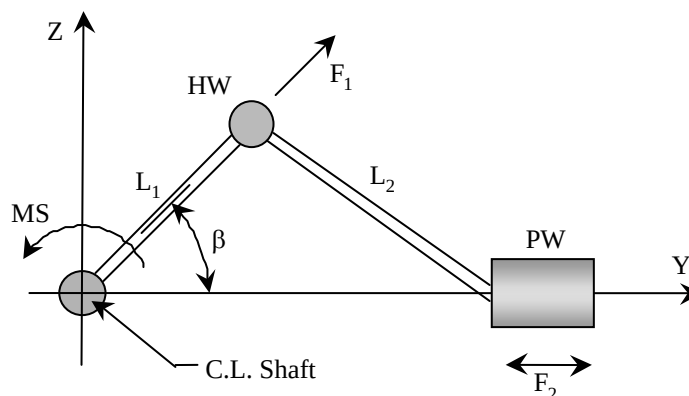
Attachment 17

Vibration Calculation



VIBRATING MACHINERY FOUNDATIONS ON PILES

Reciprocating Machine Forces



For one cylinder:

$$F_x(t) = 0$$

$$F_y(t) = \{(HW + PW)(L_1)(MS)^2 \cos [(MS)(t) + \beta] + (PW / L_2)(L_1 MS)^2 \cos 2 [(MS)(t) + \beta]\} / \text{gravity}$$

$$F_z(t) = (HW)(L_1)(MS)^2 \sin [(MS)(t) + \beta] / \text{gravity}$$

Where:

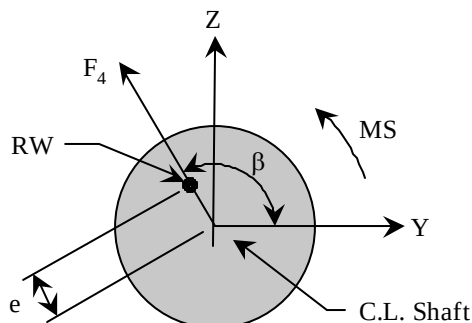
- HW = weight of hinge (kips)
- PW = weight of piston (kips)
- L₁ = length from shaft to hinge (ft)
- L₂ = length from hinge to piston (ft)
- gravity = 32.2 (ft/sec²)
- MS = machine speed (rad/sec)
- t = time (sec)
- F_x, F_y, F_z = forces in each coordinate direction (kips)
- β = crank angle (rad)

For multiple cylinders, the individual piston forces are combined using the appropriate distances between cylinders and the appropriate crank angles. These combined forces are usually provided midway between the cylinders by the machine supplier.



VIBRATING MACHINERY FOUNDATIONS ON PILES

Centrifugal Machine Forces



For one rotor:

$$F_4 = (RW)(e)(MS)^2 / 12(\text{gravity})$$

or

$$F_4 = (RW)(Q_g)(MS) / 12$$

$$F_x(t) = 0$$

$$F_y(t) = F_4 \cos [(MS)(t) + \beta]$$

$$F_z(t) = F_4 \sin [(MS)(t) + \beta]$$

Where:

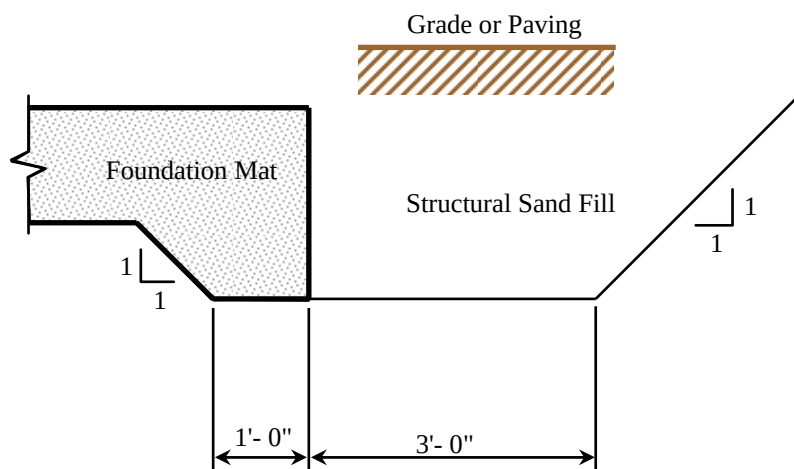
F_4	= inertia force of unbalanced rotor (kips)
RW	= rotor weight (kips)
e	= eccentricity of rotor (in)
Q_g	= measure of quality grade of rotor (in/sec)
gravity	= 32.2 (ft/sec ²)
MS	= machine speed (rad/sec)
t	= time (sec)
F_x, F_y, F_z	= forces in each coordinate direction (kips)
β	= crank angle (rad)

Multiple rotors should be handled independently. Each rotor can be analyzed separately and the results added together. Alternately, forces from each rotor can be combined prior to analysis using crank angles in worst case orientations. For this method, the speeds of each rotor must match.

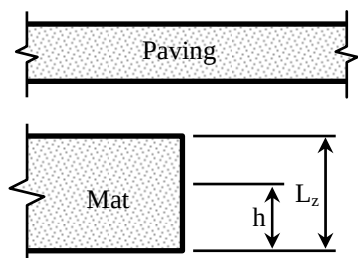


VIBRATING MACHINERY FOUNDATIONS ON PILES

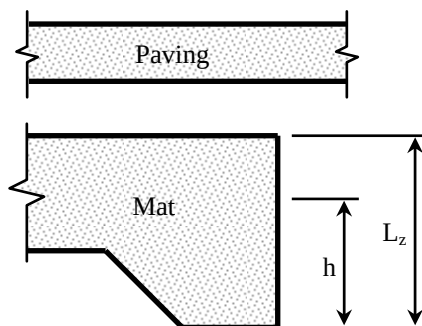
Embedment Details



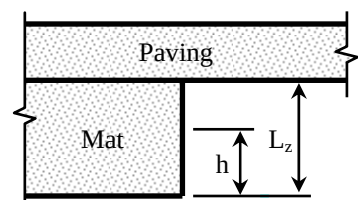
(a)



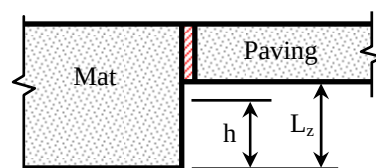
(b)



(c)



(d)

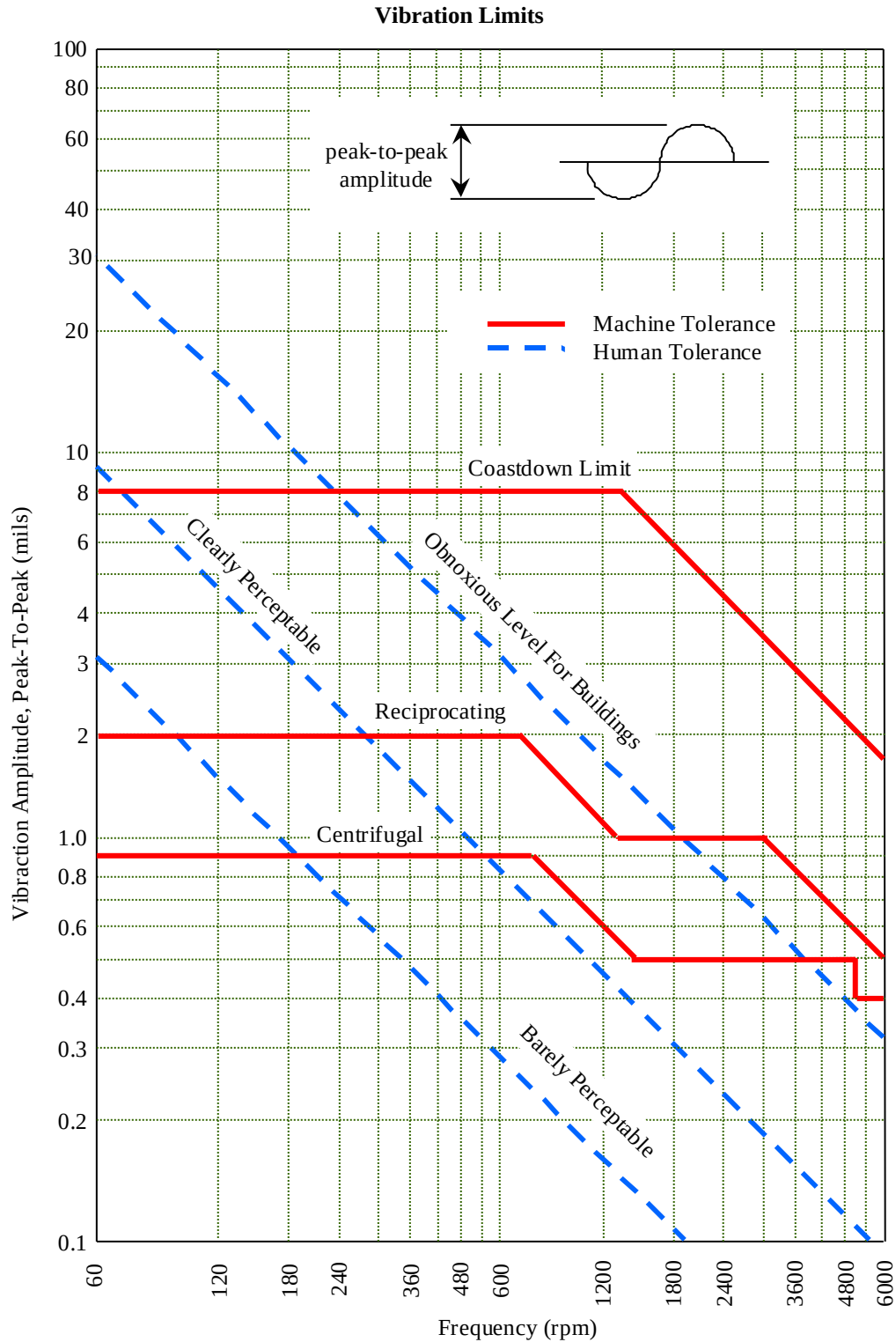


(e)

Effective Embedded Depth, $h = (2/3)(L_z)$



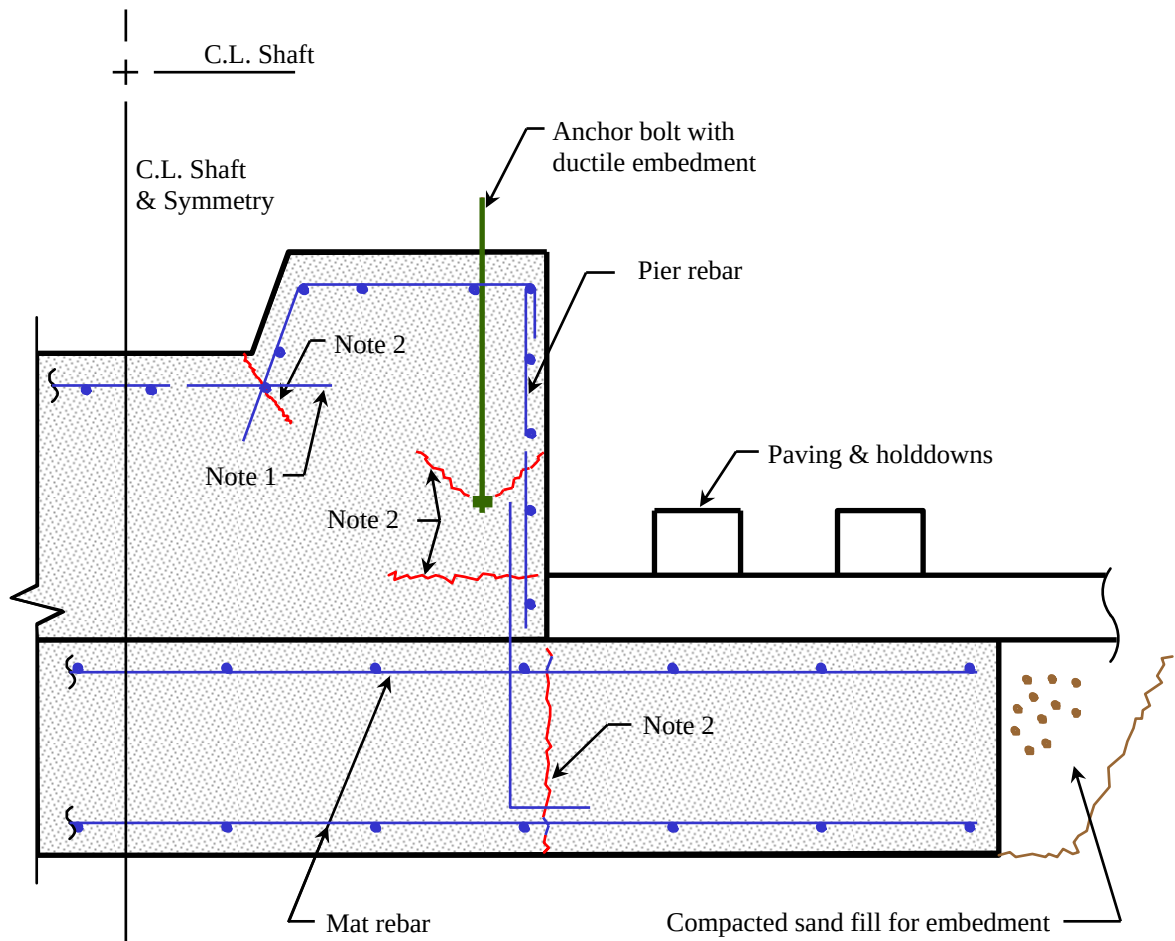
VIBRATING MACHINERY FOUNDATIONS ON PILES





VIBRATING MACHINERY FOUNDATIONS ON PILES

Reinforcing Details



Notes:

1. Develop all reinforcing, especially at re-entrant corners where pier cracking can originate.
2. Potential cracking to be resisted by concrete and reinforcing.



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation

Machine: (centrifugal machine)

Machine weight = 22,180 lb (C.G. at point #1)

Rotor weight = 1,365 lb (at point #2)

speed = 3,565 rpm (373 rad/sec)

Piles: (round timber piles, Southern Pine)

radius = 6 inches

length = 50 feet

static capacity = 20 tons (40 kips)

modulus of elasticity, $E_p = 1,500$ ksi (216,000 ksf)

shear modulus, $G_p = 93,750$ psi (13,500 ksf)

density, $W_p = 40$ pcf

Soil Properties:

soft clay:

shear modulus, $G_s = 5,000$ psi (720 ksf)

density, $W_s = 110$ pcf

poisson's ratio, $\mu_s = 0.45$

purchased sand backfill:

shear modulus, $G_f = 15,000$ psi (2,160 ksf)

density, $W_f = 125$ pcf

poisson's ratio, $\mu_f = 0.30$

Miscellaneous:

Dynamic forces not provided by vendor

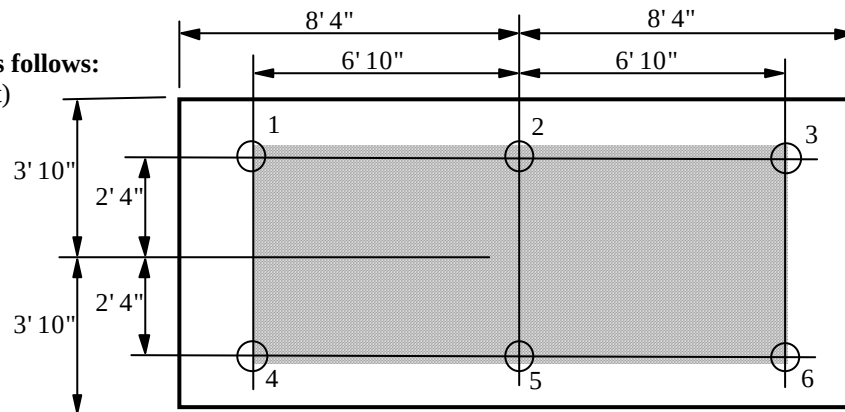
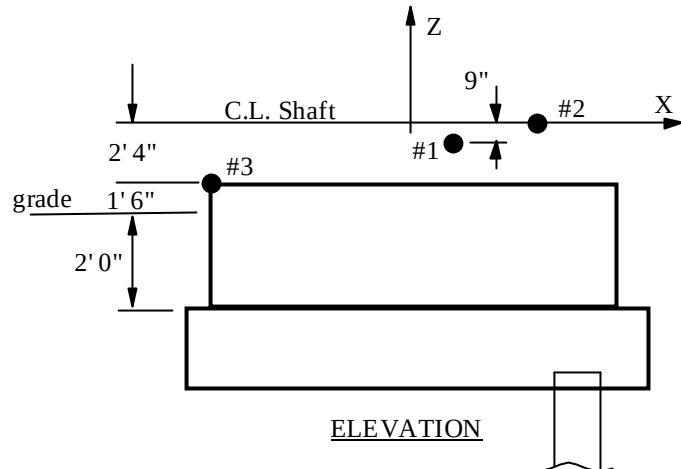
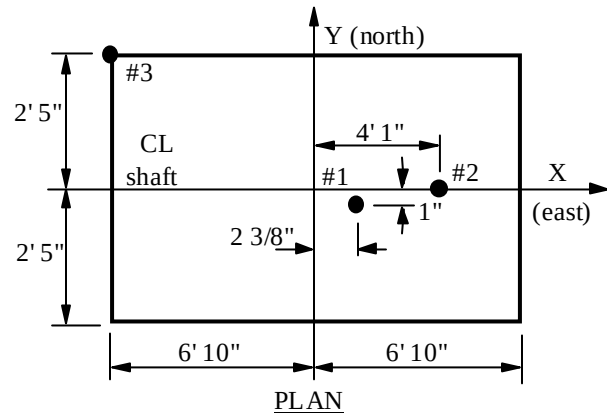
Use computed radiation damping only

Use attachment #4 for allowable amplitudes

$f'_c = 3,000$ psi

Try pile cap and piles as follows:

(cap thickness, $L_z = 2.0$ ft)



PILE LAYOUT PLAN



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation

Pile Cap Thickness:

verify assumed thickness, $L_z = 2' 0''$

pier cantilever, $d = [(16.67 \text{ ft pile cap}) - (13.67 \text{ ft pier})] / 2 = 1.5 \text{ ft}$

elastic modulus of concrete, $E_c = 57,000 \sqrt{f'_c} = 57,000 \sqrt{3,000 \text{ psi}} = 3,122,000 \text{ psi}$

elastic modulus of soil, $E_s = 2 (G_s)(1 + \mu_s) = 2 (5,000 \text{ psi})(1 + 0.45) = 14,500 \text{ psi}$

$$\begin{aligned} [E_c / E_s][L_z / d]^3 &\geq 1 && \text{(Equation 6)} \\ [(3,122,000 \text{ psi}) / (14,500 \text{ psi})][(2.0 \text{ ft}) / (1/5 \text{ ft})]^3 &\geq 1 \\ 490 > 1 &\text{ OK, rigid} && \underline{\text{USE 2' 0" thickness}} \end{aligned}$$

Determine Dynamic Loads:

use approximate formula & $Q_g = 0.2 \text{ in/sec}$

force = $(RW)(Q_g)(MS) / 12 (\text{gravity}) = (1,365 \text{ lb})(0.2 \text{ in/sec})(373 \text{ rad/sec}) / 12(32.2 \text{ ft/sec}^2) = 264 \text{ lb}$

$$F_x = F_x^v = 0 \text{ lb}$$

$$F_y = F_y^v = 264 \text{ lb}$$

$$F_z = F_z^v = 264 \text{ lb}$$

Select Amplitude Measurement Locations:

Use point #3 for human tolerance:

$$X = -6.833 \text{ ft}$$

$$Y = +2.417 \text{ ft}$$

$$Z = -2.333 \text{ ft}$$

Select two points along the shaft for machine tolerance:

East end of shaft:

$$X = +6.833 \text{ ft}$$

$$Y = 0$$

$$Z = 0$$

West end of shaft:

$$X = -6.833 \text{ ft}$$

$$Y = 0$$

$$Z = 0$$

Analysis Options:

For embedment, use $2/3$ of mat thickness, $h = 2/3 (2.0 \text{ ft}) = 1.333 \text{ ft}$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation

First Run: (soil properties as given)

$W_s = 110$ pcf
 $G_s = 5,000$ psi (720 ksf)
 $\mu_s = 0.45$

1. Mass ratio = $3.99 < 5$, low but check remaining criteria

2. Pile load = 18.74 kips $< 0.5 * 40$ kips, OK

3. Center of gravity offsets:

north/south = $100 * (0.02 \text{ ft} / 16.67 \text{ ft mat}) = 0.1\% < 5\%$, OK
east/west = $100 * (0.04 \text{ ft} / 7.67 \text{ ft mat}) = 0.5\% < 5\%$, OK

4. Frequency ratios:

mode:	primary:
1 (vertical)	$2.66 > 1.2$, OK
2 (torsional)	$2.58 > 1.2$, OK
3 (transverse)	$3.61 > 1.2$, OK
4 (transverse)	$1.63 > 1.2$, OK
5 (longitudinal)	$3.44 > 1.2$, OK
6 (longitudinal)	$1.45 > 1.2$, OK

5. Amplitudes:

point #3	amplitude (mils)
N/S (y axis)	0.06, barely perceptible, OK
E/W (x axis)	0.02
vertical (z axis)	0.06

east end of shaft	amplitude (mils)
N/S (y axis)	$0.08 < 0.5$ mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

west end of shaft	amplitude (mils)
N/S (y axis)	$0.08 < 0.5$ mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

6. Coastdown:

maximum reported coastdown amplitude is 0.10 mils at 2451 rpm, OK by inspection



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation

Second Run: (soil 20% stronger)

$$W_s = 110 \text{ pcf}$$

$$G_s = 5,000 \text{ psi} * 1.2 = 6,000 \text{ psi (864 ksf)}$$

$$\mu_s = 0.45 / 1.2 = 0.37$$

1. Mass ratio, same as run #1, OK
2. Pile load, same as run #1, OK
3. Center of gravity offsets, same as run #1, OK

4. Frequency ratios:

mode:	primary:
1 (vertical)	2.58 > 1.2, OK
2 (torsional)	2.52 > 1.2, OK
3 (transverse)	3.60 > 1.2, OK
4 (transverse)	1.60 > 1.2, OK
5 (longitudinal)	3.44 > 1.2, OK
6 (longitudinal)	1.41 > 1.2, OK

5. Amplitudes:

point #3	amplitude (mils)
N/S (y axis)	0.06, barely perceptible, OK
E/W (x axis)	0.02
vertical (z axis)	0.06

east end of shaft	amplitude (mils)
N/S (y axis)	0.08 < 0.5 mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

west end of shaft	amplitude (mils)
N/S (y axis)	0.08 < 0.5 mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

Third Run: (soil 20% weaker)

$$W_s = 110 \text{ pcf}$$

$$G_s = 5,000 \text{ psi} / 1.2 = 4,167 \text{ psi (600 ksf)}$$

$$\mu_s = 0.45 * 1.2 = 0.53$$

A poisson's ratio of 0.53 is unrealistic for an elastic material. Use a maximum value of 0.47 for a very weak clay.

1. Mass ratio, same as run #1, OK
2. Pile load, same as run #1, OK



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation

3. Center of gravity offsets, same as run #1, OK

4. Frequency ratios:

mode:	primary:
1 (vertical)	2.73 > 1.2, OK
2 (torsional)	2.71 > 1.2, OK
3 (transverse)	3.61 > 1.2, OK
4 (transverse)	1.68 > 1.2, OK
5 (longitudinal)	3.47 > 1.2, OK
6 (longitudinal)	1.50 > 1.2, OK

5. Amplitudes:

point #3	amplitude (mils)
N/S (y axis)	0.05, barely perceptible, OK
E/W (x axis)	0.01
vertical (z axis)	0.06

east end of shaft	amplitude (mils)
N/S (y axis)	0.07 < 0.5 mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

west end of shaft	amplitude (mils)
N/S (y axis)	0.08 < 0.5 mils, OK
E/W (x axis)	0.02
vertical (z axis)	0.04

Anchor Bolt Check:

Note: For brevity, anchor bolt design is not shown here. See Guideline 000.215.1207, "Anchor Bolt Design Criteria," for details of anchor bolt design.

Pier Reinforcing Design:

USE #5 at 12 inches each way

Mat Reinforcing Design:

mat cantilever $\approx [(16.67 \text{ ft mat}) - (13.67 \text{ ft pier})] / 2 = 1.5 \text{ ft}$

By Inspection, a 1.5 ft cantilever and a 2 ft thick mat will not require more than nominal reinforcing.

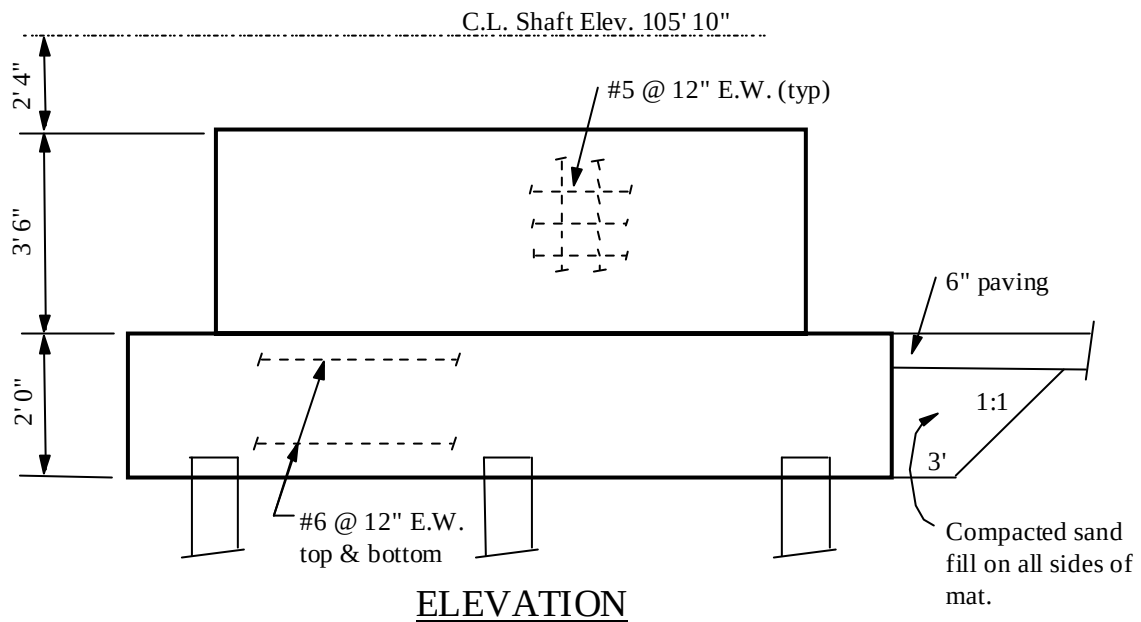
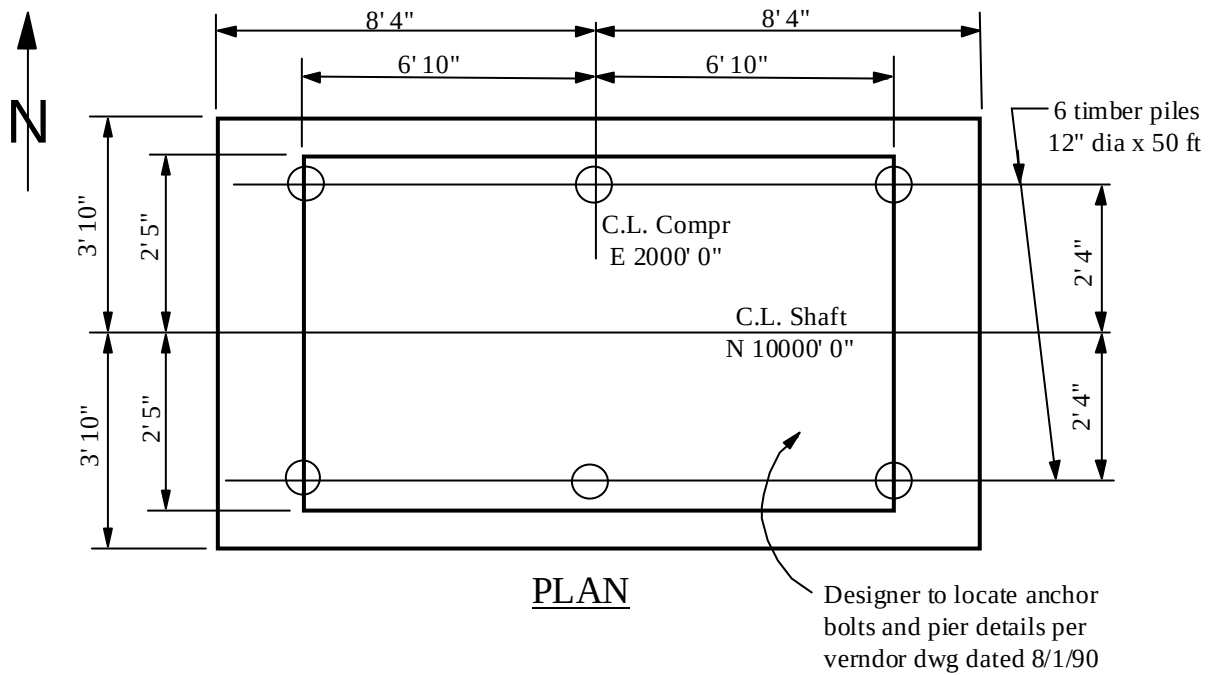
$$A_s = (0.0018)(24 \text{ in depth})(12 \text{ in unit width}) = 0.52 \text{ in}^2$$

USE #5 at 12 inches each way, top and bottom



VIBRATING MACHINERY FOUNDATIONS ON PILES

Sample Design of Centrifugal Machine Foundation





VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

The current version of PVAP uses a slightly different method in the computation of pile group factors and embedment impedance from that described in Attachment 10. Results are not exact.

FLUOR Date: 8/24/96
Contract: 215-1235
PVAP (rev: 2.1c, 12/15/98) By: Bounds
Page: 1
Technical Guideline, Sample Design
Soil Properties as Given
PStiff program was used to compute individual pile impedance

Machine Forces:

Machine Data-	Location of Forces-
Centrifugal machine.	N/S coordinate = N 0.000 ft
Speed = 3565 rpm	E/W coordinate = E 4.083 ft
	Elevation = 0.000 ft
Direction-	Primary-
N/S force	0.26 K
E/W force	0.00 K
Vertical force	0.26 K
Moment about N/S axis	0.00 K-ft
Moment about E/W axis	0.00 K-ft
Torsional moment	0.00 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec ² -ft)
The Machine	S 0.083	E 0.198	-0.750	22.180	0 0 0

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #3	N 2.417	W 6.833	-2.333
East end	N 0.000	E 6.833	0.000
West end	N 0.000	W 6.833	0.000



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

FLUOR

Date: 8/24/96

Contract: 215-1235

PVAP (rev: 2.1c, 12/15/98)

By: Bounds

Page: 2

Technical Guideline, Sample Design

Soil Properties as Given

PStiff program was used to compute individual pile impedance

Foundation Data:

- * Foundation geometry will be checked by the analysis.
- * Group stiffness and damping will be computed from individual pile data.
- * Piles will be located as specified.
- * Pile caps will be located as specified.
- * Embedment effects will be included.

Elevations-		Embedment-	
Grade elevation	= -3.83 ft	Effective height	= 1.33 ft
T.O. pile elevation	= -7.83 ft	Centroid elev	= -7.17 ft

Geotechnical:	soil:	backfill:
Unit weight	110 lb/ft ³	125 lb/ft ³
Shear Modulus	720 ksf	2160 ksf
Poisson's ratio	0.50	0.30

Individual Pile Impedance:

Pile diameter	= 12.00 in
Pile E * I	= 10604.00 K-ft ²
Internal damping	= 0.00 %

@ center of stiffness:	stiffness:	damping:
variation factor	1.00	1.00
horizontal	6405 K/ft	14.20 K-sec/ft
vertical	19420 K/ft	26.50 K-sec/ft
rocking	11997 K-ft/rad	7.60 K-ft-sec/rad
torsion	1658 K-ft/rad	0.20 K-ft-sec/rad
cross term	-6412 K/rad	-8.60 K-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

FLUOR

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PVAP (rev: 2.1c, 12/15/98)

By: Bounds

Page: 3

Technical Guideline, Sample Design

Soil Properties as Given

PStiff program was used to compute individual pile impedance

Typical Pile Layout:

		E/W piles			
		E/W coord	E/W spacing		
			<---->		
N		0	0	0	0
		^			
		^			
N/S piles	N/S spacing	-0 ^ v -0	0	0	0
			0	0	0
	N/S coord	-0	0	0	0
			PLAN-Pile Group		

Pile Group Data:

N/S piles: (ea)	N/S space: (ft)	E/W piles: (ea)	E/W space: (ft)	N/S coord: (ft)	E/W coord: (ft)
2	4.667	3	6.833	S 2.333	W 6.833

Typical Pier & Pile Cap Dimensioning:

[illegible]



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

coord---***** Elev---*****
PLAN ELEV
FLUOR Date: 8/24/96
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Technical Guideline, Sample Design
Soil Properties as Given
PStiff program was used to compute individual pile impedance

Pile Cap Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pile Cap	7.667	16.667	2.000	S 3.833	W 8.333	-7.833

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pier	4.833	13.667	3.500	S 2.417	W 6.833	-5.833

Results of Mass Computation:

Center of gravity:		Weight & Mass:	
N/S coordinate =	S 0.017 ft	Total weight =	110.63 K
E/W coordinate =	E 0.040 ft	Machine weight =	22.18 K
elevation =	-4.472 ft	Mass =	3.44 K-sec2/ft
		Mass ratio =	3.99 to 1
Footing data:		Mass moments @ CG:	
Maximum pile load =	18.74 K	about N/S axis =	77 K-sec2-ft
Minimum pile load =	18.14 K	about E/W axis =	30 K-sec2-ft
		about Vert axis =	71 K-sec2-ft
N/S eccentricity =	0.02 ft		
E/W eccentricity =	0.04 ft		

Pile Group Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	14748 K/ft	33 K-sec/ft
E/W translation	13125 K/ft	29 K-sec/ft
vertical	50861 K/ft	69 K-sec/ft
about N/S axis	3698853 K-ft/rad	4995 K-ft-sec/rad
about E/W axis	706459 K-ft/rad	911 K-ft-sec/rad
torsional	1415403 K-ft/rad	3117 K-ft-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

FLUOR

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By: Bounds

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Technical Guideline, Sample Design

Soil Properties as Given

PStiff program was used to compute individual pile impedance

Embedment Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	11613 K/ft	236 K-sec/ft
E/W translation	11613 K/ft	236 K-sec/ft
vertical	7774 K/ft	165 K-sec/ft
about N/S axis	124657 K-ft/rad	4578 K-ft-sec/rad
about E/W axis	100471 K-ft/rad	1881 K-ft-sec/rad
torsional	0 K-ft/rad	0 K-ft-sec/rad

Total Impedance Results:

Center of stiffness:	Pile group factors:
N/S coordinate = N 0.001 ft	N/S translation = 2.61
E/W coordinate = W 0.000 ft	E/W translation = 2.93
Elevation = -7.833 ft	Vert translation = 2.29

@ center of stiffness:	stiffness:	damping:
N/S translation	26362 K/ft	269 K-sec/ft
E/W translation	24738 K/ft	265 K-sec/ft
vertical	58635 K/ft	234 K-sec/ft
about N/S axis	4230404 K-ft/rad	10248 K-ft-sec/rad
about E/W axis	1232096 K-ft/rad	3509 K-ft-sec/rad
torsional	1415430 K-ft/rad	3117 K-ft-sec/rad

Results of Modal Analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	712	780	1247	1347	1989	2244
Resonant freq (rpm)	989	1035	1342	1381	2188	2451
Resonant Freq. Ratio	3.61	3.44	2.66	2.58	1.63	1.45
Dynamic Magnification	0.04	0.05	0.14	0.17	0.41	0.56
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	-0.01	-0.72	0.00
E/W trans (ft)	0.00	1.00	-0.00	-0.00	0.00	0.50
Vertical trans (ft)	-0.00	0.00	1.00	0.00	0.01	0.02
about N/S axis (rad)	0.00	-0.02	0.00	0.00	0.00	1.00
about E/W axis (rad)	0.08	0.00	0.00	-0.00	1.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	1.00	0.00	0.00



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

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PVAP (rev: 2.1c, 12/15/98)

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By: Bounds

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Technical Guideline, Sample Design

Soil Properties as Given

PStiff program was used to compute individual pile impedance

Double Amplitudes at point #3

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.01	0.00	0.00
E/W translation	0.00	0.00	0.00
Vert translation	0.00	0.00	0.01
Rocking about N/S axis	0.00	0.01	0.02
Rocking about E/W axis	0.02	0.00	0.02
Torsional	0.02	0.01	0.00

Total @ 3565 rpm	0.06	0.02	0.06
Coastdown @ 2451 rpm	0.07	0.02	0.08
Coastdown @ 2188 rpm	0.07	0.02	0.08
Coastdown @ 1381 rpm	0.08	0.02	0.05
Coastdown @ 1342 rpm	0.08	0.02	0.05
Coastdown @ 1035 rpm	0.04	0.01	0.03
Coastdown @ 989 rpm	0.03	0.01	0.02

Double Amplitudes at East end

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.01	0.00	0.00
E/W translation	0.00	0.00	0.00
Vert translation	0.00	0.00	0.01
Rocking about N/S axis	0.00	0.02	0.02
Rocking about E/W axis	0.04	0.00	0.00
Torsional	0.02	0.00	0.00

Total @ 3565 rpm	0.08	0.02	0.04
Coastdown @ 2451 rpm	0.10	0.02	0.05
Coastdown @ 2188 rpm	0.10	0.02	0.05
Coastdown @ 1381 rpm	0.09	0.01	0.03
Coastdown @ 1342 rpm	0.09	0.01	0.03
Coastdown @ 1035 rpm	0.04	0.00	0.02
Coastdown @ 989 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #1 (Soil Properties As Given)

FLUOR

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Technical Guideline, Sample Design

Soil Properties as Given

PStiff program was used to compute individual pile impedance

Double Amplitudes at West end

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation		0.01	0.00	0.00
E/W translation		0.00	0.00	0.00
Vert translation		0.00	0.00	0.01
Rocking about N/S axis		0.00	0.02	0.02
Rocking about E/W axis		0.04	0.00	0.00
Torsional		0.02	0.00	0.00
		-----	-----	-----
Total @	3565 rpm	0.08	0.02	0.04
Coastdown @	2451 rpm	0.10	0.02	0.05
Coastdown @	2188 rpm	0.10	0.02	0.05
Coastdown @	1381 rpm	0.09	0.01	0.03
Coastdown @	1342 rpm	0.09	0.01	0.03
Coastdown @	1035 rpm	0.04	0.00	0.02
Coastdown @	989 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

The current version of PVAP uses a slightly different method in the computation of pile group factors and embedment impedance from that described in Attachment 10. Results are not exact.

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Technical Guideline, Sample Design
Soil Properties 20% Stronger
PStiff program was used to compute individual pile impedance

Machine Forces:

Machine Data-	Location of Forces-
Centrifugal machine.	N/S coordinate = N 0.000 ft
Speed = 3565 rpm	E/W coordinate = E 4.083 ft
	Elevation = 0.000 ft
Direction-	Primary-
N/S force	0.26 K
E/W force	0.00 K
Vertical force	0.26 K
Moment about N/S axis	0.00 K-ft
Moment about E/W axis	0.00 K-ft
Torsional moment	0.00 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec ² -ft)
The Machine	S 0.083	E 0.198	-0.750	22.180	0 0 0

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #3	N 2.417	W 6.833	-2.333
East end	N 0.000	E 6.833	0.000
West end	N 0.000	W 6.833	0.000



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

FLUOR

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

PStiff program was used to compute individual pile impedance

Foundation Data:

- * Foundation geometry will be checked by the analysis.
- * Group stiffness and damping will be computed from individual pile data.
- * Piles will be located as specified.
- * Pile caps will be located as specified.
- * Embedment effects will be included.

Elevations-

Grade elevation = -3.83 ft

T.O. pile elevation = -7.83 ft

Embedment-

Effective height = 1.33 ft

Centroid elev = -7.17 ft

Geotechnical:

Unit weight

Shear Modulus

Poisson's ratio

soil:

110 lb/ft³

864 ksf

0.50

backfill:

125 lb/ft³

2160 ksf

0.30

Individual Pile Impedance:

Pile diameter = 12.00 in
Pile E * I = 10604.00 K-ft²
Internal damping = 0.00 %

@ center of stiffness:

variation factor

horizontal

vertical

rocking

torsion

cross term

stiffness:

1.00

6733 K/ft

20949 K/ft

12108 K-ft/rad

1825 K-ft/rad

-6574 K/rad

damping:

1.00

13.80 K-sec/ft

27.40 K-sec/ft

7.20 K-ft-sec/rad

0.20 K-ft-sec/rad

-8.20 K-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

FLUOR

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

PStiff program was used to compute individual pile impedance

Typical Pile Layout:

			E/W piles		
		E/W coord	E/W spacing		
N			<---->		
		0	0	0	0
^					
^					
	-0	0	0	0	0
N/S	^				
piles					
	N/S				
	spacing v				
	-0	0	0	0	0
	N/S				
	coord-0	0	0	0	0
			PLAN-Pile Group		

File Group Data:

N/S piles: (ea)	N/S space: (ft)	E/W piles: (ea)	E/W space: (ft)	N/S coord: (ft)	E/W coord: (ft)
2	4.667	3	6.833	S 2.333	W 6.833

Typical Pier & Pile Cap Dimensioning:

[illegible]



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

PLAN	ELEV
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Soil Properties 20% Stronger	
PStiff program was used to compute individual pile impedance	

Pile Cap Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pile Cap	7.667	16.667	2.000	S 3.833	W 8.333	-7.833

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pier	4.833	13.667	3.500	S 2.417	W 6.833	-5.833

Results of Mass Computation:

Center of gravity:		Weight & Mass:	
N/S coordinate =	S 0.017 ft	Total weight =	110.63 K
E/W coordinate =	E 0.040 ft	Machine weight =	22.18 K
elevation =	-4.472 ft	Mass =	3.44 K-sec ² /ft
		Mass ratio =	3.99 to 1
Footing data:		Mass moments @ CG:	
Maximum pile load =	18.74 K	about N/S axis =	77 K-sec ² -ft
Minimum pile load =	18.14 K	about E/W axis =	30 K-sec ² -ft
		about Vert axis =	71 K-sec ² -ft
N/S eccentricity =	0.02 ft		
E/W eccentricity =	0.04 ft		

Pile Group Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	15504 K/ft	32 K-sec/ft
E/W translation	13797 K/ft	28 K-sec/ft
vertical	54865 K/ft	72 K-sec/ft
about N/S axis	3985074 K-ft/rad	5160 K-ft-sec/rad
about E/W axis	757080 K-ft/rad	938 K-ft-sec/rad
torsional	1488378 K-ft/rad	3029 K-ft-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

PStiff program was used to compute individual pile impedance

Embedment Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	11613 K/ft	236 K-sec/ft
E/W translation	11613 K/ft	236 K-sec/ft
vertical	7774 K/ft	165 K-sec/ft
about N/S axis	124657 K-ft/rad	4578 K-ft-sec/rad
about E/W axis	100471 K-ft/rad	1881 K-ft-sec/rad
torsional	0 K-ft/rad	0 K-ft-sec/rad

Total Impedance Results:

Center of stiffness:	Pile group factors:
N/S coordinate = N 0.001 ft	N/S translation = 2.61
E/W coordinate = W 0.000 ft	E/W translation = 2.93
Elevation = -7.833 ft	Vert translation = 2.29

@ center of stiffness:	stiffness:	damping:
N/S translation	27117 K/ft	268 K-sec/ft
E/W translation	25410 K/ft	265 K-sec/ft
vertical	62639 K/ft	237 K-sec/ft
about N/S axis	4530756 K-ft/rad	10388 K-ft-sec/rad
about E/W axis	1297781 K-ft/rad	3509 K-ft-sec/rad
torsional	1488407 K-ft/rad	3029 K-ft-sec/rad

Results of Modal Analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	723	791	1289	1381	2040	2321
Resonant freq (rpm)	991	1038	1383	1412	2231	2525
Resonant Freq. ratio	3.60	3.44	2.58	2.52	1.60	1.41
Dynamic Magnification	0.04	0.05	0.15	0.18	0.44	0.62
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	-0.01	-0.71	0.00
E/W trans (ft)	0.00	1.00	-0.00	-0.00	0.00	0.48
Vertical trans (ft)	-0.00	0.00	1.00	0.00	0.01	0.02
about N/S axis (rad)	0.00	-0.02	0.00	0.00	0.00	1.00
about E/W axis (rad)	0.08	0.00	0.00	-0.00	1.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	1.00	0.00	0.00



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

PStiff program was used to compute individual pile impedance

Double Amplitudes at point #3

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation		0.02	0.00	0.00
E/W translation		0.00	0.00	0.00
Vert translation		0.00	0.00	0.01
Rocking about N/S axis		0.00	0.01	0.02
Rocking about E/W axis		0.02	0.00	0.02
Torsional		0.02	0.01	0.00
		-----	-----	-----
Total @	3565 rpm	0.06	0.02	0.06
Coastdown @	2525 rpm	0.07	0.02	0.08
Coastdown @	2231 rpm	0.07	0.02	0.08
Coastdown @	1412 rpm	0.08	0.03	0.05
Coastdown @	1383 rpm	0.08	0.03	0.05
Coastdown @	1038 rpm	0.04	0.01	0.03
Coastdown @	991 rpm	0.03	0.01	0.02

Double Amplitudes at East end

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation		0.02	0.00	0.00
E/W translation		0.00	0.00	0.00
Vert translation		0.00	0.00	0.01
Rocking about N/S axis		0.00	0.02	0.02
Rocking about E/W axis		0.04	0.00	0.00
Torsional		0.02	0.00	0.00
		-----	-----	-----
Total @	3565 rpm	0.08	0.02	0.04
Coastdown @	2525 rpm	0.10	0.02	0.05
Coastdown @	2231 rpm	0.10	0.02	0.05
Coastdown @	1412 rpm	0.09	0.01	0.04
Coastdown @	1383 rpm	0.09	0.01	0.03
Coastdown @	1038 rpm	0.04	0.00	0.02
Coastdown @	991 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #2 (Soil 20 Percent Stronger)

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

PStiff program was used to compute individual pile impedance

Double Amplitudes at West end

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation		0.02	0.00	0.00
E/W translation		0.00	0.00	0.00
Vert translation		0.00	0.00	0.01
Rocking about N/S axis		0.00	0.02	0.02
Rocking about E/W axis		0.04	0.00	0.00
Torsional		0.02	0.00	0.00
		-----	-----	-----
Total @	3565 rpm	0.08	0.02	0.04
Coastdown @	2525 rpm	0.10	0.02	0.05
Coastdown @	2231 rpm	0.10	0.02	0.05
Coastdown @	1412 rpm	0.10	0.01	0.04
Coastdown @	1383 rpm	0.09	0.01	0.03
Coastdown @	1038 rpm	0.04	0.00	0.02
Coastdown @	991 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

The current version of PVAP uses a slightly different method in the computation of pile group factors and embedment impedance from that described in Attachment 10. Results are not exact.

FLUOR
PVAP (rev: 2.1c, 12/15/98)
Technical Guideline, Sample Design
Soil Properties 20% Weaker
PStiff program was used to compute individual pile impedance

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Machine Forces:

Machine Data-	Location of Forces-
Centrifugal machine.	N/S coordinate = N 0.000 ft
Speed = 3565 rpm	E/W coordinate = E 4.083 ft
	Elevation = 0.000 ft
Direction-	Primary-
N/S force	0.26 K
E/W force	0.00 K
Vertical force	0.26 K
Moment about N/S axis	0.00 K-ft
Moment about E/W axis	0.00 K-ft
Torsional moment	0.00 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec2-ft)
The Machine	S 0.083	E 0.198	-0.750	22.180	0 0 0

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #3	N 2.417	W 6.833	-2.333
East end	N 0.000	E 6.833	0.000
West end	N 0.000	W 6.833	0.000



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

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Soil Properties 20% Weaker
PStiff program was used to compute individual pile impedance

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Foundation Data:

- * Foundation geometry will be checked by the analysis.
- * Group stiffness and damping will be computed from individual pile data.
- * Piles will be located as specified.
- * Pile caps will be located as specified.
- * Embedment effects will be included.

Elevations-		Embedment-	
Grade elevation	= -3.83 ft	Effective height	= 1.33 ft
T.O. pile elevation	= -7.83 ft	Centroid elev	= -7.17 ft

Geotechnical:	soil:	backfill:
Unit weight	110 lb/ft ³	125 lb/ft ³
Shear Modulus	600 ksf	2160 ksf
Poisson's ratio	0.50	0.30

Individual Pile Impedance:

Pile diameter = 12.00 in
Pile E * I = 10604.00 K-ft²
Internal damping = 0.00 %

@ center of stiffness:	stiffness:	damping:
variation factor	1.00	1.00
horizontal	5780 K/ft	13.80 K-sec/ft
vertical	18009 K/ft	25.60 K-sec/ft
rocking	11681 K-ft/rad	7.80 K-ft-sec/rad
torsion	1507 K-ft/rad	0.30 K-ft-sec/rad
cross term	-6038 K/rad	-8.50 K-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

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Technical Guideline, Sample Design

Soil Properties 20% Weaker

PStiff program was used to compute individual pile impedance.

Typical Pile Layout:

		E/W piles			
		E/W coord	E/W spacing		
			<---->		
N		0	0	0	0
^					
^					
		-0	0	0	0
N/S		^			
piles	N/S				
	spacing	v			
		-0	0	0	0
	N/S				
	coord-0	0	0	0	0
			PLAN-Pile	Group	

Pile Group Data:

N/S piles: (ea)	N/S space: (ft)	E/W piles: (ea)	E/W space: (ft)	N/S coord: (ft)	E/W coord: (ft)
2	4.667	3	6.833	S 2.333	W 6.833

Typical Pier & Pile Cap Dimensioning:

[illegible]



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

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Pile Cap Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pile Cap	7.667	16.667	2.000	S 3.833	W 8.333	-7.833

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Pier	4.833	13.667	3.500	S 2.417	W 6.833	-5.833

Results of Mass Computation:

Center of gravity:		Weight & Mass:	
N/S coordinate =	S 0.017 ft	Total weight =	110.63 K
E/W coordinate =	E 0.040 ft	Machine weight =	22.18 K
elevation =	-4.472 ft	Mass =	3.44 K-sec ² /ft
		Mass ratio =	3.99 to 1
Footing data:		Mass moments @ CG:	
Maximum pile load =	18.74 K	about N/S axis =	77 K-sec ² -ft
Minimum pile load =	18.14 K	about E/W axis =	30 K-sec ² -ft
		about Vert axis =	71 K-sec ² -ft
N/S eccentricity =	0.02 ft		
E/W eccentricity =	0.04 ft		

Pile Group Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	13309 K/ft	32 K-sec/ft
E/W translation	11844 K/ft	28 K-sec/ft
vertical	47166 K/ft	67 K-sec/ft
about N/S axis	3433439 K-ft/rad	4828 K-ft-sec/rad
about E/W axis	658464 K-ft/rad	883 K-ft-sec/rad
torsional	1277353 K-ft/rad	3030 K-ft-sec/rad



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

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Technical Guideline, Sample Design

Soil Properties 20% Weaker

PStiff program was used to compute individual pile impedance

Embedment Impedance Results:

@ center of stiffness:	stiffness:	damping:
N/S translation	11613 K/ft	236 K-sec/ft
E/W translation	11613 K/ft	236 K-sec/ft
vertical	7774 K/ft	165 K-sec/ft
about N/S axis	124657 K-ft/rad	4578 K-ft-sec/rad
about E/W axis	100471 K-ft/rad	1881 K-ft-sec/rad
torsional	0 K-ft/rad	0 K-ft-sec/rad

Total Impedance Results:

Center of stiffness:	Pile group factors:
N/S coordinate = N 0.001 ft	N/S translation = 2.61
E/W coordinate = W 0.000 ft	E/W translation = 2.93
Elevation = -7.833 ft	Vert translation = 2.29

@ center of stiffness:	stiffness:	damping:
N/S translation	24922 K/ft	268 K-sec/ft
E/W translation	23457 K/ft	265 K-sec/ft
vertical	54940 K/ft	232 K-sec/ft
about N/S axis	3935438 K-ft/rad	10068 K-ft-sec/rad
about E/W axis	1152763 K-ft/rad	3466 K-ft-sec/rad
torsional	1277377 K-ft/rad	3030 K-ft-sec/rad

Results of Modal Analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	697	760	1208	1280	1923	2164
Resonant freq (rpm)	986	1027	1304	1313	2124	2372
Resonant Freq. ratio	3.61	3.47	2.73	2.71	1.68	1.50
Dynamic Magnification	0.04	0.05	0.13	0.15	0.37	0.51
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	-0.01	-0.71	0.00
E/W trans (ft)	0.00	1.00	-0.00	-0.01	0.00	0.50
Vertical trans (ft)	-0.00	0.00	1.00	0.00	0.01	0.02
about N/S axis (rad)	0.00	-0.02	0.00	0.00	0.00	1.00
about E/W axis (rad)	0.08	0.00	0.00	-0.00	1.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	1.00	0.00	0.00



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

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Technical Guideline, Sample Design

Soil Properties 20% Weaker

PStiff program was used to compute individual pile impedance

Double Amplitudes at point #3

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.01	0.00	0.00
E/W translation	0.00	0.00	0.00
Vert translation	0.00	0.00	0.01
Rocking about N/S axis	0.00	0.01	0.02
Rocking about E/W axis	0.02	0.00	0.02
Torsional	0.02	0.01	0.00

Total @ 3565 rpm	0.05	0.01	0.06
Coastdown @ 2372 rpm	0.07	0.02	0.08
Coastdown @ 2124 rpm	0.07	0.02	0.08
Coastdown @ 1313 rpm	0.08	0.02	0.05
Coastdown @ 1304 rpm	0.08	0.02	0.05
Coastdown @ 1027 rpm	0.04	0.01	0.03
Coastdown @ 986 rpm	0.04	0.01	0.03

Double Amplitudes at East end

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.01	0.00	0.00
E/W translation	0.00	0.00	0.00
Vert translation	0.00	0.00	0.01
Rocking about N/S axis	0.00	0.01	0.02
Rocking about E/W axis	0.04	0.00	0.00
Torsional	0.02	0.00	0.00

Total @ 3565 rpm	0.07	0.02	0.04
Coastdown @ 2372 rpm	0.09	0.02	0.05
Coastdown @ 2124 rpm	0.10	0.02	0.04
Coastdown @ 1313 rpm	0.09	0.01	0.03
Coastdown @ 1304 rpm	0.09	0.01	0.03
Coastdown @ 1027 rpm	0.05	0.00	0.02
Coastdown @ 986 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

PVAP Analysis, Run #3 (Soil 20 Percent Weaker)

FLUOR

Date: 8/24/96

PVAP (rev: 2.1c, 12/15/98)

Contract: 215-1235

By: Bounds

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Technical Guideline, Sample Design

Soil Properties 20% Weaker

PStiff program was used to compute individual pile impedance

Double Amplitudes at West end

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.01	0.00	0.00
E/W translation	0.00	0.00	0.00
Vert translation	0.00	0.00	0.01
Rocking about N/S axis	0.00	0.01	0.02
Rocking about E/W axis	0.04	0.00	0.00
Torsional	0.02	0.00	0.00

Total @ 3565 rpm	0.08	0.02	0.04
Coastdown @ 2372 rpm	0.09	0.02	0.05
Coastdown @ 2124 rpm	0.10	0.02	0.05
Coastdown @ 1313 rpm	0.09	0.01	0.03
Coastdown @ 1304 rpm	0.09	0.01	0.03
Coastdown @ 1027 rpm	0.05	0.00	0.02
Coastdown @ 986 rpm	0.04	0.00	0.02



VIBRATING MACHINERY FOUNDATIONS ON PILES

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Analysis Procedure

A General Considerations

A1 The Equation of Motion

- A1a The vibration analysis of compressor foundations involves solving the following standard dynamic equation: (Several basic dynamics references are listed in Section K1.)

$$\begin{array}{ll} M y'' + C_t y' + K_t y = F(t) & \text{for translation} \quad (\text{Equation 10-1a}) \\ I \phi'' + C_r \phi' + K_r \phi = M(t) & \text{for rocking} \quad (\text{Equation 10-1b}) \end{array}$$

where,

M = mass (K-sec²/ft)
 I = mass moment of inertia (K-sec²-ft)
 C_t = damping constant (K-sec/ft)
 C_r = rotational damping constant (ft-K-sec/rad)
 K_t = stiffness (K/ft)
 K_r = rotational stiffness (ft-K/rad)
 $F(t)$ = dynamic force (kips)
 $M(t)$ = dynamic moment (ft-K)
 y'' = acceleration (ft/sec²)
 y' = velocity (ft/sec)
 y = displacement (feet)
 ϕ'' = rotational acceleration (rad/sec²)
 ϕ' = rotational velocity (rad/sec)
 ϕ = rotational displacement (radians)

- A1b Three of each of the above equations are needed to describe the 3 translation and 3 rocking modes of vibration. Forces are determined at the foundation-machine-soil center of gravity.
- A1c The first term, mass times acceleration, is the inertia force. Details of mass computations are provided in Section B.
- A1d The next two terms, damping times velocity and stiffness times displacement, represent forces from the soil and are also termed the soil impedance. Details of the impedance calculation are provided in Sections D, E, and F.
- A1e The dynamic force is normally a combination of sinusoidal functions. This force is preferably provided by the machine supplier. Alternately, standard criteria can be used as described in the main guideline.

A2 Equation Solutions

- A2a Natural Frequencies - The use of frequency dependent impedance implies that natural frequencies cannot be directly determined. The method indicated in Sections G5 and H5 use an iteration where a trial frequency is assumed, the impedance is determined, and a resulting frequency is computed from undamped conditions. The correct natural frequency is found when the trial and resulting frequencies match.



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A2b Forced response - The equations for acceleration, velocity, and displacement provided in Sections H and I are derived using the Complex Method (References 2d and 4c). This method is suitable for either frequency dependent or independent soil impedance. It is also a direct solution in that natural frequencies and mode shapes are not a part of the solution sequence. Details of this solution method are provided in Sections H and I.

A3 *Solution Notes*

A3a The solution method provided in this attachment takes advantage of independent vibration modes. Instead of having to solve 6 coupled equations, a decoupling takes place when the system center of gravity is directly above the mat centerlines. With this decoupling, vertical translation and torsional rocking are solved as single degree of freedom modes. Translation and rocking in each horizontal direction are solved as coupled modes with 2 degrees of freedom. The alignment of the center of gravity over the mat centerline is computed in Section B and checked according to criteria provided in the main guideline.

A3b A rigid foundation is assumed. The 6 vibration modes described above and the soil impedance criteria presented in this attachment depend on a rigid foundation. Otherwise, a finite element analysis modeling the machine, pier, mat, and possibly the soil may be required for an accurate analysis. Criteria for checking the mat rigidity is presented in the main guideline.

A3c Phase angles are neglected in this procedure. The direction of reciprocating machine forces are usually known, but may be subject to change at a later date. The direction of centrifugal machine forces cannot be predicted, normally worst case orientations are used. Amplitudes are conservatively computed in this procedure by adding peak values from each mode.

A3d A reference coordinate system is required. Any reasonably located origin will do. An origin at the centerline and base of the foundation mat is often used. An alternate would be to use an origin at the bottom of baseplate in order to minimize location changes when mat or pier sizes are revised. In this technical attachment, an X axis parallel to the shaft, a Y axis perpendicular to the shaft, and a vertical Z axis will be utilized as shown in Attachment 11.

A4 *Software*

Computer programs are available which are capable of performing most or all of the calculations described in this attachment. Though some time is required to input data and to print results, the analysis usually requires only a matter of seconds. Since performing these calculations by hand is quite tedious and time consuming, use of a computer is strongly recommended. Several computer programs are listed in the reference section. It should be noted that exact solution methods will vary for particular programs.

B *Mass Calculations*

B1 Mass properties can be calculated using standard equations. In the following procedure, the machine weight, total weight, mass moments, and center of gravity are computed simultaneously from the reference origin described in Section A10. The vibration calculation (refer to Attachment 17) uses a slight variation in order to locate the mat under the center of gravity.

TW = Foundation, machine, and soil weight (kips)



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MW = Machine weight (kips)
 I_x, I_y, I_z = Mass moments of inertia, about each axis (K-sec²-ft)
 CG_x, CG_y, CG_z = Center of gravity location with respect to the reference origin (ft)

It is assumed that the machine and foundation are symmetric enough to ignore cross mass moments of inertia. Connected pipe could contribute to the mass if it vibrates rigidly with the machine. However, connected pipe is usually insignificant compared to the total machine foundation mass.

- B2 The concrete foundation, soil, and the machine are divided into a series of components using the following definitions. Attachment 11 provides a pictorial definition of the dimensional terms.

CW = Weight of component (kips)
 $\partial_x, \partial_y, \partial_z$ = Distance from the origin to the component's center in each axis (ft)
 L_x, L_y, L_z = Dimensions of the component in each axis (ft)
 E_x, E_y, E_z = Component weight times distance from origin (ft-K)
 Q_x, Q_y, Q_z = Component mass moment of inertia, about component CG (K-sec²-ft)
 J_x, J_y, J_z = Component's translated mass moment of inertia (K-sec²-ft)
 W_c = Density of concrete (K/ft³)
 W_f = Density of compacted fill material (K/ft³)
gravity = Acceleration of gravity (32.2 ft/sec²)

- B3 The equations given in the following sections for component properties are standard formulae for rectangular blocks. Nonrectangular blocks can be used if the proper equations for weight (CW) and mass moment (Q_x, Q_y, Q_z) are substituted. Attachment 12 provides some criteria for nonrectangular blocks.

- B4 The concrete foundation and soil above the mat are divided into a series of components with the following properties and calculated values:

$CW = (L_x)(L_y)(L_z)(W_c)$	Concrete components	(Equation 10-2a)
$CW = (L_x)(L_y)(L_z)(W_f)$	Soil components	(Equation 10-2b)

$E_x = CW (\partial_x)$	(Equation 10-3a)
$E_y = CW (\partial_y)$	(Equation 10-3b)
$E_z = CW (\partial_z)$	(Equation 10-3c)

$J_x = CW [\partial_y^2 + \partial_z^2]$	(Equation 10-4a)
$J_y = CW [\partial_x^2 + \partial_z^2]$	(Equation 10-4b)
$J_z = CW [\partial_x^2 + \partial_y^2]$	(Equation 10-4c)

$Q_x = CW [L_y^2 + L_z^2] / 12$	(Equation 10-5a)
$Q_y = CW [L_x^2 + L_z^2] / 12$	(Equation 10-5b)
$Q_z = CW [L_x^2 + L_y^2] / 12$	(Equation 10-5c)

- B5 Machine components are now included with the following properties:

CW, Q_x, Q_y, Q_z = provided by the machine supplier



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Equations 10-3a-c and 10-4a-c are used to compute needed properties about the origin.

- B6 The center of gravity can now be computed by summing individual computed values:

$$\begin{array}{ll} MW = \sum (CW) & \text{Machine components only} \\ TW = \sum (CW) & \text{Concrete, soil, and machine components} \end{array}$$

$$CG_x = \sum (E_x) / TW \quad (\text{Equation 10-6a})$$

$$CG_y = \sum (E_y) / TW \quad (\text{Equation 10-6b})$$

$$CG_z = \sum (E_z) / TW \quad (\text{Equation 10-6c})$$

- B7 The mass moments are computed by summing individual computed values, adjusting for the actual center of gravity, and dividing by gravity.

$$I_x = [\sum (Q_x) + \sum (J_x) - TW (CG_y^2 + CG_z^2)] / \text{gravity} \quad (\text{Equation 10-7a})$$

$$I_y = [\sum (Q_y) + \sum (J_y) - TW (CG_x^2 + CG_z^2)] / \text{gravity} \quad (\text{Equation 10-7b})$$

$$I_z = [\sum (Q_z) + \sum (J_z) - TW (CG_x^2 + CG_y^2)] / \text{gravity} \quad (\text{Equation 10-7c})$$

- B8 The mass and mass ratio can now be computed. The mass ratio is compared using the acceptance criteria given in the main guideline.

$$\begin{array}{ll} \text{mass} & = \text{Translational mass (K-sec}^2/\text{ft)} \\ \text{MR} & = \text{Mass ratio} \end{array}$$

$$\text{mass} = TW / \text{gravity} \quad (\text{Equation 10-8})$$

$$\text{MR} = (TW - MW) / MW \quad (\text{Equation 10-9})$$

C Pile and Pile Cap Properties

- C1 Properties of the pile pattern are needed:

$$\begin{array}{ll} N & = \text{Total number of piles (ea)} \\ \partial_x, \partial_y & = \text{Distance from origin to a particular pile (ft)} \\ CP_x, CP_y & = \text{Distance from origin to centroid of pile pattern (ft)} \\ \sum x^2, \sum y^2 & = \text{Summed square of distances to pile centroid (ft}^2\text{)} \end{array}$$

Properties are computed by summing values from each pile location:

$$CP_x = \sum (\partial_x) / N \quad (\text{Equation 10-10a})$$

$$CP_y = \sum (\partial_y) / N \quad (\text{Equation 10-10b})$$

$$\sum x^2 = \sum (\partial_x - CP_x)^2 \quad (\text{Equation 10-11a})$$

$$\sum y^2 = \sum (\partial_y - CP_y)^2 \quad (\text{Equation 10-11b})$$



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- C2 Alignment offsets are now computed to confirm the assumptions of Section A6. Comparisons are made using the acceptance criteria provided in the Narrative.

$$\begin{aligned} OS_x, OS_y &= \text{CG offset in each direction (\%)} \\ L_x, L_y &= \text{Mat dimensions (ft)} \end{aligned}$$

$$OS_x = \text{abs} \{ 100 (CG_x - CP_x) / L_x \} \quad (\text{Equation 10-12a})$$

$$OS_y = \text{abs} \{ 100 (CG_y - CP_y) / L_y \} \quad (\text{Equation 10-12b})$$

- C3 The pile load must be kept low to ensure an elastic and predictable response. Pile loads will be computed and compared using the acceptance criteria provided in the Narrative.

$$PL = \text{Maximum pile load (kips)}$$

$$PL = TW / N \quad (\text{Equation 10-13})$$

D Pile Impedance Due To Individual Piles

- D1 For the majority of cases, uniform soil conditions will be encountered. The impedance equations, from Novak, provided in this section are valid for uniform soil conditions. Only very modest variations in soil properties can be accommodated. Attachment 13 is provided for use in hand calculations as an aid to the listed equations.

Impedance results for nonuniform conditions vary widely. Because variations with frequency also change, effective soil properties should not be used to approximate layered soil. There are 2 options for obtaining accurate impedance values:

- From the soil consultant: This is the preferred option if the consultant is experienced at the evaluation of dynamic soil impedance. The soil consultant should be more knowledgeable of site soil conditions and may have specialized software or evaluation methods.
- From alternate criteria: Reference 3d provides an extension of Novak's criteria for layered soil. This extended procedure is too complex for hand calculation. Reference Programs 7a and 7b are capable of performing this analysis.

- D2 Apply Novak's equations (References 3a, 3b, and 3c) to compute stiffness and damping for an individual pile. Attachment 13 may be used to determine stiffness and damping values for selected pile types using the dimensionless pile frequency computed in Section D3. A second set of impedance values must be determined for reciprocating machines with secondary forces. For detailed hand calculations, standard mathematical tables may be used to evaluate Bessel functions.

$$K_{tx}^i, K_{ty}^i, K_{tz}^i = \text{Pile translation stiffness values (K/ft)}$$

$$K_{rx}^i, K_{ry}^i, K_{rz}^i = \text{Pile rotational stiffness values (ft-K/rad)}$$

$$K_{cx}^i, K_{cy}^i = \text{Pile cross stiffness values (K/rad)}$$

$$C_{tx}^i, C_{ty}^i, C_{tz}^i = \text{Pile translation damping values (K-sec/ft)}$$

$$C_{rx}^i, C_{ry}^i, C_{rz}^i = \text{Pile rotational damping values (ft-K-sec/rad)}$$



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Analysis Procedure

C_{cx}^i, C_{cy}^i = Pile cross damping values (K-sec/rad)

D3 Compute dimensionless pile properties:

A_o = Pile dimensionless ratio
 S_1 = Dimensionless parameter for vertical motion (complex)
 S_2 = Dimensionless parameter for torsional motion (complex)
 S_h = Dimensionless parameter for horizontal and rocking motion (complex)
 MS = Machine speed (rad/sec)
 R_p = Pile radius (ft)
 W_s = Soil density (K/ft³)
 G_s = Soil shear modulus (K/ft²)
 μ_s = Poisson's ratio for soil
 q = Dimensionless value
 X_o = Dimensionless value
 $H_0(x)$ = Bessel function of the third kind, of order zero, evaluated at x (complex)
 $H_1(x)$ = Bessel function of the third kind, of order one, evaluated at x (complex)
 $H_2(x)$ = Bessel function of the third kind, of order two, evaluated at x (complex)

$$A_o = (MS)(R_p) \sqrt{\frac{W_s}{(G_s)(\text{gravity})}} \quad (\text{Equation 10-14})$$

$$q = \frac{2(1 - \mu_s)}{1 - 2(\mu_s)} \quad (\text{Equation 10-15})$$

$$X_o = A_o / \sqrt{q} \quad (\text{Equation 10-16})$$

$$S_1 = 2\pi A_o \frac{H_1(A_o)}{H_0(A_o)} \quad (\text{Equation 10-17a})$$

$$S_2 = 4\pi - 2\pi A_o \frac{H_0(A_o)}{H_1(A_o)} \quad (\text{Equation 10-17b})$$

$$S_h = 2\pi A_o \frac{(1/\sqrt{q}) H_2(A_o) H_1(A_o) + H_2(A_o) H_1(X_o)}{H_0(A_o) H_2(X_o) + H_2(X_o) H_0(A_o)} \quad (\text{Equation 10-17c})$$

D4 Compute complex frequency parameters:

Ω_v = Frequency parameter for vertical motion (complex)
 Ω_x, Ω_y = Frequency parameter for horizontal and rocking motion (complex)
 Ω_z = Frequency parameter for torsional motion (complex)
 L_p = Length of pile (ft)



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Analysis Procedure

I_{px}, I_{py} = Pile moment of inertia (ft⁴)
 I_{pz} = Pile polar moment of inertia (ft⁴)
 W_p = Pile material density (K/ft³)
 E_p = Pile modulus of elasticity (K/ft²)
 A_p = Pile Area (ft²)

$$\Omega_v = \sqrt{\frac{L_p^2}{E_p A_p} \left[\frac{W_p A_p M S^2}{\text{gravity}} - G_s S_1 \right]} \quad (\text{Equation 10-18a})$$

$$\Omega_x = \sqrt[4]{\frac{L_p^2}{E_p I_{px}} \left[\frac{W_p A_p M S^2}{\text{gravity}} - G_s S_h \right]} \quad (\text{Equation 10-18b})$$

$$\Omega_y = \sqrt[4]{\frac{L_p^2}{E_p I_{py}} \left[\frac{W_p A_p M S^2}{\text{gravity}} - G_s S_h \right]} \quad (\text{Equation 10-18c})$$

$$\Omega_z = \sqrt{\frac{L_p^2}{E_p I_{pz}} \left[\frac{W_p I_{pz} M S^2}{\text{gravity}} - G_s R_p S_2 \right]} \quad (\text{Equation 10-18d})$$

D5 Compute stiffness and damping coefficients. The equations shown are valid for friction piles with a fixed top connection to the pile cap and homogenous soil properties. The stiffness and damping coefficients are evaluated as the real and imaginary components respectively, of the complex formula on the right side of the equation.

S_{tx}^i, S_{tz}^i = Pile translation stiffness coefficient
 S_{rx}^i, S_{rz}^i = Pile rotational stiffness coefficient
 S_{cx}^i, S_{cy}^i = Pile cross stiffness coefficient

N_{tx}^i, N_{tz}^i = Pile translation damping coefficient
 N_{rx}^i, N_{rz}^i = Pile rotational damping coefficient
 N_{cx}^i, N_{cy}^i = Pile cross damping coefficient

i = Complex number coefficient

$$S_{tx}^i + i (N_{tx}^i) = -(\Omega_x)^3 \frac{\cos(\Omega_x) \sinh(\Omega_x) + \sin(\Omega_x) \cosh(\Omega_x)}{1 + \cos(\Omega_x) \cosh(\Omega_x)} \quad (\text{Equation 10-19a})$$

$$S_{ty}^i + i (N_{ty}^i) = -(\Omega_y)^3 \frac{\cos(\Omega_y) \sinh(\Omega_y) + \sin(\Omega_y) \cosh(\Omega_y)}{1 + \cos(\Omega_y) \cosh(\Omega_y)} \quad (\text{Equation 10-19b})$$

$$S_{tz}^i + i (N_{tz}^i) = \Omega_v \tan(\Omega_v) \quad (\text{Equation 10-19c})$$



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$$S_{rx}^i + i (N_{rx}^i) = \Omega_y \frac{\cos(\Omega_y) \sinh(\Omega_y) - \sin(\Omega_y) \cosh(\Omega_y)}{1 + \cos(\Omega_y) \cosh(\Omega_y)} \quad (\text{Equation 10-19d})$$

$$S_{ry}^i + i (N_{ry}^i) = \Omega_x \frac{\cos(\Omega_x) \sinh(\Omega_x) - \sin(\Omega_x) \cosh(\Omega_x)}{1 + \cos(\Omega_x) \cosh(\Omega_x)} \quad (\text{Equation 10-19e})$$

$$S_{rz}^i + i (N_{rz}^i) = \Omega_z \tan(\Omega_z) \quad (\text{Equation 10-19f})$$

$$S_{cx}^i + i (N_{cx}^i) = -(\Omega_x)^2 \frac{\sin(\Omega_x) \sinh(\Omega_x)}{1 + \cos(\Omega_x) \cosh(\Omega_x)} \quad (\text{Equation 10-19g})$$

$$S_{cy}^i + i (N_{cy}^i) = -(\Omega_y)^2 \frac{\sin(\Omega_y) \sinh(\Omega_y)}{1 + \cos(\Omega_y) \cosh(\Omega_y)} \quad (\text{Equation 10-19h})$$

D6 Pile stiffness values are computed from the following:

$$K_{tx}^i = E_p I_x S_{tx}^i / L_p^3 \quad (\text{Equation 10-20a})$$

$$K_{ty}^i = E_p I_y S_{ty}^i / L_p^3 \quad (\text{Equation 10-20b})$$

$$K_{tz}^i = E_p A_p S_{tz}^i / L_p \quad (\text{Equation 10-20c})$$

$$K_{rx}^i = E_p I_x S_{rx}^i / L_p \quad (\text{Equation 10-20d})$$

$$K_{ry}^i = E_p I_y S_{rx}^i / L_p \quad (\text{Equation 10-20e})$$

$$K_{rz}^i = E_p I_{xy} S_{rz}^i / L_p \quad (\text{Equation 10-20f})$$

$$K_{cx}^i = E_p I_y S_{cx}^i / L_p \quad (\text{Equation 10-20g})$$

$$K_{cy}^i = E_p I_x S_{cy}^i / L_p \quad (\text{Equation 10-20h})$$

D7 Pile damping values are computed from the following:

$$C_{tx}^i = E_p I_x N_{tx}^i / W_s L_p^3 \quad (\text{Equation 10-21a})$$

$$C_{ty}^i = E_p I_y N_{ty}^i / W_s L_p^3 \quad (\text{Equation 10-21b})$$

$$C_{tz}^i = E_p A_p N_{tz}^i / W_s L_p \quad (\text{Equation 10-21c})$$

$$C_{rx}^i = E_p I_x N_{rx}^i / W_s L_p \quad (\text{Equation 10-21d})$$

$$C_{ry}^i = E_p I_y N_{rx}^i / W_s L_p \quad (\text{Equation 10-21e})$$



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$$C_{rz}^i = E_p I_{xy} N_{rz}^i / W_s L_p \quad (\text{Equation 10-21f})$$

$$C_{cx}^i = E_p I_y N_{cx}^i / W_s L_p \quad (\text{Equation 10-21g})$$

$$C_{cy}^i = E_p I_x N_{cy}^i / W_s L_p \quad (\text{Equation 10-21h})$$

E Group Impedance at Base of Pile Cap

- E1 The group interaction criteria presented in this section, from Poulos, is based on static conditions. Static criteria has been used widely in the past due to the lack of dynamic interaction criteria. It is known that interaction values for dynamic loads vary; however, generally accepted criteria has not yet been defined. Research into this aspect of dynamic design continues.

Less information is available concerning rocking interaction. The criteria presented to determine rocking interaction is only a suggested procedure based on a logical application of translation interaction factors.

- E2 Use Poulos' criteria (Reference 4a) to compute pile interaction:

$$\begin{aligned} GP_{tx}, GP_{ty}, GP_{tz} &= \text{Translation group reduction factors} \\ GP_{rx}, GP_{ry}, GP_{rz} &= \text{Rotation group reduction factors} \end{aligned}$$

- E3 Compute stiffness parameters:

$$\begin{aligned} K_v &= \text{Stiffness ratio for vertical interaction} \\ K_h &= \text{Stiffness ratio for horizontal interaction} \\ R_a &= \text{Pile solidity ratio} \\ A_p &= \text{Area of pile (ft}^2\text{)} \\ I_p &= \text{Pile moment of inertia (ft}^4\text{)} \end{aligned}$$

$$R_a = A_p / \pi R_p^2 \quad (\text{Equation 10-22})$$

$$K_v = E_p R_a / E_s \quad (\text{Equation 10-23})$$

$$K_h = E_p I_p / E_s L_p^4 \quad (\text{Equation 10-24})$$

- E4 Determine 2 pile interaction values using Attachment 15 for vertical interaction and Attachment 16 for horizontal interaction.

$$\begin{aligned} \alpha_v &= \text{Two pile interaction factor for vertical motion} \\ \alpha_h(\beta) &= \text{Two pile interaction factor for horizontal motion evaluated at departure angle } \beta \\ S_p &= \text{Spacing between 2 piles (ft)} \end{aligned}$$

For each unique spacing, S_p , between piles, three interaction factors should be determined: α_v , $\alpha_h(0^\circ)$, and $\alpha_h(90^\circ)$.

- E5 Compute averaged interaction values for translation. A generic description is provided below. Refer to the



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vibration calculation example for further clarification.

1. Create a table for each pile in the pile pattern. With some experience, symmetry in pile patterns can be used to reduce the total number of tables required. The pile selected for each table will be called the "reference pile." Enter a line in each table for every pile in the pattern, including the reference pile. Each line will then designate a "target pile."
2. For each line, determine the spacing between reference and target piles and the angles relative to the X and Y axes. Computed angles should be between 0 and 90 degrees. Where the reference and target piles are the same, spacing and angles will not be needed.
3. All necessary group factors have been determined in Step E4 above. For each line, enter the correct vertical and horizontal factors in separate columns in the table. Horizontal factors are applied by ratioing between departure angles of 0 degrees and 90 degrees. Where the reference and target piles are the same, the interaction factor is one.

4. Sum the 3 columns of interaction factors in each table.

α_x = Summation of X axis interaction factors
 α_y = Summation of Y axis interaction factors
 α_z = Summation of Z axis interaction factors

5. The total interaction factors are computed by averaging summed interaction factors for all reference piles.

$$GP_{tx} = \sum (\alpha_x) / N \quad \text{(Equation 10-25a)}$$

$$GP_{ty} = \sum (\alpha_y) / N \quad \text{(Equation 10-25b)}$$

$$GP_{tz} = \sum (\alpha_z) / N \quad \text{(Equation 10-25c)}$$

- E6 Compute averaged interaction values for rotation. There is no accepted procedure for determining rocking interaction factors. The suggested procedure listed below is similar to that used for translation except that considerable judgement must be used to include appropriate interaction conditions. Refer to the vibration calculation example for further clarification.

1. Include only reference and target pile combinations that are on the same side of the rocking axis.
2. Include only reference and target pile combinations with distances to the rocking axis that differ by less than a factor of 1/2 to 2.
3. For torsional rotation, include only reference and target pile combinations that form an angle of 22 degrees or less with the torsional axis.
4. Vertical interaction factors, α_z , are used for rocking about the X and Y axes. Horizontal interaction factors, α_x and α_y , are used for torsional interaction.



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5. Total interaction factors are computed by averaging summed interaction factors using the square of distances to the rocking axis.

$$GP_{rx} = \frac{\sum[(\alpha_z)(y)^2]}{\sum y^2} \quad (\text{Equation 10-26a})$$

$$GP_{ry} = \frac{\sum[(\alpha_z)(x)^2]}{\sum x^2} \quad (\text{Equation 10-26b})$$

$$GP_{rz} = \frac{\sum[(\alpha_z)(y)^2 + (\alpha_z)(x)^2]}{\sum x^2 + \sum y^2} \quad (\text{Equation 10-26c})$$

- E7 Compute group pile stiffness and damping at the pile centroid. A second set of stiffness and damping values is required for reciprocating machines with secondary forces using twice the machine speed.

K_{tx}^c, K_{tz}^c = Group translation stiffness values (K/ft)

K_{rx}^c, K_{rz}^c = Group rotational stiffness values (K-ft/rad)

K_{cx}^c, K_{cy}^c = Group cross stiffness values (K/rad)

C_{tx}^c, C_{tz}^c = Group translation damping values (K-sec/ft)

C_{rx}^c, C_{rz}^c = Group rotational damping values (K-ft-sec/rad)

C_{cx}^c, C_{cy}^c = Group cross damping values (K-sec/rad)

- E8 Group stiffness values are computed from the following:

$$K_{tx}^c = (K_{tx}^i)(N) / GP_{tx} \quad (\text{Equation 10-27a})$$

$$K_{ty}^c = (K_{ty}^i)(N) / GP_{ty} \quad (\text{Equation 10-27b})$$

$$K_{tz}^c = (K_{tz}^i)(N) / GP_{tz} \quad (\text{Equation 10-27c})$$

$$K_{rx}^c = [(K_{rx}^i)(N) + (K_{tz}^i)(\sum y^2)] / GP_{rx} \quad (\text{Equation 10-27d})$$

$$K_{ry}^c = [(K_{ry}^i)(N) + (K_{tz}^i)(\sum x^2)] / GP_{ry} \quad (\text{Equation 10-27e})$$

$$K_{rz}^c = [(K_{rz}^i)(N) + (K_{tx}^i)(\sum y^2) + (K_{ty}^i)(\sum x^2)] / GP_{rz} \quad (\text{Equation 10-27f})$$

$$K_{cx}^c = (K_{cx}^i)(N) \quad (\text{Equation 10-27g})$$

$$K_{cy}^c = (K_{cy}^i)(N) \quad (\text{Equation 10-27h})$$

- E9 Group damping values are computed from the following:

$$C_{tx}^c = (C_{tx}^i)(N) / GP_{tx} \quad (\text{Equation 10-28a})$$



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Analysis Procedure

$$C_{ty}^c = (C_{ty}^i)(N) / GP_{ty} \quad (\text{Equation 10-28b})$$

$$C_{tz}^c = (C_{tz}^i)(N) / GP_{tz} \quad (\text{Equation 10-28c})$$

$$C_{rx}^c = [(C_{rx}^i)(N) + (C_{tz}^i)(\Sigma y^2)] / GP_{rx} \quad (\text{Equation 10-28d})$$

$$C_{ry}^c = [(C_{ry}^i)(N) + (C_{tz}^i)(\Sigma x^2)] / GP_{ry} \quad (\text{Equation 10-28e})$$

$$C_{rz}^c = [(C_{rz}^i)(N) + (C_{tx}^i)(\Sigma y^2) + (C_{ty}^i)(\Sigma x^2)] / GP_{rx} \quad (\text{Equation 10-28f})$$

$$C_{cx}^c = (C_{cx}^i)(N) \quad (\text{Equation 10-28g})$$

$$C_{cy}^c = (C_{cy}^i)(N) \quad (\text{Equation 10-28h})$$

F Soil Impedance Due to Pile Cap Embedment

F1 The majority of published design criteria for embedment impedance is by Novak. Other published reports are available; however, none are as comprehensive. In the following listing of Novak's equations, simplified versions are not used due to a limited frequency range. Attachment 14 is provided for use in hand calculations as an aid to the listed equations.

F2 Apply Novak's equations (References 5a, 5b, and 5c) to compute stiffness and damping along the sides of the mat:

$$\begin{aligned} K_{tx}^e, K_{ty}^e, K_{tz}^e &= \text{Embedment translational stiffnesses (K/ft)} \\ K_{rx}^e, K_{ry}^e, K_{rz}^e &= \text{Embedment rotational stiffnesses, about each axis (ft-K/rad)} \end{aligned}$$

$$\begin{aligned} C_{tx}^e, C_{ty}^e, C_{tz}^e &= \text{Embedment translational damping (K-sec/ft)} \\ C_{rx}^e, C_{ry}^e, C_{rz}^e &= \text{Embedment rotational damping, about each axis (ft-K-sec/rad)} \end{aligned}$$

F3 An effective mat embedment depth should be used in order to compute a conservative value of stiffness and damping (refer to Attachment 03).

$$h = \text{Effective mat embedment (ft)}$$

$$h = (2 / 3)(\text{underground thickness of mat}) \quad (\text{Equation 10-29})$$

F4 Dimensionless frequencies are used to compute the impedance coefficients defined in the next section. A second set of dimensionless frequencies is required for reciprocating machines with secondary forces using twice the machine speed.

$$\begin{aligned} A_t &= \text{Dimensionless frequency for translation} \\ A_x, A_y, A_z &= \text{Dimensionless frequency for rotation} \\ G_f &= \text{Shear modulus of compacted fill material (K/ft}^2\text{)} \\ R_t &= \text{Equivalent mat radii for translation (ft)} \\ R_x, R_y, R_z &= \text{Equivalent mat radii for rotation, about each axis (ft)} \\ L_x &= \text{Pile cap dimension along X axis (ft)} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

L_y = Pile cap dimension along Y axis (ft)

$$R_t = \sqrt{(L_x)(L_y)/\pi} \quad (\text{Equation 10-30a})$$

$$R_x = \sqrt[4]{(L_x)(L_y)^3/3\pi} \quad (\text{Equation 10-30b})$$

$$R_y = \sqrt[4]{(L_y)(L_x)^3/3\pi} \quad (\text{Equation 10-30c})$$

$$R_z = \sqrt[4]{(L_x)(L_y)[(L_x)^2 + (L_y)^2]/6\pi} \quad (\text{Equation 10-30d})$$

$$A_t = (MS)(R_t)\sqrt{W_f/(G_f)(\text{gravity})} \quad (\text{Equation 10-31a})$$

$$A_x = (MS)(R_x)\sqrt{W_f/(G_f)(\text{gravity})} \quad (\text{Equation 10-31b})$$

$$A_y = (MS)(R_y)\sqrt{W_f/(G_f)(\text{gravity})} \quad (\text{Equation 10-31c})$$

$$A_z = (MS)(R_z)\sqrt{W_f/(G_f)(\text{gravity})} \quad (\text{Equation 10-31d})$$

F5 Stiffness and damping coefficients are determined next. Coefficients may be computed from the following criteria or may be taken directly from the precomputed coefficients provided in Attachment 14. Since Bessel functions and complex arithmetic are involved, direct calculations are not recommended. A second set of coefficients and resulting impedance must be determined for reciprocating machines with secondary forces

$S_{tx}^e, S_{ty}^e, S_{tz}^e$ = Embedment translation stiffness coefficient
 $S_{rx}^e, S_{ry}^e, S_{rz}^e$ = Embedment rocking stiffness coefficient, about each axis

$N_{tx}^e, N_{ty}^e, N_{tz}^e$ = Embedment translation damping coefficient
 $N_{rx}^e, N_{ry}^e, N_{rz}^e$ = Embedment rocking damping coefficient, about each axis

X_o = Dimensionless frequency
 f = Dimensionless variable
 g = Complex variable
 μ_f = Poisson's ratio for compacted fill material

$H_0(x)$ = Bessel function of the third kind, or order 0, evaluated at x (complex)
 $H_1(x)$ = Bessel function of the third kind, or order 1, evaluated at x (complex)
 $H_2(x)$ = Bessel function of the third kind, or order 2, evaluated at x (complex)

$$f = \sqrt{2(1 - \mu_f)/1 - 2(\mu_f)} \quad (\text{Equation 10-32})$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

$$X_o = A_t / f \quad (\text{Equation 10-33})$$

$$g = \frac{f (H_2(A_t))(H_1(X_o)) + (H_2(X_o))(A_t)}{(H_0(A_t))(H_2(X_o)) + (H_0(X_o))(H_2(A_t))} \quad (\text{Equation 10-34})$$

$$S_{tx}^e + iN_{tx}^e = S_{ty}^e + iN_{ty}^e = 2 \pi (A_t)(g) \quad (\text{Equation 10-35a})$$

$$S_{tz}^e + iN_{tz}^e = 2 \pi (A_t) (H_1(A_t)) / (H_0(A_t)) \quad (\text{Equation 10-35b})$$

$$S_{rx}^e + iN_{rx}^e = \pi - \pi (A_x) (H_0(A_x)) / (H_1(A_x)) \quad (\text{Equation 10-35c})$$

$$S_{ry}^e + iN_{ry}^e = \pi - \pi (A_y) (H_0(A_y)) / (H_1(A_y)) \quad (\text{Equation 10-35d})$$

$$S_{rz}^e + iN_{rz}^e = 4 \pi - 2 \pi (A_z) (H_0(A_z)) / (H_1(A_z)) \quad (\text{Equation 10-35e})$$

F6 The stiffness and damping values that make up embedment impedance can now be computed from the following formulae:

$$K_{tx}^e = (S_{tx}^e)(G_f)(h) \quad (\text{Equation 10-36a})$$

$$K_{ty}^e = (S_{ty}^e)(G_f)(h) \quad (\text{Equation 10-36b})$$

$$K_{tz}^e = (S_{tz}^e)(G_f)(h) \quad (\text{Equation 10-36c})$$

$$K_{rx}^e = \left[S_{rx}^e + \left(\frac{h}{R_x} \right)^2 \left(\frac{S_{ty}^e}{12} \right) \right] (G_f)(h)(R_x)^2 \quad (\text{Equation 10-36d})$$

$$K_{ry}^e = \left[S_{ry}^e + \left(\frac{h}{R_y} \right)^2 \left(\frac{S_{tx}^e}{12} \right) \right] (G_f)(h)(R_y)^2 \quad (\text{Equation 10-36e})$$

$$K_{rz}^e = (S_{rz}^e)(G_f)(h)(R_z)^2 \quad (\text{Equation 10-36f})$$

$$C_{tx}^e = (N_{tx}^e)(G_f)(h) / MS \quad (\text{Equation 10-37a})$$

$$C_{ty}^e = (N_{ty}^e)(G_f)(h) / MS \quad (\text{Equation 10-37b})$$

$$C_{tz}^e = (N_{tz}^e)(G_f)(h) / MS \quad (\text{Equation 10-37c})$$

$$C_{rx}^e = \left[N_{rx}^e + \left(\frac{h}{R_x} \right)^2 \left(\frac{N_{ty}^e}{12} \right) \right] \frac{(G_f)(h)(R_x)^2}{MS} \quad (\text{Equation 10-37d})$$

$$C_{ry}^e = \left[N_{ry}^e + \left(\frac{h}{R_y} \right)^2 \left(\frac{N_{tx}^e}{12} \right) \right] \frac{(G_f)(h)(R_y)^2}{MS} \quad (\text{Equation 10-37e})$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

$$C_{rz}^e = (N_{rz}^e)(G_f)(h)(R_z)^2 / MS \quad (\text{Equation 10-37f})$$

G Forces at the Center of Gravity

G1 All dynamic forces must be determined at the center of gravity in order to apply the basic dynamic equation described in Section A1. Mass and mass moments of inertia have already been computed at the CG in Section B7. Impedance and machine forces will be resolved at the CG in the following sections.

G2 Compute total stiffness and damping at the center of gravity. A second set of stiffness and damping values must be computed for reciprocating machines with secondary forces.

$$\begin{aligned} K_{tx}, K_{ty}, K_{tz} &= \text{Translational stiffness at CG (K/ft)} \\ K_{rx}, K_{ry}, K_{rz} &= \text{Rotational stiffness at CG, about each axis (ft-K/rad)} \\ K_{cx}, K_{cy} &= \text{Cross stiffness at CG (K/rad)} \\ C_{tx}, C_{ty}, C_{tz} &= \text{Translational damping at CG (K-sec/ft)} \\ C_{rx}, C_{ry}, C_{rz} &= \text{Rotational damping at CG, about each axis (ft-K-sec/rad)} \\ C_{cx}, C_{cy} &= \text{Cross damping at CG (K-sec/rad)} \end{aligned}$$

G3 Two dimensions need to be determined first.

$$\begin{aligned} \partial_c &= \text{Height from CG to bottom of mat (ft)} \\ \partial_e &= \text{Height from CG to center of embedment (ft)} \end{aligned}$$

$$\partial_c = \text{MatHt} - \text{CG}_z \quad (\text{Equation 10-38})$$

$$\partial_e = \partial_c - 0.5 (h) \quad (\text{Equation 10-39})$$

G4 Total stiffness values are computed from the following:

$$K_{tx} = K_{tx}^c + K_{tx}^e \quad (\text{Equation 10-40a})$$

$$K_{ty} = K_{ty}^c + K_{ty}^e \quad (\text{Equation 10-40b})$$

$$K_{tz} = K_{tz}^c + K_{tz}^e \quad (\text{Equation 10-40c})$$

$$K_{rx} = K_{rx}^c + K_{rx}^e + (K_{ty}^c)(\partial_c)^2 + (K_{ty}^e)(\partial_e)^2 \quad (\text{Equation 10-40d})$$

$$K_{ry} = K_{ry}^c + K_{ry}^e + (K_{tx}^c)(\partial_c)^2 + (K_{tx}^e)(\partial_e)^2 \quad (\text{Equation 10-40e})$$

$$K_{rz} = K_{rz}^c + K_{rz}^e \quad (\text{Equation 10-40f})$$

$$K_{cx} = - (K_{ty}^c)(\partial_c) - (K_{ty}^e)(\partial_e) \quad (\text{Equation 10-40g})$$

$$K_{cy} = - (K_{tx}^c)(\partial_c) - (K_{tx}^e)(\partial_e) \quad (\text{Equation 10-40h})$$

G5 Total damping values are computed from the following:

$$C_{tx} = C_{tx}^c + C_{tx}^e \quad (\text{Equation 10-41a})$$

$$C_{ty} = C_{ty}^c + C_{ty}^e \quad (\text{Equation 10-41b})$$

$$C_{tz} = C_{tz}^c + C_{tz}^e \quad (\text{Equation 10-41c})$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

$$C_{rx} = C_{rx}^c + C_{rx}^e + (C_{ty}^c)(\partial_c)^2 + (C_{ty}^e)(\partial_e)^2 \quad (\text{Equation 10-41d})$$

$$C_{ry} = C_{ry}^c + C_{ry}^e + (C_{tx}^c)(\partial_c)^2 + (C_{tx}^e)(\partial_e)^2 \quad (\text{Equation 10-41e})$$

$$C_{rz} = C_{rz}^c + C_{rz}^e \quad (\text{Equation 10-41f})$$

$$C_{cx} = - (C_{ty}^c)(\partial_c) - (C_{ty}^e)(\partial_e) \quad (\text{Equation 10-41g})$$

$$C_{cy} = - (C_{tx}^c)(\partial_c) - (C_{tx}^e)(\partial_e) \quad (\text{Equation 10-41h})$$

- G6 Normally, unbalance forces are provided along the shaft, usually at a rotor or between cylinders. The unbalanced forces and moments must then be translated to the center of gravity. For reciprocating machines, this calculation must be performed for primary loads, and again for secondary loads.

$$\begin{aligned} F_x, F_y, F_z &= \text{Unbalanced force at CG (kips)} \\ M_x, M_y, M_z &= \text{Unbalanced moment at CG (ft-K)} \end{aligned}$$

$$\begin{aligned} F_x^v, F_y^v, F_z^v &= \text{Force provided by supplier (kips)} \\ M_x^v, M_y^v, M_z^v &= \text{Moment provided by supplier (ft-kips)} \end{aligned}$$

$$\partial_x, \partial_y, \partial_z = \text{Distance from origin to location of supplier forces in each axis (ft)}$$

$$F_x = F_x^v \quad (\text{Equation 10-42a})$$

$$F_y = F_y^v \quad (\text{Equation 10-42b})$$

$$F_z = F_z^v \quad (\text{Equation 10-42c})$$

$$M_x = M_x^v + F_y^v [\text{abs}(\partial_z - CG_z)] + F_z^v [\text{abs}(\partial_y - CG_y)] \quad (\text{Equation 10-42d})$$

$$M_y = M_y^v + F_x^v [\text{abs}(\partial_z - CG_z)] + F_z^v [\text{abs}(\partial_x - CG_x)] \quad (\text{Equation 10-42e})$$

$$M_z = M_z^v + F_x^v [\text{abs}(\partial_y - CG_y)] + F_y^v [\text{abs}(\partial_x - CG_x)] \quad (\text{Equation 10-42f})$$

H Single DOF Analysis

- H1 A single degree of freedom analysis is used for vertical translation and for torsional rocking. (Reference 2d)

- H2 The analysis for vertical translation, designated mode #1, computes the following variables:

$$\begin{aligned} \text{Mode \#1:} & \quad (\text{Z axis translation}) \\ NF_1 &= \text{Vertical natural frequency (rad/sec)} \\ FR_1 &= \text{Vertical frequency ratio} \\ \Delta_z &= \text{Vertical translation at CG (ft)} \end{aligned}$$

- H3 Compute 2 quantities:

$$a_1, a_2 = \text{Impedance variables (K/ft)}$$

$$a_1 = (K_{tz}) - (\text{mass})(MS)^2 \quad (\text{Equation 10-43})$$

$$a_2 = (C_{tz})(MS) \quad (\text{Equation 10-44})$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

H4 Compute amplitude at the center of gravity:

$$\Delta_z = F_z / \sqrt{(a_1)^2 + (a_2)^2} \quad (\text{Equation 10-45})$$

H5 Iterate to find the natural frequency:

1. Select a trial frequency, NF_1 .
2. Determine S_{tz}^c and K_{tz}^c from the criteria given in Section D.
3. If embedded, determine S_{tz}^e and K_{tz}^e from the criteria given in Section E.
4. Compute K_{tz} from the equations given in Section F4.
5. Compute NF_1 :

$$NF_1 = \sqrt{K_{tz} / \text{mass}} \quad (\text{Equation 10-46})$$

6. Repeat the above steps until the trial and computed frequencies match.

H6 Compute the frequency ratio and compare using the acceptance criteria provided in the main guideline.

$$FR_1 = MS / NF_1 \quad (\text{Equation 10-47})$$

H7 The analysis for Z axis rocking or torsion, designated mode #2, computes the following variables:

Mode #2: (Z axis rocking)
 NF_2 = Torsional natural frequency (rad/sec)
 FR_2 = Torsional frequency ratio
 ϕ_z = Torsional rotation at CG (ft)

H8 The computations described in Sections G3 through G7 are repeated for mode #2 with the following substitutions:

I_z	for mass	(mass)
K_{tz}	for K_{tz}	(stiffness)
C_{tz}	for C_{tz}	(damping)
M_z	for F_z	(unbalanced load)
ϕ_z	for Δ_z	(amplitude)
NF_2	for NF_1	(natural frequency)
FR_2	for FR_1	(frequency ratio)

H9 Secondary amplitudes must be computed for reciprocating machines with a secondary force. The above steps are repeated with twice the machine speed and with secondary values replacing all primary values.

I Two DOF Analysis

I1 A two degree of freedom analysis is used in each horizontal direction for coupled translation and rocking. (Reference 4c)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

- I2 The analysis for lateral translation and rocking, designated Modes #3 and #4, computes the following variables:

Modes #3 and #4: (Y axis translation & X axis rocking)
 NF_3, NF_4 = Natural frequencies (rad/sec)
 FR_3, FR_4 = Frequency ratios
 Δ_y = Translation at CG (ft)
 ϕ_x = Rotation at CG (rad)

- I3 Compute 6 quantities:

a_1, a_2 = Impedance variables (ft-K²)
 b_1, b_2, c_1, c_2 = Impedance variables (kips²)

$$a_1 = [(K_{rx}) - (I_x)(MS)^2](F_y) - (K_{cx})(M_x) \quad \text{(Equation 10-48)}$$

$$a_2 = (C_{rx})(MS)(F_y) - (C_{cx})(MS)(M_x) \quad \text{(Equation 10-49)}$$

$$b_1 = [(K_{ty}) - (\text{mass})(MS)^2](M_x) - (K_{cx})(F_y) \quad \text{(Equation 10-50)}$$

$$b_2 = (C_{ty})(MS)(M_x) - (C_{cx})(MS)(F_y) \quad \text{(Equation 10-51)}$$

$$c_1 = (\text{mass})(I_x)(MS)^4 + (K_{ty})(K_{rx}) - (K_{cx})^2 - [(\text{mass})(K_{rx}) + (I_x)(K_{ty}) + (C_{ty})(C_{rx}) - (C_{cx})^2](MS)^2 \quad \text{(Equation 10-52)}$$

$$c_2 = [(C_{ty})(K_{rx}) + (C_{rx})(K_{ty}) - 2(C_{cx})(K_{cx})](MS) - [(\text{mass})(C_{rx}) + (I_x)(C_{ty})](MS)^3 \quad \text{(Equation 10-53)}$$

- I4 Compute amplitudes:

$$\Delta_y = \sqrt{(a_1)^2 + (a_2)^2 / (c_1)^2 + (c_2)^2} \quad \text{(Equation 10-54)}$$

$$\phi_x = \sqrt{(b_1)^2 + (b_2)^2 / (c_1)^2 + (c_2)^2} \quad \text{(Equation 10-55)}$$

- I5 Iterate to find the natural frequency:

1. Select a trial frequency, NF.
2. Determine S_{ty}^c , S_{rx}^c , K_{ty}^c , and K_{rx}^c from the criteria given in Section D.
3. If embedded, determine S_{ty}^e , S_{rx}^e , K_{ty}^e , and K_{rx}^e from the criteria given in Section E.
4. Compute K_{ty} , K_{rx} , and K_{cx} from the equations given in Section F4.
5. Compute NF_3 and NF_4 :

b = Frequency variable (1/sec²)
 c = Frequency variable (1/sec⁴)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

$$b = (K_{ty} / \text{mass}) + (K_{rx} / I_x) \quad (\text{Equation 10-56})$$

$$c = (K_{ty})(K_{rx}) - (K_{cx})^2 / (\text{mass})(I_x) \quad (\text{Equation 10-57})$$

$$NF_3 = 0.5(b) - 0.5\sqrt{(b)^2 - 4(c)} \quad (\text{Equation 10-58})$$

$$NF_4 = 0.5(b) + 0.5\sqrt{(b)^2 - 4(c)} \quad (\text{Equation 10-59})$$

6. Repeat the above steps until the trial and either of both computed frequencies match.

I6 Compute the frequency ratios and compare using the acceptance criteria provided in the main guideline.

$$FR_3 = MS / NF_3 \quad (\text{Equation 10-60a})$$

$$FR_4 = MS / NF_4 \quad (\text{Equation 10-60b})$$

I7 The analysis for lateral translation and rocking, designated modes #5 and #6, computes the following variables:

Modes #5 and #6: (X axis translation and Y axis rocking)
 NF_5, NF_6 = Natural frequencies (rad/sec)
 FR_5, FR_6 = Frequency ratios
 Δ_x = Translation at CG (ft)
 ϕ_y = Rotation at CG (rad)

I8 The computations described in Sections H3 through H7 are repeated for Modes #5 and #6 with the following substitutions:

I_y	for I_x	(mass moment of inertia)
K_{tx}	for K_{ty}	(translation stiffness)
K_{ry}	for K_{rx}	(rocking stiffness)
C_{tx}	for C_{ty}	(translation damping)
C_{ry}	for C_{rx}	(rocking damping)
F_x	for F_y	(unbalanced force)
M_y	for M_x	(unbalanced moment)
Δ_x	for Δ_y	(translation)
ϕ_y	for ϕ_x	(rotation)
NF_5, NF_6	for NF_3, NF_4	(natural frequency)
FR_5, FR_6	for FR_3, FR_4	(frequency ratio)

I9 Secondary amplitudes must be computed for reciprocating machines with secondary loads. The above steps are repeated with twice the machine speed and with secondary values replacing all primary values.



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

J Double Amplitudes at Selected Locations

- J1 Double amplitudes (peak to peak) are computed at a selected point on the foundation. A comparison is then made using the acceptance criteria provided in the main guideline.

DA_x, DA_y, DA_z = Double, or peak to peak, amplitude, at specified location (mils)
 PT_x, PT_y, PT_z = Distance from origin to selected point (ft)
 D_x, D_y, D_z = Distance from CG to selected point, along each axis (ft)

$$D_x = \text{abs} [PT_x - CG_x] \quad (\text{Equation 10-61a})$$

$$D_y = \text{abs} [PT_y - CG_y] \quad (\text{Equation 10-61b})$$

$$D_z = \text{abs} [PT_z - CG_z] \quad (\text{Equation 10-61c})$$

$$DA_x = \{\Delta_x + D_y (\phi_z) + D_z (\phi_y)\} (24000) \quad (\text{Equation 10-62a})$$

$$DA_y = \{\Delta_y + D_x (\phi_z) + D_z (\phi_x)\} (24000) \quad (\text{Equation 10-62b})$$

$$DA_z = \{\Delta_z + D_x (\phi_y) + D_y (\phi_x)\} (24000) \quad (\text{Equation 10-62c})$$

- J2 For reciprocating machines, amplitudes must be computed for secondary forces and added to those obtained for primary forces.

K NOMENCLATURE (including variables used elsewhere in the Guideline)

K1 *Superscripts and Subscripts*

1 = Vertical vibration mode
 2 = Torsional vibration mode
 3, 4 = Coupled Y axis translation and X axis rocking vibration modes
 5, 6 = Coupled X axis translation and Y axis rocking vibration modes
 c = Mat centroid quantity
 e = Embedment quantity
 f = Fill quantity
 h = Horizontal
 i = Individual pile quantity
 p = Pile quantity
 r = Rotation
 s = Soil quantity (subscript), Static quantity (superscript)
 t = Translation
 v = Supplier provided load
 x = Coordinate axis parallel to shaft
 y = Coordinate axis perpendicular to shaft
 z = Vertical coordinate axis

K2 *Variables*

a_1, a_2 = Impedance variables (K/ft)
 A_o = Pile dimensionless frequency
 A_p = Area of pile (ft²)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

A_t	= Dimensionless frequency for translation
A_x, A_y, A_z	= Dimensionless frequencies for rotation about each axis
A_{crank}	= Area of piston head (in ²)
A_{head}	= Area of piston head on crank side (in ²)
b	= Frequency variable (1/sec ²)
b_1, b_2	= Impedance variables (K/ft)
B	= Cylinder bore diameter (in)
c	= Frequency variable (1/sec ⁴)
c_1, c_2	= Impedance variables (K/ft)
$C_{tx}..C_{tz}$	= Translation damping at CG (K-sec/ft)
$C_{rx}..C_{rz}$	= Rotational damping at CG (ft-K-sec/rad)
C_{cx}, C_{cy}	= Cross damping at CG (K-sec/rad)
$C_{tx}^c..C_{tz}^c$	= Base translational damping (K-sec/ft)
$C_{rx}^c..C_{rz}^c$	= Base rotational damping (ft-K-sec/rad)
$C_{cx}..C_{cy}$	= Base cross damping (K-sec/rad)
$C_{tx}^e..C_{tz}^e$	= Embedment translational damping (K-sec/ft)
$C_{rx}^e..C_{rz}^e$	= Embedment rotational damping (ft-K-sec/rad)
$C_{tx}^i..C_{tz}^i$	= Pile translation damping (K-sec/ft)
$C_{rx}^i..C_{rz}^i$	= Pile rotational damping (ft-K-sec/rad)
$C_{cx}^i..C_{cy}^i$	= Pile cross damping (K-sec/rad)
CG_x, CG_y, CG_z	= Center of gravity location with respect to the referenced origin (ft)
CS	= Coastdown speed (rad/sec)
CW	= Weight of component (kips)
d	= Mat cantilever beyond face of pier, in either direction (ft)
D	= Rod diameter (in)
D_x, D_y, D_z	= Distance from CG to selected point, along each axis (ft)
DA_x, DA_y, DA_z	= Double, or peak to peak, amplitude at specified location (mils)
e	= Eccentricity of rotor (ft)
E_x, E_y, E_z	= Component weight times distance from origin (ft-K)
E_c	= Modulus of elasticity for concrete (K/in ²)
E_s	= Modulus of elasticity for soil (K/in ²)
f	= Embedment variable
F_x, F_y, F_z	= Unbalanced force at CG (kips)
F_x^v, F_y^v, F_z^v	= Force provided by supplier (kips)
F_1	= Inertia force of hinge (kips)
F_2	= Primary inertia force of piston (kips)
F_3	= Secondary inertia force of piston (kips)
F_4	= Inertia force of unbalanced rotor (kips)
F_{cr}	= Correction factor
F_{fdn}	= Lateral force on foundation, tributary to cylinder (kips)
F_{red}	= Reduction factor, use 2.0 unless better data is available
$FR_1..FR_6$	= Frequency ratio



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

g	= Embedment variable (complex)
G_f	= Shear modulus of compacted fill material (K/ft ²)
G_s	= Shear modulus of soil (K/ft ²)
$GP_{tx}..GP_{tz}$	= Translation group reduction factors
$GP_{rx}..GP_{rz}$	= Rotation group reduction factors
GradeHt	= Height from origin to grade (ft)
gravity	= Acceleration of gravity (32.2 ft/sec ²)
h	= Effective mat embedment (ft)
$H_0(x)$	= Bessel function of the third kind, or order 0, evaluated at x .
$H_1(x)$	= Bessel function of the third kind, or order 1, evaluated at x .
$H_2(x)$	= Bessel function of the third kind, or order 2, evaluated at x .
HW	= Weight of hinge (kips)
i	= Complex number coefficient ($\sqrt{-1}$)
I_p, I_{px}, I_{py}	= Pile moment of inertia (ft ⁴)
I_{pz}	= Pile polar moment of inertia (ft ⁴)
I_x, I_y, I_z	= Mass moment of inertia, about each axis (K-sec ² -ft)
J_x, J_y, J_z	= Component's translated mass moment of inertia (K-sec ² -ft)
K_v	= Stiffness ratio for vertical interaction
K_h	= Stiffness ratio for horizontal interaction
$K_{tx}..K_{tz}$	= Translation stiffness at CG (K/ft)
$K_{rx}..K_{rz}$	= Rotational stiffness at CG (ft-K/rad)
K_{cx}, K_{cy}	= Cross stiffness at CG (K/rad)
$K_{tx}^c..K_{tz}^c$	= Base translational stiffness (K/ft)
$K_{rx}^c..K_{rz}^c$	= Base rotational stiffness (ft-K/rad)
$K_{cx}^c..K_{cy}^c$	= Base cross stiffness (K/rad)
$K_{tx}^e..K_{tz}^e$	= Embedment translational stiffness (K/ft)
$K_{rx}^e..K_{rz}^e$	= Embedment rotational stiffness (ft-K/rad)
$K_{tx}^i..K_{tz}^i$	= Pile translational stiffness (K/ft)
$K_{rx}^i..K_{rz}^i$	= Pile rotational stiffness (ft-K/rad)
$K_{cx}^i..K_{cy}^i$	= Pile cross stiffness (K/rad)
L	= Length (ft)
L_1	= Length from shaft to hinge (ft)
L_2	= Length from hinge to piston (ft)
L_x, L_y, L_z	= Dimensions of foundation component (ft)
M_x, M_y, M_z	= Unbalanced moment at CG (ft-kips)
M_x^v, M_y^v, M_z^v	= Moment provided by supplier (ft-kips)
mass	= Translational mass (K-sec ² /ft)
MatHt	= Height from origin to bottom of mat (ft)
MR	= Mass ratio
MS	= Machine speed (rad/sec)
MW	= Machine weight (kips)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

$N_{tx}^e .. N_{tz}^e$	= Embedment translation damping coefficient
$N_{rx}^e .. N_{rz}^e$	= Embedment rotational damping coefficient
$N_{tx}^i .. N_{tz}^i$	= Pile translation damping coefficient
$N_{rx}^i .. N_{rz}^i$	= Pile rotational damping coefficient
$N_{cx}^i .. N_{cy}^i$	= Pile cross damping coefficient
$NF_1 .. NF_6$	= Natural frequency (rad/sec)
OS_x, OS_y	= Center of gravity offset in each direction (percent)
PL	= Maximum pile load (kips)
P_{crank}	= Instantaneous crank pressure (ksi)
P_{head}	= Instantaneous head pressure (ksi)
PT_x, PT_y, PT_z	= Distance from origin to selected point (ft)
PW	= Weight of piston (kips)
q	= Dimensionless value
Q_g	= Measure of rotor quality grade (in/sec)
Q_x, Q_y, Q_z	= Component mass moment of inertia about component CG (K-sec ² -ft)
R_a	= Pile solidity ratio
R_p	= Pile radius (ft)
R_t	= Equivalent mat radius for translation (ft)
R_x, R_y, R_z	= Equivalent mat radius for rotation, about each axis (ft)
RW	= Weight of rotor (kips)
R_1, R_2	= Combined foundation ratios
S_1	= Dimensionless parameter for vertical motion (complex)
S_2	= Dimensionless parameter for torsional motion (complex)
S_h	= Dimensionless parameter for horizontal & rocking motion (complex)
S_p	= Spacing between two piles (ft)
$S_{tx}^e .. S_{tz}^e$	= Embedment translation stiffness coefficient
$S_{rx}^e .. S_{rz}^e$	= Embedment rotational stiffness coefficient
$S_{tx}^i .. S_{tz}^i$	= Pile translation stiffness coefficient
$S_{rx}^i .. S_{rz}^i$	= Pile rotational stiffness coefficient
$S_{cx}^i .. S_{cy}^i$	= Pile cross stiffness coefficient
t	= Time (sec)
TW	= Foundation, machine, and soil weight (kips)
W_c	= Density of concrete (K/ft ³)
W_f	= Density of compacted fill material (K/ft ³)
W_p	= Density of pile (K/ft ³)
W_s	= Density of soil (K/ft ³)
X_o	= Dimensionless frequency



VIBRATING MACHINERY FOUNDATIONS ON PILES

Analysis Procedure

α_v	= Two pile interaction factor for vertical motion
$\alpha_h(x)$	= Two pile interaction factor for horizontal motion at angle x
β	= Crank angle (rad)
Ω_v	= Frequency parameter for vertical motion (complex)
Ω_x, Ω_y	= Frequency parameter for horizontal & rocking motion (complex)
Ω_z	= Frequency parameter for torsional motion (complex)
∂_c	= Height from CG to bottom of mat (ft)
∂_e	= Height from CG to center of embedment (ft)
$\partial_x, \partial_y, \partial_z$	= Distance from origin to center of selected location (ft)
$\Delta_x, \Delta_y, \Delta_z$	= Translation at CG (ft)
ϕ_x, ϕ_y, ϕ_z	= Rotation at CG (rad)
μ_f	= Poisson's ratio for compacted fill material
μ_s	= Poisson's ratio for soil

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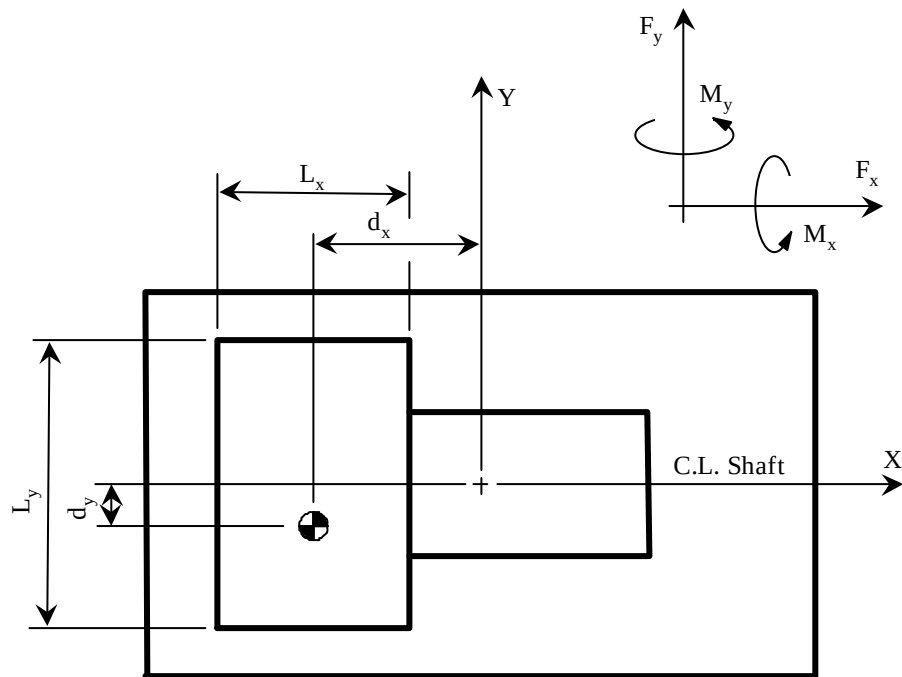
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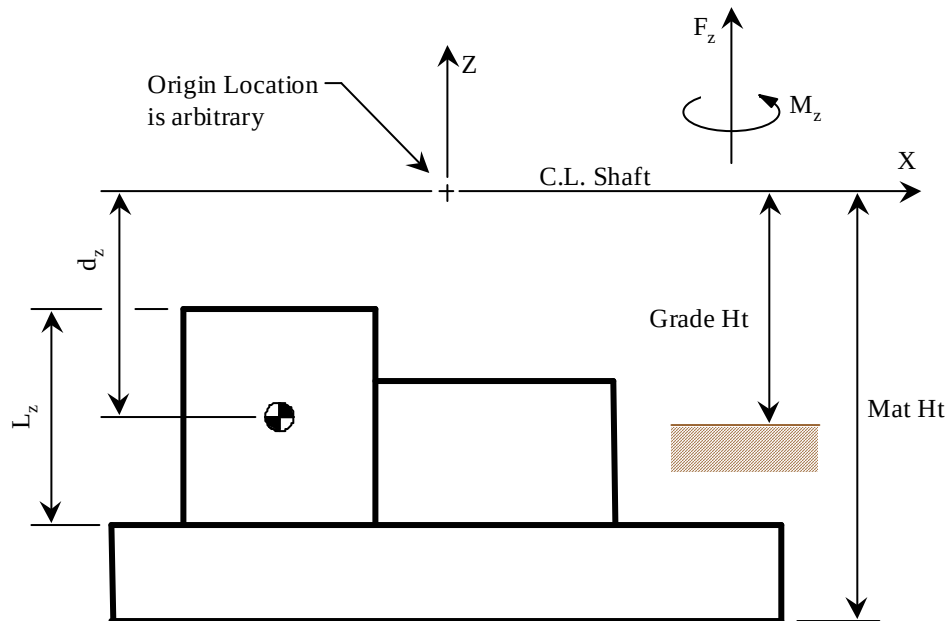


VIBRATING MACHINERY FOUNDATIONS ON PILES

Directional Nomenclature



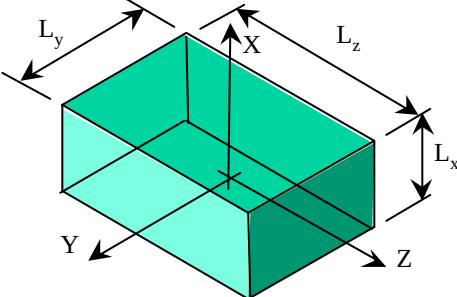
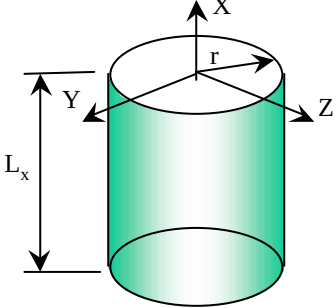
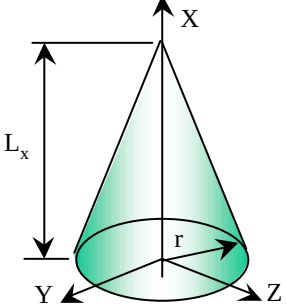
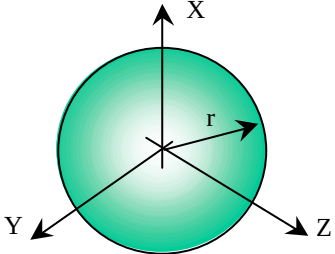
FOUNDATION PLAN



ELEVATION

VIBRATING MACHINERY FOUNDATIONS ON PILES

Mass Moment of Inertia

	Rectangular Prism $CW = (\text{Density})(L_x)(L_y)(L_z)$ $Q_x = (CW)[(L_y)^2 + (L_z)^2] / 12$ $Q_y = (CW)[(L_x)^2 + (L_z)^2] / 12$ $Q_z = (CW)[(L_x)^2 + (L_y)^2] / 12$
	Circular Cylinder $CW = (\text{Density})(\pi)(r)^2 (L_x) / 4$ $Q_x = (CW)(r)^2 / 2$ $Q_y = (CW)[3(r)^2 + (L_x)^2] / 12$ $Q_z = Q_y$
	Circular Cone $CW = (\text{Density})(\pi)(r)^2 (L_x) / 3$ $Q_x = 3(CW)(r)^2 / 10$ $Q_y = 3(CW)[.25(r)^2 + (L_x)^2] / 5$ $Q_z = Q_y$
	Sphere $CW = (\text{Density})(4\pi)(r)^2 / 3$ $Q_x = 3(CW)(r)^2 / 5$ $Q_y = Q_x$ $Q_z = Q_x$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Typical Pile Impedence Coefficients

Typical Timber Pile Stiffness and Damping (Refer to Attachment 10, Section D2)

Friction Pile: Southern Pine, 12" diameter, 50' long

Soil: $W_s = 110$ pcf, $\mu_s = 0.45$, $G_s = 5,000$ psi

A_0^i	K_{tx}^i static	K_{ty}^i static	K_{rx}^i static	K_{ry}^i static	K_{cz}^i static	C_{tx}^i static	C_{ty}^i static	C_{rx}^i static	C_{ry}^i static	C_{cz}^i static
0.00										
0.02	3944	13095	9688	1733	-4390	60.0	134.7	48.1	0.0	-43.9
0.04	4374	14101	10062	1731	-4719	39.6	85.9	29.5	0.0	-28.0
0.06	4666	14782	10311	1729	-4942	31.7	67.2	22.4	0.0	-21.8
0.08	4893	15314	10505	1726	-5115	27.3	57.0	18.7	0.1	-18.5
0.10	5080	15759	10666	1723	-5260	24.5	50.4	16.2	0.1	-16.3
0.12	5241	16145	10805	1719	-5384	22.5	45.8	14.5	0.1	-14.8
0.14	5381	16489	10929	1716	-5495	21.0	42.4	13.3	0.1	-13.7
0.16	5506	16800	11041	1712	-5595	19.8	39.7	12.3	0.1	-12.8
0.18	5618	17087	11144	1707	-5686	18.9	37.5	11.5	0.1	-12.1
0.20	5720	17352	11240	1703	-5770	18.1	35.7	10.9	0.1	-11.5
0.22	5813	17600	11330	1699	-5849	17.5	34.2	10.3	0.2	-11.0
0.24	5898	17834	11414	1694	-5922	16.9	32.9	9.8	0.2	-10.6
0.26	5977	18056	11495	1690	-5992	16.4	31.7	9.4	0.2	-10.2
0.28	6049	18267	11572	1686	-6057	16.0	30.7	9.1	0.2	-9.9
0.30	6117	18468	11645	1681	-6120	15.6	29.9	8.8	0.2	-9.6
0.32	6179	18662	11716	1677	-6179	15.3	29.1	8.5	0.2	-9.4
0.34	6238	18847	11784	1672	-6237	15.0	28.4	8.2	0.2	-9.2
0.36	6292	19026	11850	1668	-6291	14.8	27.7	8.0	0.2	-9.0
0.38	6343	19199	11915	1664	-6344	14.5	27.2	7.8	0.2	-8.8
0.40	6390	19366	11977	1660	-6395	14.3	26.6	7.6	0.2	-8.6
0.42	6435	19529	12038	1656	-6445	14.1	26.1	7.4	0.3	-8.5
0.44	6476	19686	12097	1652	-6493	13.9	25.7	7.3	0.3	-8.3
0.46	6515	19839	12155	1648	-6539	13.8	25.3	7.1	0.3	-8.2
0.48	6551	19989	12212	1644	-6585	13.6	24.9	7.0	0.3	-8.1
0.50	6585	20134	12268	1640	-6629	13.5	24.5	6.9	0.3	-7.9
0.52	6616	20276	12323	1637	-6672	13.4	24.2	6.7	0.3	-7.8
0.54	6646	20415	12377	1633	-6714	13.2	23.8	6.6	0.3	-7.7
0.56	6673	20550	12430	1630	-6755	13.1	23.5	6.5	0.3	-7.7
0.58	6699	20683	12482	1627	-6795	13.0	23.3	6.4	0.3	-7.6
0.60	6723	20812	12534	1624	-6834	12.9	23.0	6.3	0.3	-7.5
0.62	6745	20940	12585	1621	-6873	12.8	22.7	6.2	0.3	-7.4
0.64	6765	21064	12635	1618	-6911	12.8	22.5	6.2	0.3	-7.3
0.66	6784	21187	12685	1615	-6949	12.7	22.3	6.1	0.3	-7.3
0.68	6801	21307	12734	1613	-6986	12.6	22.0	6.0	0.3	-7.2
0.70	6818	21424	12783	1610	-7022	12.5	21.8	5.9	0.3	-7.1
0.72	6832	21540	12831	1608	-7058	12.5	21.6	5.9	0.4	-7.1
0.74	6846	21654	12879	1605	-7093	12.4	21.4	5.8	0.4	-7.0
0.76	6858	21765	12926	1603	-7128	12.3	21.2	5.7	0.4	-7.0
0.78	6869	21875	12973	1601	-7162	12.3	21.1	5.7	0.4	-6.9
0.80	6879	21983	13020	1599	-7197	12.2	20.9	5.6	0.4	-6.9
0.82	6888	22090	13066	1597	-7230	12.2	20.7	5.5	0.4	-6.8
0.84	6895	22194	13112	1595	-7264	12.1	20.6	5.5	0.4	-6.8
0.86	6902	22297	13158	1594	-7297	12.1	20.4	5.4	0.4	-6.7
0.88	6908	22399	13204	1592	-7330	12.1	20.3	5.4	0.4	-6.7
0.90	6913	22499	13249	1590	-7363	12.0	20.1	5.3	0.4	-6.7
0.92	6918	22597	13294	1589	-7395	12.0	20.0	5.3	0.4	-6.6
0.94	6921	22694	13339	1588	-7427	12.0	19.9	5.2	0.4	-6.6
0.96	6924	22790	13383	1586	-7459	11.9	19.8	5.2	0.4	-6.6
0.98	6926	22884	13428	1585	-7491	11.9	19.6	5.1	0.4	-6.5
1.00	6927	22977	13472	1584	-7523	11.9	19.5	5.1	0.4	-6.5



VIBRATING MACHINERY FOUNDATIONS ON PILES

Typical Pile Impedence Coefficients

Typical Concrete Pile Stiffness and Damping (Refer to Attachment 10, Section D2)

Friction Pile: 4,000 psi, 14" square, 50' long

Soil: $W_s = 110$ pcf, $\mu_s = 0.45$, $G_s = 5,000$ psi

A_0^i 0.00	K_{tx}^i static	K_{ty}^i static	K_{rx}^i static	K_{ry}^i static	K_{tz}^i static	C_{tx}^i static	C_{ty}^i static	C_{rz}^i static	C_{ry}^i static	C_{tz}^i static
0.02	6537	25659	44161	19808	-12066	131.1	403.1	288.6	0.2	-159.0
0.04	7243	27956	45851	19787	-12964	86.6	253.1	177.0	0.4	-101.3
0.06	7715	29474	46969	19758	-13564	69.2	196.1	135.1	0.7	-79.2
0.08	8077	30629	47826	19722	-14025	59.7	165.2	112.4	0.9	-67.1
0.10	8368	31565	48531	19680	-14402	53.5	145.4	98.0	1.1	-59.3
0.12	8611	32352	49133	19633	-14723	49.2	131.7	87.9	1.3	-53.9
0.14	8818	33029	49661	19583	-15001	46.0	121.4	80.5	1.5	-49.9
0.16	8994	33620	50133	19529	-15246	43.4	113.5	74.7	1.7	-46.7
0.18	9146	34142	50560	19472	-15465	41.4	107.2	70.1	1.9	-44.2
0.20	9278	34606	50952	19413	-15662	39.8	102.0	66.3	2.1	-42.2
0.22	9391	35021	51314	19351	-15841	38.4	97.7	63.2	2.3	-40.5
0.24	9488	35393	51651	19288	-16003	37.2	94.1	60.5	2.5	-39.0
0.26	9571	35728	51967	19224	-16152	36.2	91.0	58.2	2.6	-37.8
0.28	9641	36030	52265	19158	-16289	35.4	88.3	56.2	2.8	-36.7
0.30	9699	36303	52547	19092	-16415	34.6	85.9	54.5	3.0	-35.7
0.32	9746	36548	52816	19024	-16531	33.9	83.8	53.0	3.1	-34.9
0.34	9782	36769	53072	18956	-16639	33.3	82.0	51.6	3.3	-34.2
0.36	9810	36968	53319	18888	-16739	32.8	80.4	50.3	3.4	-33.5
0.38	9828	37146	53556	18819	-16832	32.3	78.9	49.2	3.6	-32.9
0.40	9837	37304	53785	18749	-16918	31.9	77.6	48.2	3.7	-32.4
0.42	9839	37444	54007	18680	-16998	31.5	76.4	47.3	3.9	-31.9
0.44	9833	37567	54222	18610	-17073	31.2	75.3	46.5	4.0	-31.4
0.46	9820	37675	54433	18540	-17143	30.9	74.3	45.7	4.1	-31.0
0.48	9800	37767	54638	18471	-17208	30.6	73.4	45.0	4.3	-30.7
0.50	9774	37844	54839	18401	-17269	30.3	72.6	44.4	4.4	-30.4
0.52	9741	37908	55037	18331	-17327	30.1	71.8	43.8	4.5	-30.0
0.54	9702	37958	55231	18261	-17380	29.9	71.1	43.2	4.6	-29.8
0.56	9658	37996	55423	18192	-17431	29.7	70.5	42.7	4.7	-29.5
0.58	9608	38022	55612	18122	-17478	29.5	69.9	42.2	4.8	-29.3
0.60	9552	38036	55800	18053	-17523	29.3	69.4	41.7	5.0	-29.0
0.62	9491	38039	55985	17983	-17565	29.2	68.9	41.3	5.1	-28.8
0.64	9426	38032	56170	17914	-17605	29.1	68.5	40.9	5.2	-28.7
0.66	9356	38014	56353	17845	-17643	28.9	68.1	40.5	5.3	-28.5
0.68	9281	37987	56536	17776	-17679	28.8	67.7	40.1	5.4	-28.3
0.70	9201	37950	56718	17708	-17714	28.7	67.4	39.8	5.5	-28.2
0.72	9118	37905	56900	17639	-17746	28.6	67.1	39.5	5.5	-28.0
0.74	9030	37851	57081	17571	-17778	28.6	66.8	39.2	5.6	-27.9
0.76	8938	37790	57263	17503	-17808	28.5	66.6	38.9	5.7	-27.8
0.78	8842	37722	57444	17435	-17837	28.4	66.4	38.6	5.8	-27.7
0.80	8742	37648	57626	17367	-17865	28.4	66.2	38.3	5.9	-27.6
0.82	8639	37568	57808	17299	-17892	28.3	66.0	38.1	6.0	-27.5
0.84	8533	37484	57991	17232	-17918	28.3	65.9	37.8	6.1	-27.4
0.86	8422	37395	58174	17164	-17943	28.2	65.8	37.6	6.1	-27.3
0.88	8309	37304	58358	17097	-17968	28.2	65.7	37.4	6.2	-27.2
0.90	8192	37210	58542	17030	-17993	28.1	65.6	37.1	6.3	-27.1
0.92	8073	37115	58728	16964	-18017	28.1	65.6	36.9	6.4	-27.0
0.94	7950	37019	58914	16897	-18041	28.1	65.5	36.7	6.5	-27.0
0.96	7824	36925	59100	16831	-18064	28.1	65.5	36.5	6.5	-26.9
0.98	7696	36832	59288	16765	-18087	28.1	65.5	36.3	6.6	-26.9
1.00	7565	36742	59477	16699	-18111	28.1	65.5	36.1	6.7	-26.8



VIBRATING MACHINERY FOUNDATIONS ON PILES

Typical Pile Impedence Coefficients

Typical Steel Pile Stiffness and Damping (Refer to Attachment 10, Section D2)

Friction Pile: HP10x42 (A36), strong axis, 100' long

Soil: $W_s = 110$ pcf, $\mu_s = 0.45$, $G_s = 5,000$ psi

A_0^i	K_{tx}^i	K_{ty}^i	K_{rx}^i	K_{ry}^i	K_{tz}^i	C_{tx}^i	C_{ty}^i	C_{rz}^i	C_{ry}^i	C_{tz}^i
0.00	static	static	static	static	static	static	static	static	static	static
0.02	5573	19070	27344	8896	-15003	78.9	181.5	126.2	0.1	-139.6
0.04	6179	20523	28397	8887	-16127	52.1	116.1	77.3	0.1	-88.9
0.06	6590	21506	29098	8875	-16885	41.6	90.9	58.9	0.2	-69.4
0.08	6908	22272	29641	8859	-17474	35.9	77.2	49.0	0.3	-58.7
0.10	7170	22908	30092	8842	-17963	32.2	68.3	42.6	0.3	-51.9
0.12	7393	23457	30481	8822	-18383	29.5	62.1	38.2	0.4	-47.1
0.14	7586	23943	30826	8801	-18755	27.6	57.5	34.9	0.5	-43.5
0.16	7758	24381	31138	8779	-19089	26.1	53.8	32.3	0.5	-40.7
0.18	7910	24780	31423	8756	-19393	24.8	50.9	30.3	0.6	-38.5
0.20	8048	25148	31688	8732	-19672	23.8	48.5	28.6	0.7	-36.6
0.22	8172	25489	31935	8707	-19931	23.0	46.5	27.2	0.7	-35.1
0.24	8285	25808	32168	8682	-20172	22.2	44.7	25.9	0.8	-33.8
0.26	8388	26108	32389	8657	-20399	21.6	43.3	24.9	0.8	-32.6
0.28	8482	26391	32599	8631	-20612	21.1	41.9	24.0	0.9	-31.6
0.30	8568	26659	32800	8605	-20814	20.6	40.8	23.2	0.9	-30.8
0.32	8647	26914	32993	8579	-21007	20.2	39.7	22.4	1.0	-30.0
0.34	8719	27157	33179	8554	-21190	19.8	38.8	21.8	1.0	-29.3
0.36	8785	27389	33359	8528	-21364	19.5	38.0	21.2	1.1	-28.6
0.38	8846	27612	33533	8502	-21532	19.2	37.2	20.7	1.1	-28.1
0.40	8901	27825	33702	8477	-21693	18.9	36.5	20.2	1.2	-27.5
0.42	8951	28030	33867	8451	-21848	18.6	35.9	19.8	1.2	-27.1
0.44	8997	28228	34027	8426	-21997	18.4	35.3	19.3	1.3	-26.6
0.46	9039	28419	34184	8402	-22142	18.2	34.8	19.0	1.3	-26.2
0.48	9076	28603	34338	8377	-22281	18.0	34.3	18.6	1.3	-25.9
0.50	9110	28781	34489	8353	-22417	17.8	33.8	18.3	1.4	-25.5
0.52	9140	28953	34636	8329	-22548	17.7	33.3	18.0	1.4	-25.2
0.54	9166	29120	34782	8306	-22676	17.5	32.9	17.7	1.4	-24.9
0.56	9190	29282	34925	8282	-22801	17.4	32.6	17.4	1.5	-24.6
0.58	9210	29439	35066	8260	-22922	17.2	32.2	17.2	1.5	-24.4
0.60	9227	29592	35205	8237	-23041	17.1	31.9	16.9	1.5	-24.1
0.62	9242	29740	35342	8215	-23157	17.0	31.5	16.7	1.6	-23.9
0.64	9253	29884	35477	8193	-23270	16.9	31.2	16.5	1.6	-23.7
0.66	9263	30024	35611	8172	-23381	16.8	30.9	16.3	1.6	-23.5
0.68	9269	30161	35744	8151	-23490	16.7	30.7	16.1	1.7	-23.3
0.70	9274	30294	35876	8130	-23597	16.6	30.4	15.9	1.7	-23.1
0.72	9276	30424	36006	8110	-23702	16.5	30.2	15.7	1.7	-22.9
0.74	9276	30550	36135	8090	-23806	16.5	29.9	15.5	1.7	-22.7
0.76	9273	30673	36263	8071	-23908	16.4	29.7	15.4	1.8	-22.6
0.78	9269	30793	36390	8051	-24008	16.3	29.5	15.2	1.8	-22.4
0.80	9263	30911	36517	8033	-24107	16.3	29.3	15.1	1.8	-22.3
0.82	9255	31026	36643	8014	-24205	16.2	29.1	14.9	1.8	-22.1
0.84	9245	31138	36767	7996	-24301	16.2	28.9	14.8	1.9	-22.0
0.86	9234	31247	36892	7978	-24397	16.1	28.7	14.7	1.9	-21.9
0.88	9221	31354	37015	7961	-24491	16.1	28.6	14.5	1.9	-21.8
0.90	9206	31459	37138	7943	-24585	16.0	28.4	14.4	1.9	-21.7
0.92	9190	31561	37261	7927	-24678	16.0	28.2	14.3	1.9	-21.6
0.94	9172	31661	37383	7910	-24770	15.9	28.1	14.2	2.0	-21.4
0.96	9153	31759	37505	7894	-24861	15.9	27.9	14.1	2.0	-21.3
0.98	9133	31854	37626	7878	-24952	15.9	27.8	14.0	2.0	-21.2
1.00	9111	31948	37747	7862	-25042	15.8	27.7	13.8	2.0	-21.2



VIBRATING MACHINERY FOUNDATIONS ON PILES

Typical Pile Impedence Coefficients

Typical Steel Pile Stiffness and Damping (Refer to Attachment 10, Section D2)

Friction Pile: HP10x42, weak axis, 100' long

Soil: $W_s = 110$ pcf, $\mu_s = 0.45$, $G_s = 5,000$ psi

A_0^i	K_{tx}^i	K_{tx}^i	K_{ry}^i	K_{rz}^i	K_{ry}^i	C_{tx}^i	C_{tx}^i	C_{ry}^i	C_{rz}^i	C_{ry}^i
0.00	static	static	static	static	static	static	static	static	static	static
0.02	4260	19070	12214	8896	-2993	60.3	181.5	56.3	0.1	-27.8
0.04	4723	20523	12684	8887	-3217	39.8	116.1	34.5	0.1	-17.7
0.06	5037	21506	12997	8875	-3369	31.8	90.9	26.3	0.2	-13.8
0.08	5281	22272	13240	8859	-3486	27.4	77.2	21.9	0.3	-11.7
0.10	5481	22908	13441	8842	-3584	24.6	68.3	19.0	0.3	-10.4
0.12	5651	23457	13614	8822	-3668	22.6	62.1	17.1	0.4	-9.4
0.14	5799	23943	13769	8801	-3742	21.1	57.5	15.6	0.5	-8.7
0.16	5930	24381	13908	8779	-3808	19.9	53.8	14.4	0.5	-8.1
0.18	6047	24780	14035	8756	-3869	19.0	50.9	13.5	0.6	-7.7
0.20	6152	25148	14154	8732	-3925	18.2	48.5	12.8	0.7	-7.3
0.22	6247	25489	14264	8707	-3976	17.6	46.5	12.1	0.7	-7.0
0.24	6333	25808	14368	8682	-4024	17.0	44.7	11.6	0.8	-6.7
0.26	6412	26108	14467	8657	-4070	16.5	43.3	11.1	0.8	-6.5
0.28	6484	26391	14561	8631	-4112	16.1	41.9	10.7	0.9	-6.3
0.30	6550	26659	14651	8605	-4153	15.7	40.8	10.3	0.9	-6.1
0.32	6610	26914	14737	8579	-4191	15.4	39.7	10.0	1.0	-6.0
0.34	6665	27157	14820	8554	-4227	15.1	38.8	9.7	1.0	-5.8
0.36	6716	27389	14900	8528	-4262	14.9	38.0	9.5	1.1	-5.7
0.38	6762	27612	14978	8502	-4296	14.6	37.2	9.2	1.1	-5.6
0.40	6804	27825	15053	8477	-4328	14.4	36.5	9.0	1.2	-5.5
0.42	6842	28030	15127	8451	-4359	14.2	35.9	8.8	1.2	-5.4
0.44	6877	28228	15199	8426	-4389	14.1	35.3	8.6	1.3	-5.3
0.46	6909	28419	15269	8402	-4417	13.9	34.8	8.5	1.3	-5.2
0.48	6938	28603	15337	8377	-4445	13.8	34.3	8.3	1.3	-5.2
0.50	6964	28781	15405	8353	-4472	13.6	33.8	8.2	1.4	-5.1
0.52	6986	28953	15471	8329	-4498	13.5	33.3	8.0	1.4	-5.0
0.54	7007	29120	15536	8306	-4524	13.4	32.9	7.9	1.4	-5.0
0.56	7025	29282	15599	8282	-4549	13.3	32.6	7.8	1.5	-4.9
0.58	7040	29439	15662	8260	-4573	13.2	32.2	7.7	1.5	-4.9
0.60	7053	29592	15724	8237	-4597	13.1	31.9	7.6	1.5	-4.8
0.62	7064	29740	15786	8215	-4620	13.0	31.5	7.5	1.6	-4.8
0.64	7073	29884	15846	8193	-4642	12.9	31.2	7.4	1.6	-4.7
0.66	7080	30024	15906	8172	-4665	12.8	30.9	7.3	1.6	-4.7
0.68	7085	30161	15965	8151	-4686	12.8	30.7	7.2	1.7	-4.6
0.70	7089	30294	16024	8130	-4708	12.7	30.4	7.1	1.7	-4.6
0.72	7090	30424	16082	8110	-4729	12.6	30.2	7.0	1.7	-4.6
0.74	7090	30550	16140	8090	-4749	12.6	29.9	6.9	1.7	-4.5
0.76	7089	30673	16197	8071	-4770	12.5	29.7	6.9	1.8	-4.5
0.78	7085	30793	16254	8051	-4790	12.5	29.5	6.8	1.8	-4.5
0.80	7081	30911	16311	8033	-4809	12.4	29.3	6.7	1.8	-4.4
0.82	7075	31026	16367	8014	-4829	12.4	29.1	6.7	1.8	-4.4
0.84	7067	31138	16422	7996	-4848	12.3	28.9	6.6	1.9	-4.4
0.86	7058	31247	16478	7978	-4867	12.3	28.7	6.6	1.9	-4.4
0.88	7048	31354	16533	7961	-4886	12.3	28.6	6.5	1.9	-4.3
0.90	7037	31459	16588	7943	-4905	12.2	28.4	6.4	1.9	-4.3
0.92	7025	31561	16643	7927	-4923	12.2	28.2	6.4	1.9	-4.3
0.94	7011	31661	16697	7910	-4942	12.2	28.1	6.3	2.0	-4.3
0.96	6997	31759	16752	7894	-4960	12.1	27.9	6.3	2.0	-4.3
0.98	6981	31854	16806	7878	-4978	12.1	27.8	6.2	2.0	-4.2
1.00	6965	31948	16860	7862	-4996	12.1	27.7	6.2	2.0	-4.2



VIBRATING MACHINERY FOUNDATIONS ON PILES

Embedment Impedance Coefficients

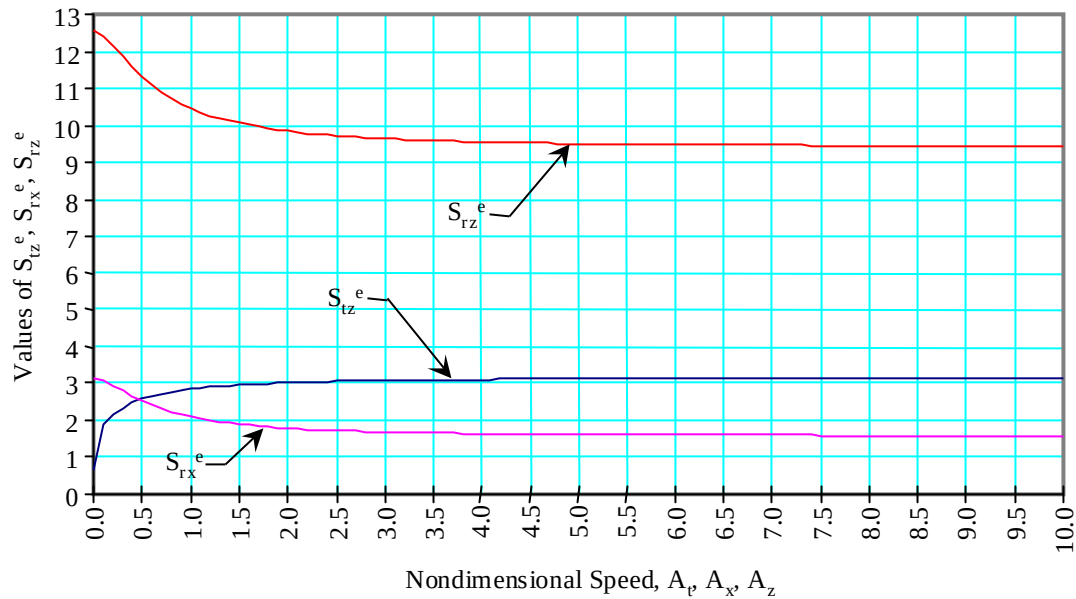


Figure 1: S_{tz}^e , S_{rx}^e , and S_{rz}^e (refer to Attachment 10, Section E5)

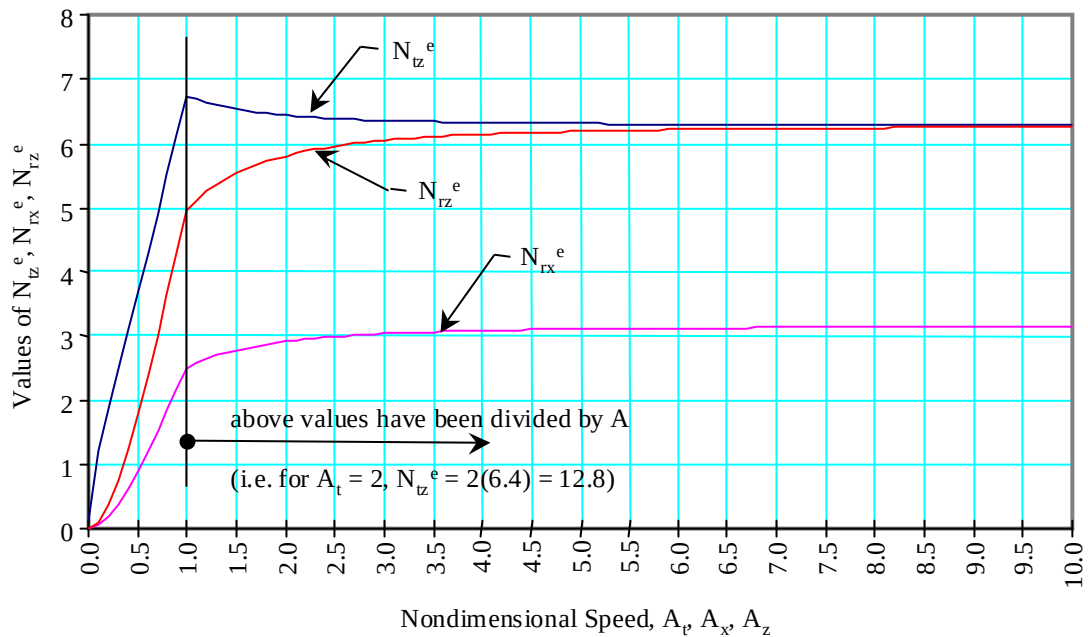


Figure 2: N_{tx}^e , N_{rx}^e , and N_{rz}^e (refer to Attachment 10, Section E5)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Embedment Impedance Coefficients

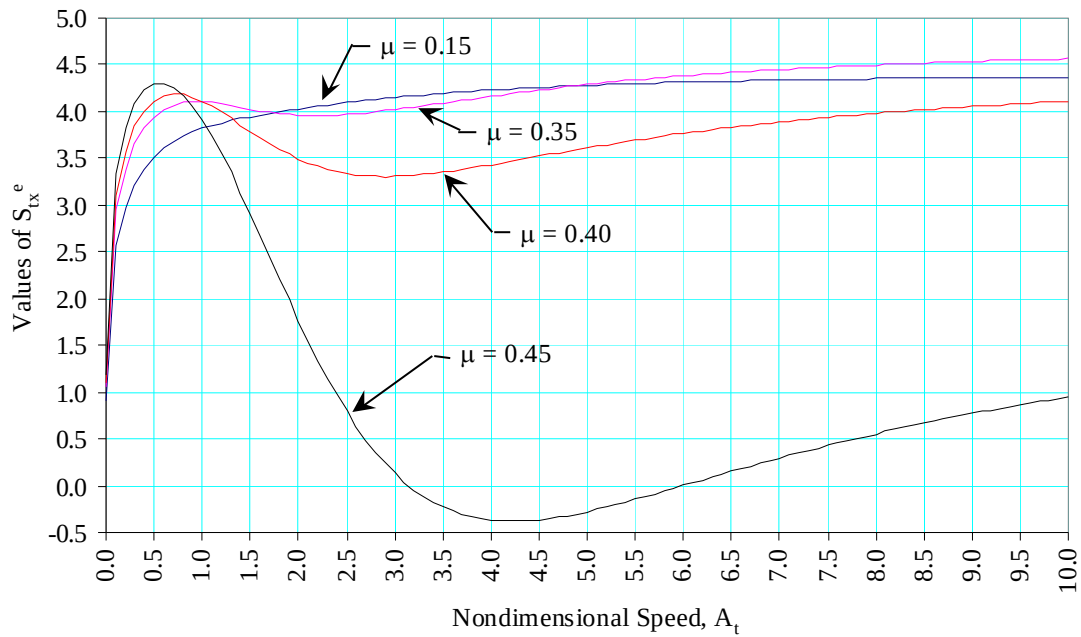


Figure 3: S_{tx}^e (refer to Attachment 10, Section E5)

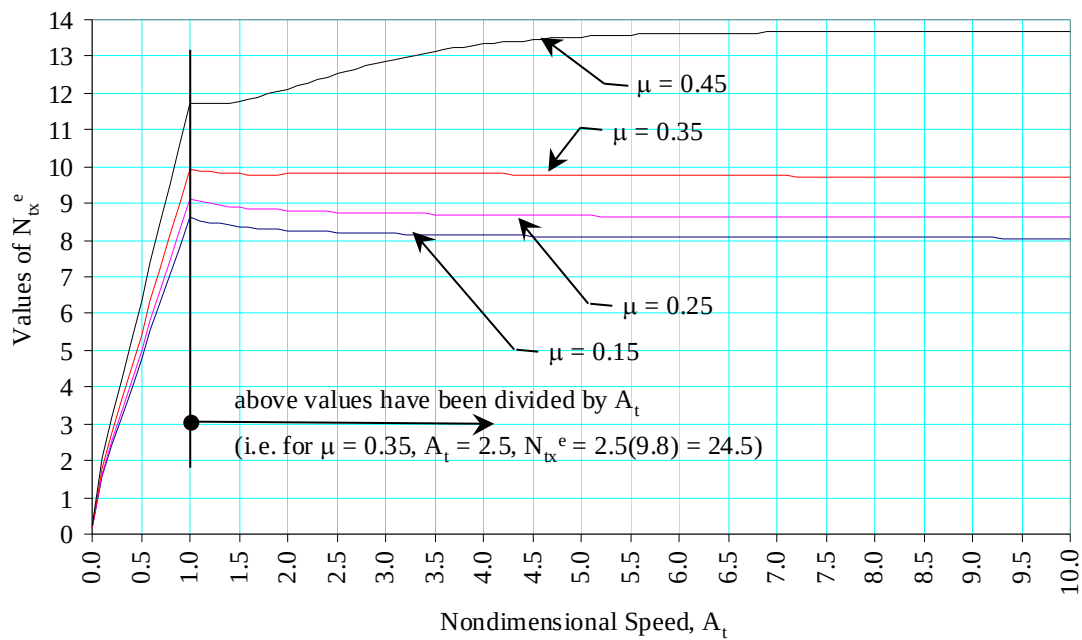


Figure 4: N_{tx}^e (refer to Attachment 10, Section E5)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vertical Group Factors

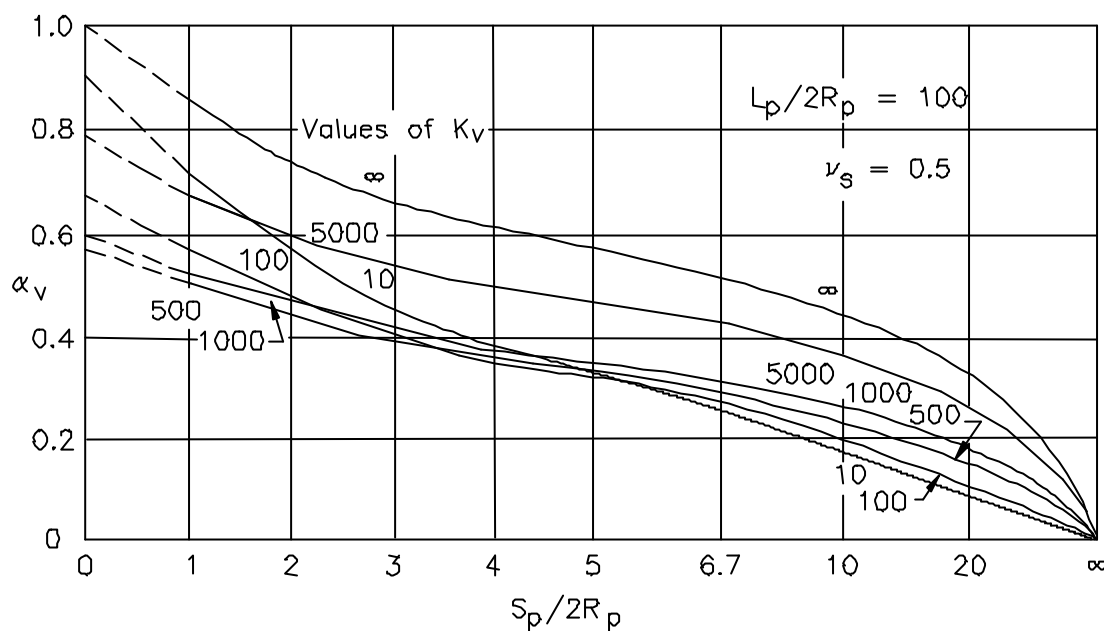


Chart A ($L_p/2R_p = 100$)

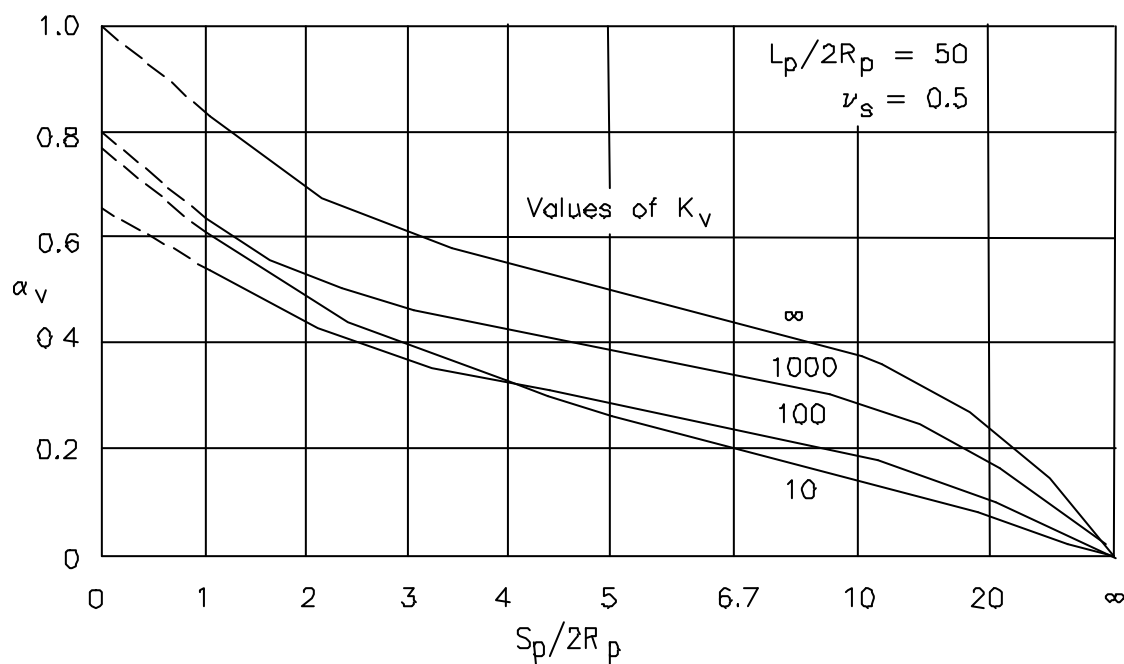


Chart B ($L_p/2R_p = 50$)



VIBRATING MACHINERY FOUNDATIONS ON PILES

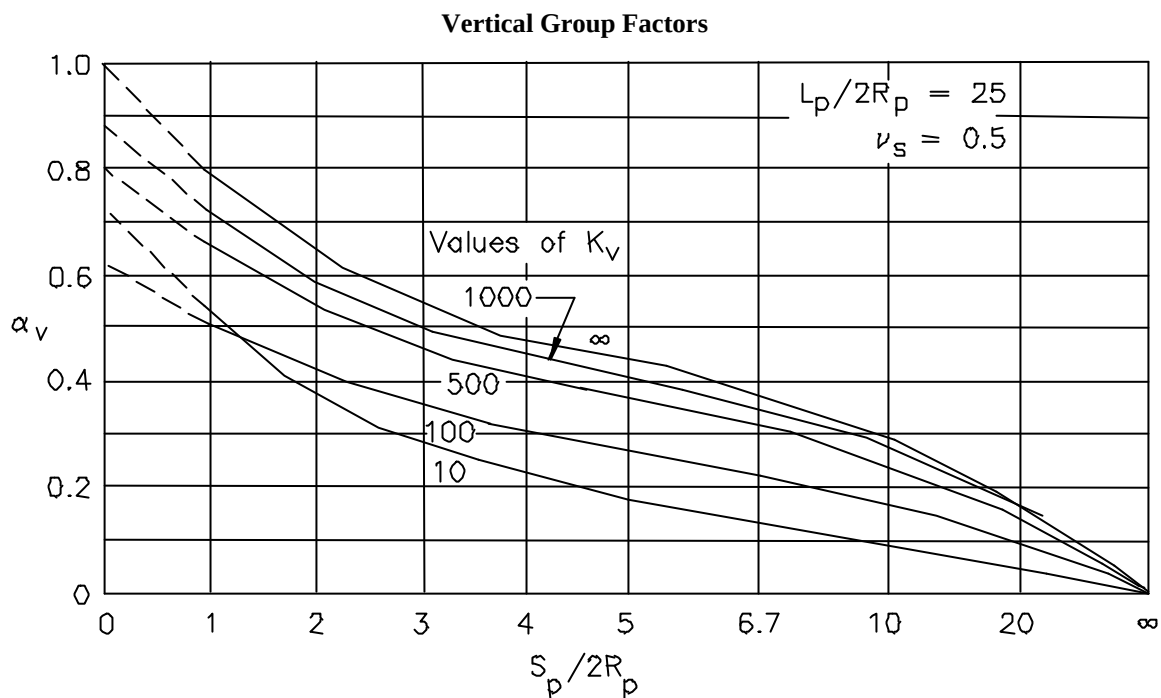


Chart C ($L_p/2R_p = 25$)

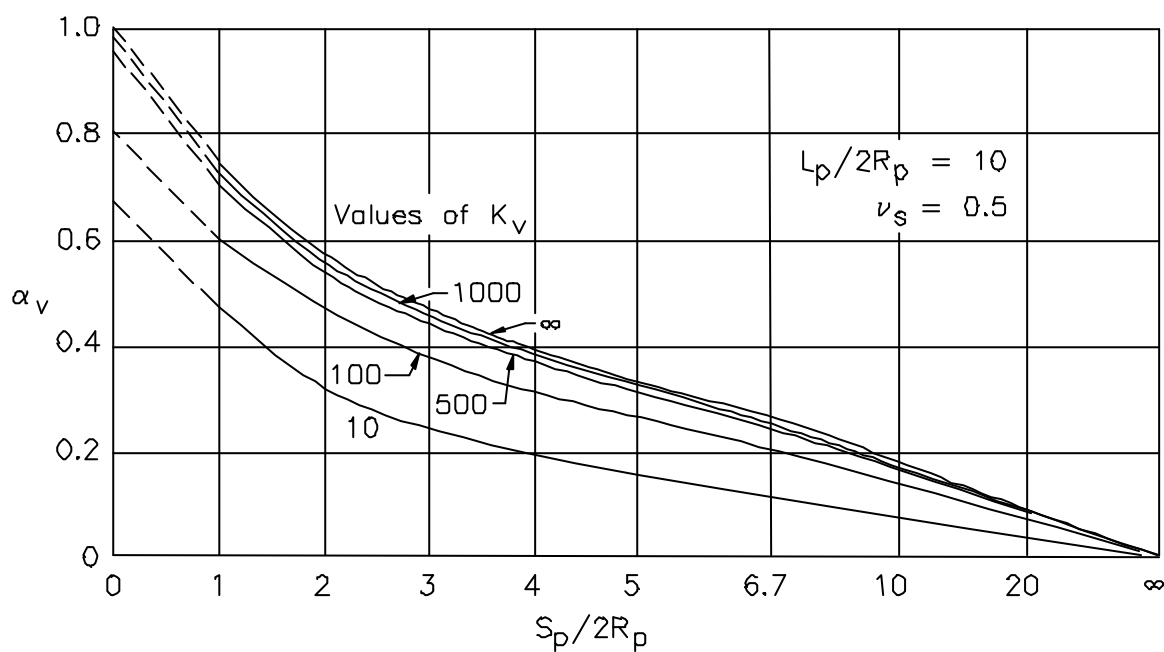


Chart D ($L_p/2R_p = 10$)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Horizontal Group Factors

Refer to Attachment 10, Section E4

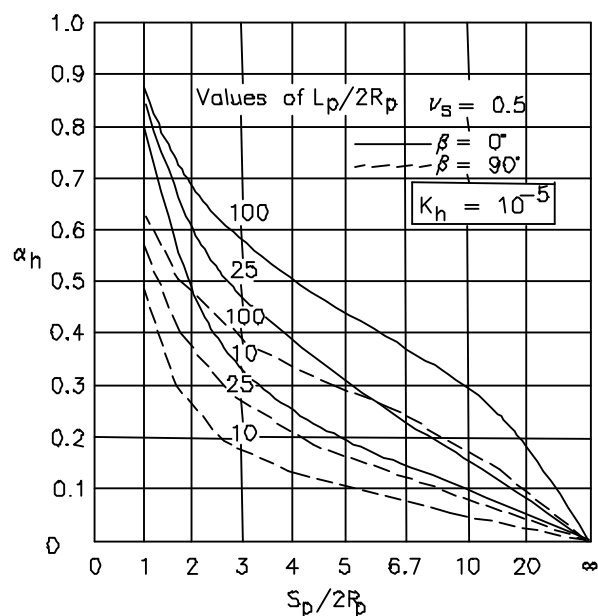


Chart A ($K_h = 10^{-5}$)

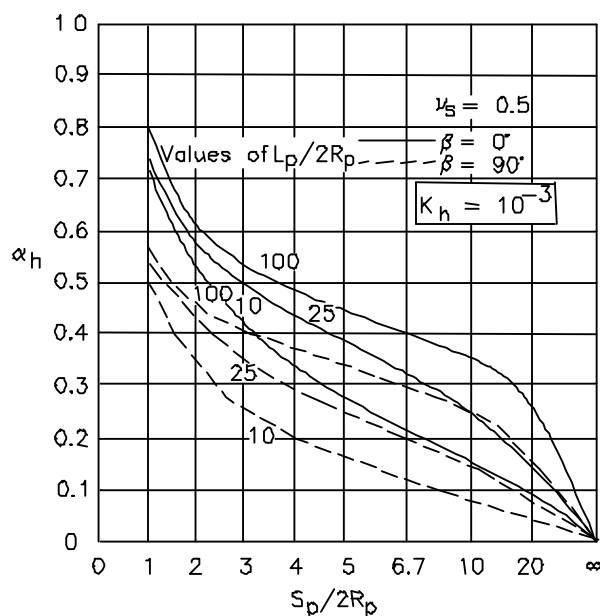


Chart B ($K_h = 10^{-3}$)

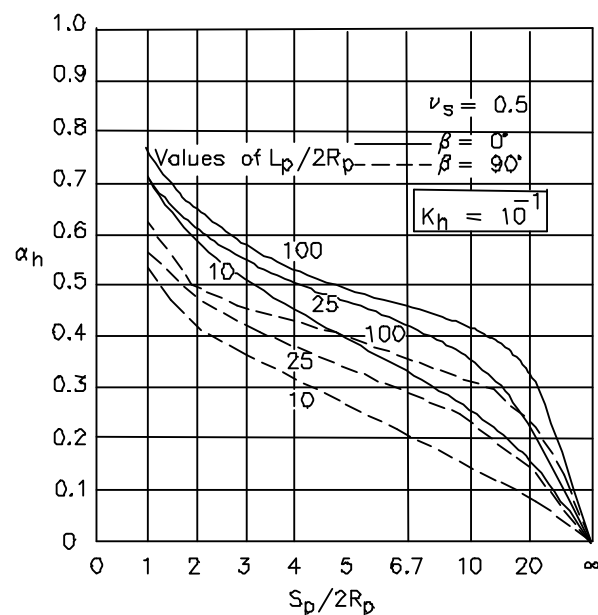


Chart C ($K_h = 10^{-1}$)

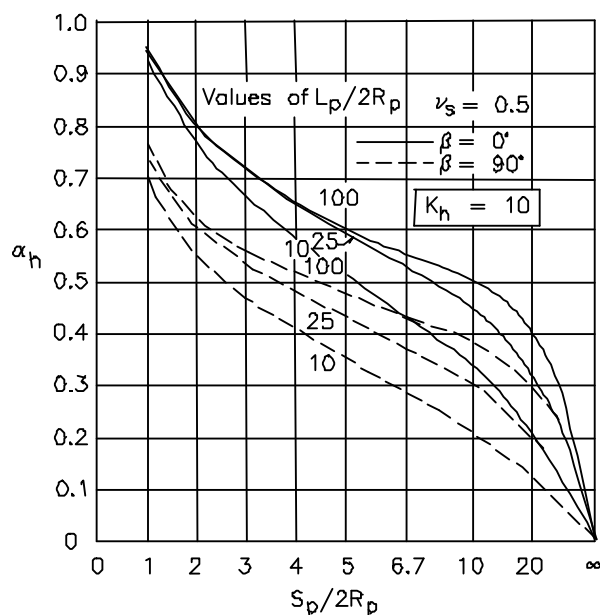


Chart D ($K_h = 10$)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Machine: (centrifugal machine)

Machine weight, MW = 22,180 lb (C.G. at point #1)
 Rotor weight, RW = 1365 lb (at point #2)
 speed, MS = 3565 rpm (373 rad/sec)

Piles: (round timber piles, Southern Pine)

radius, $R_o = 6$ inches
 length, $L = 50$ feet
 static capacity = 20 tons (40 kips)
 modulus of elasticity, $E_p = 1,500$ ksi (216,000 ksf)
 shear modulus, $G_p = 93,750$ psi (13,500 ksf)
 density, $W_p = 40$ pcf

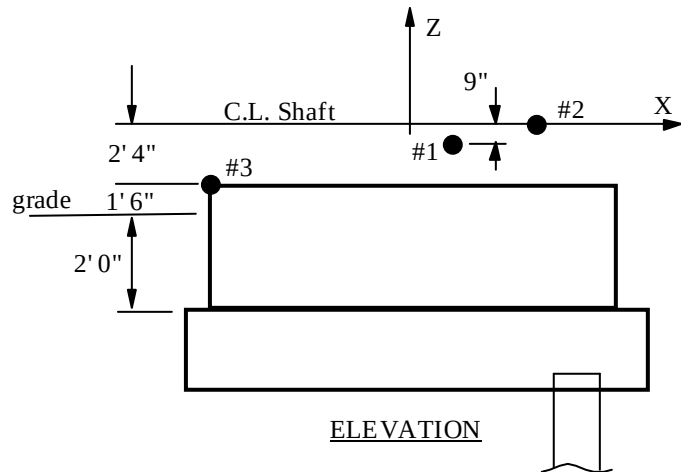
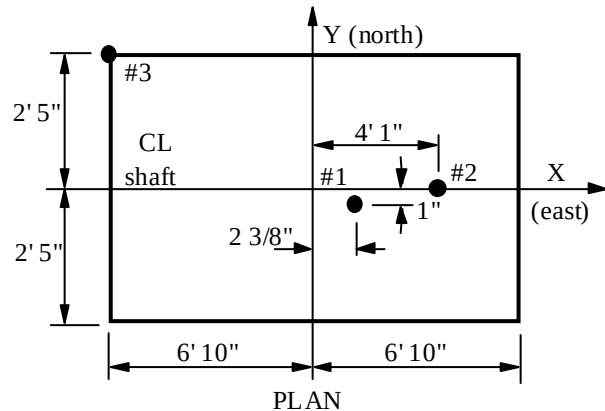
Soil Properties:

soft clay:
 shear modulus, $G_s = 5,000$ psi (720 ksf)
 density, $W_s = 110$ pcf
 poisson's ratio, $\mu_s = 0.45$

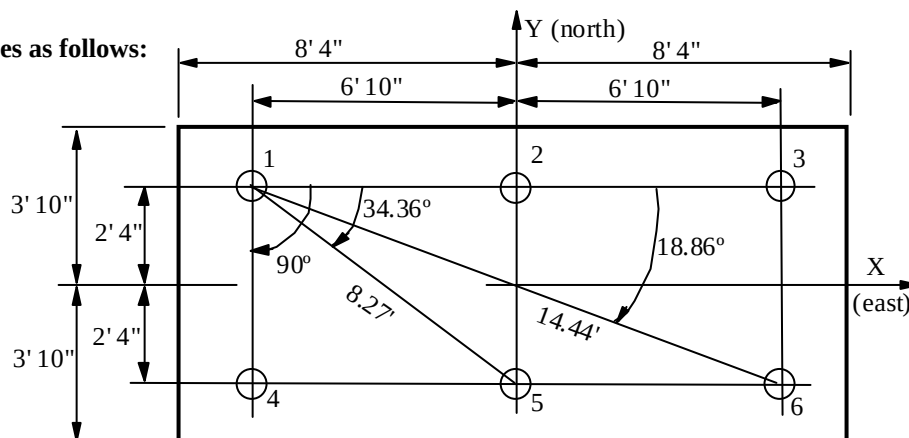
purchased sand backfill:
 shear modulus, $G_f = 15,000$ psi (2,160 ksf)
 density, $W_s = 125$ pcf
 poisson's ratio, $\mu_s = 0.30$

Miscellaneous:

Dynamic forces not provided by vendor
 Use computed radiation damping only
 Use Attachment 4 for allowable amplitudes
 $f'_c = 3,000$ psi



Try pile cap and piles as follows:





VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Pile Cap Thickness: (ref: Narrative)

This rigidity check can be skipped if the answer is obvious

verify assumed thickness, $L_z = 2' 0''$

pier cantilever, $d = [(16.67 \text{ ft pile cap}) - (13.67 \text{ ft pier})] / 2 = 1.5 \text{ ft}$

elastic modulus of concrete, $E_c = 57,000 \sqrt{f'_c} = 57,000 \sqrt{3,000 \text{ psi}} = 3,122,000 \text{ psi}$ (ACI 318-99, Section 8.5.1)

elastic modulus of soil, $E_s = 2 \text{ (shear modulus)}(1 + \text{poisson's ratio}) = 2 (5,000 \text{ psi})(1 + 0.45) = 14,500 \text{ psi}$

$$\frac{[E_c / E_s][L_z / d]^3}{[(3,122,000 \text{ psi}) / (14,500 \text{ psi})][(2.0 \text{ ft}) / (1/5 \text{ ft})]^3} \geq 1 \quad (\text{Equation 6})$$

510 > 1 OK, rigid USE $L_z = 2' 0''$

Compute Total Center of Gravity: (ref: Attachment 10:B4, B5, B6)

A.G. machine pier weight = (150 pcf)(13.67 ft)(4.83 ft)(1.5 ft) = 14,856 lb

U.G. machine pier weight = (150 pcf - 125 pcf)(13.67 ft)(4.83 ft)(2.0 ft) = 3,301 lb

soil weight = (125 pcf)(16.667 ft)(7.667 ft)(2.0 ft) = 31,946 lb

mat weight = (150 pcf)(16.667 ft)(7.667 ft)(2.0 ft) = 38,336 lb

part:	PartWt: lb	∂_x : ft	E_x : ft-lb	∂_y : ft	E_y : ft-lb	∂_z : ft	E_z : ft-lb
machine	22,180	0.2	4,392	-0.08	-1,841	-0.75	-16,635
A.G. pier	14,856	0	0	0	0	-3.08	-45,801
U.G. pier	3,301	0	0	0	0	-4.83	-15,954
soil	31,946	0	0	0	0	-4.83	-154,395
cap	38,336	0	0	0	0	-6.83	-261,950
Σ	110,619		4,392		-1,841		-494,735

$$CG_x = \Sigma(E_x) / \Sigma(\text{PartWt}) = (4,392 \text{ ft-lb}) / (110,619 \text{ lb}) = 0.040 \text{ ft} \quad (\text{Equation 10-6a})$$

$$CG_y = \Sigma(E_y) / \Sigma(\text{PartWt}) = (-1,841 \text{ ft-lb}) / (110,619 \text{ lb}) = -0.017 \text{ ft} \quad (\text{Equation 10-6b})$$

$$CG_z = \Sigma(E_z) / \Sigma(\text{PartWt}) = (-494,735 \text{ ft-lb}) / (110,619 \text{ lb}) = -4.472 \text{ ft} \quad (\text{Equation 10-6c})$$

Mass Moment and Mass Ratio: (ref: Attachment 10:B8)

$$\text{mass} = TW / \text{gravity} = (110,619 \text{ lb}) / (32.2 \text{ ft/sec}^2) = 3,435 \text{ lb-sec}^2/\text{ft} \quad (\text{Equation 10-8})$$

$$\begin{aligned} MR &= [TW - MW] / MW \\ &= [(110,619 \text{ lb}) - (22,180 \text{ lb})] / (22,180 \text{ lb}) \\ &= 3.99 > 3, \text{ OK} \end{aligned} \quad (\text{Equation 10-9})$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Mass Moment of Inertia about X axis: (ref: Attachment 10:B7)

part:	PartWt: lb	L _y : ft	L _z : ft	Q _x : lb-ft ²	∂ _y : ft	∂ _z : ft	J _x : lb-ft ²
machine	22,180	-	-	0	-0.08	-0.75	12,618
AG pier	14,856	4.83	1.5	31,667	0	-3.08	140,930
UG pier	3,301	4.83	2	7,518	0	-4.83	77,009
soil	31,946	7.67	2	167,261	0	-4.83	745,265
cap	38,336	7.67	2	200,717	0	-6.83	1,788,332
Σ	110,619			407,163			2,764,154

$$\begin{aligned}
 I_x &= \{ \Sigma (Q_x) + \Sigma (J_x) - TW [(CG_y)^2 + (CG_z)^2] \} / \text{gravity} & \text{(Equation 10-7a)} \\
 &= \frac{(407,163 \text{ lb} - \text{ft}^2) + (2,764,154 \text{ lb} - \text{ft}^2) - (110,619 \text{ lb})[(0.017 \text{ ft})^2 + (4.472 \text{ ft})^2]}{(32.2 \text{ ft/sec}^2)} \\
 &= 29,784 \text{ ft-lb-sec}^2
 \end{aligned}$$

Mass Moment of Inertia about Y axis: (ref: Attachment 10:B7)

part:	PartWt: lb	L _x : ft	L _z : ft	Q _y : lb-ft ²	∂ _x : ft	∂ _z : ft	J _y : lb-ft ²
machine	22,180	-	-	0	0.2	-0.75	1,114
AG pier	14,856	13.67	1.5	234,129	0	-3.08	140,930
UG pier	3,301	13.67	2	52,505	0	-4.83	77,009
soil	31,946	16.67	2	750,435	0	-4.83	745,265
cap	38,336	16.67	2	900,541	0	-6.83	1,788,332
Σ	110,619			1,937,610			2,764,899

$$\begin{aligned}
 I_y &= \{ \Sigma (Q_y) + \Sigma (J_y) - TW [(CG_x)^2 + (CG_z)^2] \} / \text{gravity} & \text{(Equation 10-7b)} \\
 &= \frac{(1,937,610 \text{ lb} - \text{ft}^2) + (2,764,899 \text{ lb} - \text{ft}^2) - (110,619 \text{ lb})[(0.040 \text{ ft})^2 + (4.472 \text{ ft})^2]}{(32.2 \text{ ft/sec}^2)} \\
 &= 77,332 \text{ ft-lb-sec}^2
 \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Mass Moment of Inertia about Z axis: (ref: Attachment 10:B7)

part:	PartWt: lb	L _x : ft	L _y : ft	Q _z : lb-ft ²	∂ _x : ft	∂ _y : ft	J _z : lb-ft ²
machine	22,180	-	-	0	0.2	-0.08	1,029
AG pier	14,856	13.67	4.83	260,225	0	0	0
UG pier	3,301	13.67	4.83	57,822	0	0	0
soil	31,946	16.67	7.67	896,399	0	0	0
cap	38,336	16.67	7.67	1,075,701	0	0	0
Σ	110,619			2,290,147			1,029

$$\begin{aligned}
I_z &= \{ \Sigma (Q_z) + \Sigma (J_z) - TW [(CG_x)^2 + (CG_y)^2] \} / \text{gravity} & \text{(Equation 10-7c)} \\
&= \frac{(2,290,147 \text{ lb} - \text{ft}^2) + (1,029 \text{ lb} - \text{ft}^2) - (110,619 \text{ lb})[(0.040 \text{ ft})^2 + (0.017 \text{ ft})^2]}{(32.2 \text{ ft/sec}^2)} \\
&= 71,148 \text{ ft-lb-sec}^2
\end{aligned}$$

Pile Pattern Properties: (ref: Attachment 10:C1)

number of piles, N = 6 ea

$$\Sigma x^2 = (4 \text{ ea})(6.833 \text{ ft})^2 = 186.76 \text{ ft}^2 \quad \text{(Equation 10-11a)}$$

$$\Sigma y^2 = (6 \text{ ea})(2.333 \text{ ft})^2 = 32.66 \text{ ft}^2 \quad \text{(Equation 10-11b)}$$

Compute Alignment Offsets: (ref: Attachment 10:C2)

(use outer pile dimensions, L_x = 13' 8", L_y = 4' 10")

$$\begin{aligned}
OS_x &= \text{abs} \{ 100 [CG_x - CP_x] / L_x \} & \text{(Equation 12a)} \\
&= \text{abs} \{ (100) [(0.040 \text{ ft}) - (0)] / (13.667 \text{ ft}) \} \\
&= 0.3 \% < 7\%, \text{ OK}
\end{aligned}$$

$$\begin{aligned}
OS_y &= \text{abs} \{ 100 [CG_y - CP_y] / L_y \} & \text{(Equation 12b)} \\
&= \text{abs} \{ (100) [-0.017 \text{ ft}] / (4.833 \text{ ft}) \} \\
&= 0.4 \% < 7\%, \text{ OK}
\end{aligned}$$

Compute Pile Load: (ref: Attachment 10: C3)

$$\begin{aligned}
PL &= TW / N & \text{(Equation 13)} \\
&= (110.619 \text{ kips}) / (6 \text{ ea}) \\
&= 18.4 \text{ kips} < \text{half of 40 kip static capacity} \quad \text{OK}
\end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Single Pile Impedance: (ref: Attachment 10:D2)

$$\begin{aligned}
 A_o &= (MS)(R_p) \sqrt{W_s / (G_s)(\text{gravity})} & (\text{Equation 14}) \\
 &= (373 \text{ rad/sec})(0.5 \text{ ft}) \sqrt{(110 \text{ lb/ft}^3) / (720,000 \text{ lb/ft}^2)(32.2 \text{ ft/sec}^2)} \\
 &= 0.41 \approx 0.40
 \end{aligned}$$

Select damping & stiffness from Attachment 13:

$K_{tx}^i = 6,390 \text{ k/ft}$	$C_{tx}^i = 14.3 \text{ k-sec/ft}$
$K_{ty}^i = 6,390 \text{ k/ft}$	$C_{ty}^i = 14.3 \text{ k-sec/ft}$
$K_{tz}^i = 19,366 \text{ k/ft}$	$C_{tz}^i = 26.6 \text{ k-sec/ft}$
$K_{rx}^i = 11,977 \text{ ft-k/rad}$	$C_{rx}^i = 7.6 \text{ ft-k-sec/rad}$
$K_{ry}^i = 11,977 \text{ ft-k/rad}$	$C_{ry}^i = 7.6 \text{ ft-k-sec/rad}$
$K_{rz}^i = 1,660 \text{ ft-k/rad}$	$C_{rz}^i = 0.2 \text{ ft-k-sec/rad}$
$K_{cx}^i = -6,395 \text{ k/rad}$	$C_{cx}^i = -8.6 \text{ k-sec/rad}$
$K_{cy}^i = -6,395 \text{ k/rad}$	$C_{cy}^i = -8.6 \text{ k-sec/rad}$

Compute Group Pile Efficiency:(ref: Attachment 10:E)

Determine vertical interaction factors for use with the charts in attachment 15:

$$R_a = A_p / (\pi d^2 / 4) = \text{by inspection, } 1.0 \quad (\text{Equation 22})$$

$$E_s = 2 (G_s)(1 - \mu_s) = 2 (5,000 \text{ psi})(1 - 0.45) = 14,500 \text{ psi}$$

$$K_v = (E_p)(R_a) / (E_s) = (1,500,000 \text{ psi})(1.0) / (14,500 \text{ psi}) = 103 \quad (\text{Equation 23})$$

$$L/d = (600 \text{ in}) / (12 \text{ in}) = 50, \text{ use Attachment 15, Chart b}$$

Determine horizontal interaction factors for use with the charts in attachment 16:

$$I_p = \pi (R_p)^4 / 4 = \pi (6 \text{ in})^4 / 4 = 1,018 \text{ in}^4$$

$$\begin{aligned}
 K_h &= (E_p)(I_p) / [(E_s)(L_p)^4] & (\text{Equation 24}) \\
 &= (1,500,000 \text{ psi})(1,018 \text{ in}^4) / [(14,500 \text{ psi})(600 \text{ in})^4] \\
 &= 8 \text{ E-7, use Attachment 16, Chart a}
 \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

Use the charts for two pile interaction for each unique pile spacing: (ref: Attachment 10:E4)

Piles:	S_p :	$S_p/2R_p$:	α_v :	$\alpha_h (0^\circ)$	$\alpha_h (90^\circ)$
1-1	0.00 ft	0.0	1.00	1.00	1.00
1-2	6.83 ft	6.8	0.24	0.28	0.15
1-3	13.67 ft	13.7	0.11	0.15	0.09
1-4	4.67 ft	4.7	0.29	0.35	0.22
1-5	8.27 ft	8.3	0.20	0.21	0.11
1-6	14.44 ft	14.4	0.11	0.15	0.09

Determine translation interaction factors: (ref: Attachment 10:E5)

(because of symmetry, only two typical reference piles are needed)

corner reference pile (#1 = reference)

target:	S_p :	α_v	β_x :	$\alpha_h (\beta_x)$:	β_y :	$\alpha_h (\beta_y)$:
1	0.00 ft	1.00	-	1.00	-	1.00
2	6.83 ft	0.24	0°	0.28	90°	0.15
3	13.67 ft	0.11	0°	0.15	90°	0.09
4	4.67 ft	0.29	90°	0.22	0°	0.35
5	8.27 ft	0.20	34°	0.17	56°	0.15
6	14.44 ft	<u>0.11</u>	19°	<u>0.14</u>	71°	<u>0.10</u>
		$\Sigma\alpha_z = 1.95$		$\Sigma\alpha_x = 1.96$		$\Sigma\alpha_y = 1.84$

side reference pile (#2 = reference)

target:	S_p :	α_v	β_x :	$\alpha_h (\beta_x)$:	β_y :	$\alpha_h (\beta_y)$:
1	6.83 ft	0.24	0°	0.28	90°	0.15
2	0.00 ft	1.00	-	1.00	-	1.00
3	6.83 ft	0.24	0°	0.28	90°	0.15
4	8.27 ft	0.20	34°	0.17	56°	0.15
5	4.67 ft	0.29	90°	0.22	0°	0.35
6	8.27ft	<u>0.20</u>	34°	<u>0.17</u>	56°	<u>0.15</u>
		$\Sigma\alpha_z = 2.17$		$\Sigma\alpha_x = 2.12$		$\Sigma\alpha_y = 1.95$

compute averaged group factors;

$$GP_{ix} = \Sigma(\alpha_x) / N = [(4 \text{ corner piles})(1.96) + (2 \text{ side piles})(2.12)] / (6 \text{ piles}) = 2.01 \quad (\text{Equation 10-25a})$$

$$GP_{iy} = \Sigma(\alpha_y) / N = [(4 \text{ corner piles})(1.84) + (2 \text{ side piles})(1.95)] / (6 \text{ piles}) = 1.88 \quad (\text{Equation 10-25b})$$

$$GP_{iz} = \Sigma(\alpha_z) / N = [(4 \text{ corner piles})(1.95) + (2 \text{ side piles})(2.17)] / (6 \text{ piles}) = 2.02 \quad (\text{Equation 10-25c})$$

Rocking about X axis: (ref: Attachment 10:E6)

(piles 4, 5, and 6 carry opposing load)



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

corner reference pile (#1 = reference)

target:	S _p :	α _v :
1	0.00 ft	1.00
2	6.83 ft	0.24
3	13.67 ft	<u>0.11</u>
		Σ α = 1.35

side reference pile (#2 = reference)

target:	S _p :	α _v :
1	6.83 ft	0.24
2	0.00 ft	1.00
3	6.83 ft	<u>0.24</u>
		Σ α = 1.48

$$GP_{rx} = \Sigma [(\alpha_x)(\text{distance from reference pile to X axis})^2] / \Sigma y^2 \quad (\text{Equation 10-26a})$$

$$= [(4 \text{ corner piles})(1.35)(2.333 \text{ ft})^2 + (2 \text{ side piles})(1.48)(2.333 \text{ ft})^2] / (32.66 \text{ ft}^2)$$

$$= 1.39$$

Rocking about Y axis: (ref: Attachment 10:E6)

(piles 2 and 5 carry no load; piles 3 and 6 carry opposing load)

corner reference pile (#1 = reference)

target:	S _p :	α _v :
1	0.00 ft	1.00
4	4.67 ft	<u>0.29</u>
		Σ α = 1.29

$$GP_{ry} = \Sigma [(\alpha_y)(\text{distance from reference pile to Y axis})^2] / \Sigma x^2 \quad (\text{Equation 10-26b})$$

$$= [(4 \text{ corner piles})(1.29)(6.833 \text{ ft})^2] / (186.76 \text{ ft}^2)$$

$$= 1.29$$

Torsion about Z axis: (ref: Attachment 10:E6)

(use only piles "ahead" and "behind" reference pile by less than 22 degrees)

therefore, GP_{tz} = 1.0

Combined Pile Stiffness and Damping: (ref: Attachment 10:E8, E9) (at pile centroid)

$$K_{tx}^c = (K_{tx}^i)(N) / GP_{tx} = (6,390 \text{ k/ft})(6 \text{ ea}) / 2.01 = 19,075 \text{ k/ft} \quad (\text{Equation 10-27a})$$

$$K_{ty}^c = (K_{ty}^i)(N) / GP_{ty} = (6,390 \text{ k/ft})(6 \text{ ea}) / 1.88 = 22,394 \text{ k/ft} \quad (\text{Equation 10-27b})$$

$$K_{tz}^c = (K_{tz}^i)(N) / GP_{tz} = (19,366 \text{ k/ft})(6 \text{ ea}) / 2.02 = 57,523 \text{ k/ft} \quad (\text{Equation 10-27c})$$

$$K_{rx}^c = [(K_{rx}^i)(N) + (K_{tz}^i)(\Sigma y^2)] / GP_{tx} \quad (\text{Equation 10-27d})$$

$$= [(11,977 \text{ k-ft/rad})(6 \text{ ea}) + (19,366 \text{ k/ft})(32.66 \text{ ft}^2)] / 1.39$$

$$= 506,731 \text{ k-ft/rad}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

$$\begin{aligned} K_{ry}^c &= [(K_{ry}^i)(N) + (K_{tz}^i)(\Sigma x^2)] / GP_{ry} & \text{(Equation 10-27e)} \\ &= [(11,977 \text{ k-ft/rad})(6 \text{ ea}) + (19,366 \text{ k/ft})(186.76 \text{ ft}^2)] / 1.29 \\ &= 2,859,423 \text{ k-ft/rad} \end{aligned}$$

$$\begin{aligned} K_{rz}^c &= [(K_{rz}^i)(N) + (K_{tx}^i)(\Sigma y^2) + (K_{ty}^i)(\Sigma x^2)] / GP_{rz} & \text{(Equation 10-27f)} \\ &= [(1,658 \text{ k-ft/rad})(6 \text{ ea}) + (6,390 \text{ k/ft})(32.66 \text{ ft}^2) + (6,390 \text{ k/ft})(186.76 \text{ ft}^2)] / 1.0 \\ &= 1,409,933 \text{ k-ft/rad} \end{aligned}$$

$$K_{cx}^c = K_{cy}^c = (K_{cx}^i)(N) = (-6,395 \text{ k/rad})(6 \text{ ea}) = -38,370 \text{ k/rad} \quad \text{(Equation 10-27g)}$$

$$C_{tx}^c = (C_{tx}^i)(N) / GP_{tx} = (14.3 \text{ k-sec/ft})(6 \text{ ea}) / 2.01 = 42.7 \text{ k-sec/ft} \quad \text{(Equation 10-28a)}$$

$$C_{ty}^c = (C_{ty}^i)(N) / GP_{ty} = (14.3 \text{ k-sec/ft})(6 \text{ ea}) / 1.88 = 45.6 \text{ k-sec/ft} \quad \text{(Equation 10-28b)}$$

$$C_{tz}^c = (C_{tz}^i)(N) / GP_{tz} = (26.6 \text{ k-sec/ft})(6 \text{ ea}) / 2.02 = 79.0 \text{ k-sec/ft} \quad \text{(Equation 10-28c)}$$

$$\begin{aligned} C_{rx}^c &= [(C_{rx}^i)(N) + (C_{tz}^i)(\Sigma y^2)] / GP_{rx} & \text{(Equation 10-28d)} \\ &= [(7.6 \text{ k-ft-sec/rad})(6 \text{ ea}) + (26.6 \text{ k-sec/ft})(32.66 \text{ ft}^2)] / 1.39 \\ &= 657.8 \text{ k-ft-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry}^c &= [(C_{ry}^i)(N) + (C_{tz}^i)(\Sigma x^2)] / GP_{ry} & \text{(Equation 10-28e)} \\ &= [(7.6 \text{ k-ft-sec/rad})(6 \text{ ea}) + (26.6 \text{ k-sec/ft})(186.76 \text{ ft}^2)] / 1.29 \\ &= 3,886.4 \text{ k-ft-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{rz}^c &= [(C_{rz}^i)(N) + (C_{tx}^i)(\Sigma y^2) + (C_{ty}^i)(\Sigma x^2)] / GP_{rz} & \text{(Equation 10-28f)} \\ &= [(0.2 \text{ k-ft-sec/rad})(6 \text{ ea}) + (14.3 \text{ k-sec/ft})(32.66 \text{ ft}^2) + (14.3 \text{ k-sec/ft})(186.76 \text{ ft}^2)] / 1.0 \\ &= 3,138.9 \text{ k-ft-sec/rad} \end{aligned}$$

$$C_{cx}^c = C_{cy}^c = (C_{cx}^i)(N) = (-8.6 \text{ k-sec/rad})(6 \text{ ea}) = -51.6 \text{ k-sec/rad} \quad \text{(Equation 10-28g)}$$

Stiffness & Damping of Embedded Mat: (ref: Attachment 10:F3, F4, F6)

$$h = (2/3)(\text{depth of mat}) = (2/3)(2.0 \text{ ft}) = 1.333 \text{ ft} \quad \text{(Equation 10-29)}$$

$$R_t = \sqrt{(L_x)(L_y) / \pi} = \sqrt{(16.67 \text{ ft})(7.67 \text{ ft}) / \pi} = 6.38 \text{ ft} \quad \text{(Equation 10-30a)}$$

$$R_x = \sqrt[4]{(L_x)(L_y)^3 / (3\pi)} = \sqrt[4]{(16.67 \text{ ft})(7.67 \text{ ft})^3 / (3\pi)} = 5.32 \text{ ft} \quad \text{(Equation 10-30b)}$$

$$R_y = \sqrt[4]{(L_y)(L_x)^3 / (3\pi)} = \sqrt[4]{(7.67 \text{ ft})(16.67 \text{ ft})^3 / (3\pi)} = 7.84 \text{ ft} \quad \text{(Equation 10-30c)}$$

$$\begin{aligned} R_z &= \sqrt[4]{(L_x)(L_y)[(L_x)^2 + (L_y)^2] / (6\pi)} & \text{(Equation 10-30d)} \\ &= \sqrt[4]{(16.67 \text{ ft})(7.67 \text{ ft})[(16.67 \text{ ft})^2 + (7.67 \text{ ft})^2] / (6\pi)} \\ &= 6.91 \text{ ft} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

$$\begin{aligned} A_t &= (MS)(R_t)\sqrt{(W_s/(G_f)(\text{gravity}))} && \text{(Equation 10-31a)} \\ &= (373 \text{ rad/sec})(6.38 \text{ ft})\sqrt{(0.125 \text{ k/ft}^3)/(2,160 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)} \\ &= 3.19 \approx 3.20 \end{aligned}$$

$$\begin{aligned} A_y &= (MS)(R_x)\sqrt{(W_s/(G_f)(\text{gravity}))} && \text{(Equation 10-31b)} \\ &= (373 \text{ rad/sec})(5.32 \text{ ft})\sqrt{(0.125 \text{ k/ft}^3)/(2,160 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)} \\ &= 2.66 \approx 2.65 \end{aligned}$$

$$\begin{aligned} A_y &= (MS)(R_y)\sqrt{(W_s/(G_f)(\text{gravity}))} && \text{(Equation 10-31c)} \\ &= (373 \text{ rad/sec})(7.84 \text{ ft})\sqrt{(0.125 \text{ k/ft}^3)/(2,160 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)} \\ &= 3.92 \approx 3.90 \end{aligned}$$

$$\begin{aligned} A_z &= (MS)(R_z)\sqrt{(W_s/(G_f)(\text{gravity}))} && \text{(Equation 10-31d)} \\ &= (373 \text{ rad/sec})(6.91 \text{ ft})\sqrt{(0.125 \text{ k/ft}^3)/(2,160 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)} \\ &= 3.46 \approx 3.45 \end{aligned}$$

from Attachment 14 select:

$$\begin{array}{ll} S_{tx}^e = 4.152 & N_{tx}^e = 29.653 \\ S_{ty}^e = 4.152 & N_{ty}^e = 29.653 \\ S_{tz}^e = 3.080 & N_{tz}^e = 20.326 \end{array}$$

$$\begin{array}{ll} S_{rx}^e = 1.705 & N_{rx}^e = 7.940 \\ S_{ry}^e = 1.640 & N_{ry}^e = 11.972 \\ S_{rz}^e = 9.596 & N_{rz}^e = 21.054 \end{array}$$

$$K_{tx}^e = K_{ty}^e = (S_{tx}^e)(G_f)(h) = (4.152)(2,160 \text{ k/ft}^2)(1.333 \text{ ft}) = 11,955 \text{ k/ft} \quad \text{(Equation 10-36a)}$$

$$K_{tz}^e = (S_{tz}^e)(G_f)(h) = (3.080)(2,160 \text{ k/ft}^2)(1.333 \text{ ft}) = 8,868 \text{ k/ft} \quad \text{(Equation 10-36c)}$$

$$\begin{aligned} K_{rx}^e &= \{(S_{rx}^e) + [(h)/(R_x)]^2 (S_{ty}^e) / 12\} (G_f)(h)(R_x)^2 && \text{(Equation 10-36d)} \\ &= \{(1.705) + [(1.333 \text{ ft})/(5.32 \text{ ft})]^2 (4.152) / 12\} (2,160 \text{ k/ft}^2)(1.333 \text{ ft})(5.32 \text{ ft})^2 \\ &= 140,712 \text{ ft-k/rad} \end{aligned}$$

$$\begin{aligned} K_{ry}^e &= \{(S_{ry}^e) + [(h)/(R_y)]^2 (S_{tx}^e) / 12\} (G_f)(h)(R_y)^2 && \text{(Equation 10-36e)} \\ &= \{(1.640) + [(1.333 \text{ ft})/(7.84 \text{ ft})]^2 (4.152) / 12\} (2,160 \text{ k/ft}^2)(1.333 \text{ ft})(7.84 \text{ ft})^2 \\ &= 292,012 \text{ ft-k/rad} \end{aligned}$$

$$\begin{aligned} K_{rz}^e &= (S_{rz}^e)(G_f)(h)(R_z)^2 && \text{(Equation 10-36f)} \\ &= (9.596)(2,160 \text{ k/ft}^2)(1.333 \text{ ft})(6.91 \text{ ft})^2 \\ &= 1,319,260 \text{ ft-lb/rad} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON PILES

Vibration Calculation

$$\begin{aligned} C_{tx}^e &= C_{ty}^e = (N_{tx}^e)(G_f)(h) / MS \\ &= (29.653)(2,160 \text{ k/ft}^2)(1.333 \text{ ft}) / (373 \text{ rad/sec}) \\ &= 229 \text{ k-sec/ft} \end{aligned} \quad \text{(Equation 10-37a)}$$

$$\begin{aligned} C_{tz}^e &= (N_{tz}^e)(G_f)(h) / MS \\ &= (20.326)(2,160 \text{ k/ft}^2)(1.333 \text{ ft}) / (373 \text{ rad/sec}) \\ &= 157 \text{ k-sec/ft} \end{aligned} \quad \text{(Equation 10-37c)}$$

$$\begin{aligned} C_{rx}^e &= \{ (N_{rx}^e) + [(h)/(R_x)]^2 (N_{ty}^e)/12 \} (G_f)(h)(R_x)^2 / MS \\ &= \{ (7.940) + [(1.333 \text{ ft})/(5.32 \text{ ft})]^2 (29.653) / 12 \} (2,160 \text{ k/ft}^2)(1.333 \text{ ft})(5.32 \text{ ft})^2 / (373 \text{ rad/sec}) \\ &= 1,769 \text{ ft-k-sec/rad} \end{aligned} \quad \text{(Equation 10-37d)}$$

$$\begin{aligned} C_{ry}^e &= \{ (N_{ry}^e) + [(h)/(R_y)]^2 (N_{tx}^e)/12 \} (G_f)(h)(R_y)^2 / MS \\ &= \{ (11.972) + [(1.333 \text{ ft})/(7.84 \text{ ft})]^2 (29.653) / 12 \} (2,160 \text{ k/ft}^2)(1.333 \text{ ft})(7.84 \text{ ft})^2 / (373 \text{ rad/sec}) \\ &= 5,680 \text{ ft-k-sec/rad} \end{aligned} \quad \text{(Equation 10-37e)}$$

$$\begin{aligned} C_{rz}^e &= (N_{rz}^e)(G_f)(h)(R_z)^2 / MS \\ &= (21.054)(2,160 \text{ k/ft}^2)(1.333 \text{ ft})(6.91 \text{ ft})^2 / (373 \text{ rad/sec}) \\ &= 7,760 \text{ ft-k-sec/rad} \end{aligned} \quad \text{(Equation 10-37f)}$$

Total Stiffness and Damping: (ref: Attachment 10:G3, G4, G5) (at center of gravity)

$$\partial_c = (2.333 \text{ ft}) + (1.5 \text{ ft}) + (2.0 \text{ ft}) + (2.0 \text{ ft}) - (4.472 \text{ ft}) = 3.361 \text{ ft} \quad \text{(Equation 10-38)}$$

$$\partial_e = (3.361 \text{ ft}) - (1.333 \text{ ft}) / 2 = 2.695 \text{ ft} \quad \text{(Equation 10-39)}$$

$$K_{tx} = (K_{tx}^c) + (K_{tx}^e) = (19,075 \text{ k/ft}) + (11,955 \text{ k/ft}) = 31,030 \text{ k/ft} \quad \text{(Equation 10-40a)}$$

$$K_{ty} = (K_{ty}^c) + (K_{ty}^e) = (20,394 \text{ k/ft}) + (11,955 \text{ k/ft}) = 32,349 \text{ k/ft} \quad \text{(Equation 10-40b)}$$

$$K_{tz} = (K_{tz}^c) + (K_{tz}^e) = (57,523 \text{ k/ft}) + (8,868 \text{ k/ft}) = 66,391 \text{ k/ft} \quad \text{(Equation 10-40c)}$$

$$\begin{aligned} K_{rx} &= (K_{rx}^c) + (K_{rx}^e) + (K_{ty}^c)(\partial_c)^2 + (K_{ty}^e)(\partial_e)^2 \\ &= (506,731 \text{ ft-k/rad}) + (140,712 \text{ ft-k/rad}) + (20,394 \text{ k/ft})(3.361 \text{ ft})^2 + (11,955 \text{ k/ft})(2.695 \text{ ft})^2 \\ &= 964,650 \text{ ft-k/rad} \end{aligned} \quad \text{(Equation 10-40d)}$$

$$\begin{aligned} K_{ry} &= (K_{ry}^c) + (K_{ry}^e) + (K_{tx}^c)(\partial_c)^2 + (K_{tx}^e)(\partial_e)^2 \\ &= (2,859,423 \text{ ft-k/rad}) + (292,012 \text{ ft-k/rad}) + (19,075 \text{ k/ft})(3.361 \text{ ft})^2 + (11,955 \text{ k/ft})(2.695 \text{ ft})^2 \\ &= 3,453,742 \text{ ft-k/rad} \end{aligned} \quad \text{(Equation 10-40e)}$$

$$\begin{aligned} K_{rz} &= (K_{rz}^c) + (K_{rz}^e) \\ &= (1,409,933 \text{ ft-k/rad}) + (1,319,260 \text{ ft-k/rad}) \\ &= 2,729,193 \text{ ft-k/rad} \end{aligned} \quad \text{(Equation 10-40f)}$$

$$\begin{aligned} K_{cx} &= K_{cx}^c - (K_{ty}^c)(\partial_c) - (K_{ty}^e)(\partial_e) \\ &= -38,370 \text{ k/rad} - (22,394 \text{ k/ft})(3.361 \text{ ft}) - (11,955 \text{ k/ft})(2.695 \text{ ft}) \\ &= -139,133 \text{ k/rad} \end{aligned} \quad \text{(Equation 10-40g)}$$



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$$\begin{aligned} K_{cy} &= K_{cy}^c - (K_{tx}^c)(\partial_c) - (K_{tx}^e)(\partial_e) && \text{(Equation 10-40h)} \\ &= -38,370 \text{ k/rad} - (19,075 \text{ k/ft})(3.361 \text{ ft}) - (11,955 \text{ k/ft})(2.695 \text{ ft}) \\ &= -134,700 \text{ k/rad} \end{aligned}$$

$$C_{tx} = (C_{tx}^c) + (C_{tx}^e) = (42.7 \text{ k-sec/ft}) + (229 \text{ k-sec/ft}) = 271.7 \text{ k-sec/ft} \quad \text{(Equation 10-41a)}$$

$$C_{ty} = (C_{ty}^c) + (C_{ty}^e) = (45.6 \text{ k-sec/ft}) + (229 \text{ k-sec/ft}) = 274.6 \text{ k-sec/ft} \quad \text{(Equation 10-41b)}$$

$$C_{tz} = (C_{tz}^c) + (C_{tz}^e) = (79.0 \text{ k-sec/ft}) + (157 \text{ k-sec/ft}) = 236.0 \text{ k-sec/ft} \quad \text{(Equation 10-41c)}$$

$$\begin{aligned} C_{rx} &= (C_{rx}^c) + (C_{rx}^e) + (C_{ty}^c)(\partial_c)^2 + (C_{ty}^e)(\partial_e)^2 && \text{(Equation 10-41d)} \\ &= (657.8 \text{ ft-k-sec/rad}) + (1,769 \text{ ft-k-sec/rad}) + (45.6 \text{ k-sec/ft})(3.361 \text{ ft})^2 + (229 \text{ k-sec/ft})(2.695 \text{ ft})^2 \\ &= 4,605 \text{ ft-k-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry} &= (C_{ry}^c) + (C_{ry}^e) + (C_{tx}^c)(\partial_c)^2 + (C_{tx}^e)(\partial_e)^2 && \text{(Equation 10-41e)} \\ &= (3,886.4 \text{ ft-k-sec/rad}) + (5,680 \text{ ft-k-sec/rad}) + (42.7 \text{ k-sec/ft})(3.361 \text{ ft})^2 + (229 \text{ k-sec/ft})(2.695 \text{ ft})^2 \\ &= 11,712 \text{ ft-k-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{rz} &= (C_{rz}^c) + (C_{rz}^e) && \text{(Equation 10-41f)} \\ &= (3,138.9 \text{ ft-k-sec/rad}) + (7,760 \text{ ft-k-sec/rad}) \\ &= 10,899 \text{ ft-k-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{cx} &= C_{cx}^c - (C_{ty}^c)(\partial_c) - (C_{ty}^e)(\partial_e) && \text{(Equation 10-41g)} \\ &= -51.6 \text{ k-sec/rad} - (45.6 \text{ k-sec/ft})(3.361 \text{ ft}) - (229 \text{ k-sec/ft})(2.695 \text{ ft}) \\ &= -822 \text{ k-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{cy} &= C_{cy}^c - (C_{tx}^c)(\partial_c) - (C_{tx}^e)(\partial_e) && \text{(Equation 10-41h)} \\ &= -51.6 \text{ k-sec/rad} - (42.7 \text{ k-sec/ft})(3.361 \text{ ft}) - (229 \text{ k-sec/ft})(2.695 \text{ ft}) \\ &= -812 \text{ k-sec/rad} \end{aligned}$$

Determine Primary Loads at Center of Gravity: (ref: Attachment 10:G6)

use approximate formula & $Q_g = 0.2 \text{ in/sec}$

$$\begin{aligned} \text{force} &= (RW)(Q_g)(MS) / 12 (\text{gravity}) && \text{(Attachment 02)} \\ &= (1,365 \text{ lb})(0.2 \text{ in/sec})(373 \text{ rad/sec}) / 12(32.2 \text{ ft/sec}^2) \\ &= 264 \text{ lb} \end{aligned}$$

$$F_x = F_x^v = 0 \text{ k} \quad \text{(Equation 10-42a)}$$

$$F_y = F_y^v = 0.264 \text{ k} \quad \text{(Equation 10-42b)}$$

$$F_z = F_z^v = 0.264 \text{ k} \quad \text{(Equation 10-42c)}$$

$$\begin{aligned} M_x &= M_x^v + F_y^v [\text{abs}(\partial_z - CG_z)] + F_z^v [\text{abs}(\partial_y - CG_y)] && \text{(Equation 10-42d)} \\ &= (0 \text{ ft-k}) + (0.264 \text{ k}) [\text{abs}(0 - 4.472 \text{ ft})] + (0.264 \text{ k}) [\text{abs}(0 - 0.017 \text{ ft})] \\ &= 1.185 \text{ ft-k} \end{aligned}$$

$$\begin{aligned} M_y &= M_y^v + F_x^v [\text{abs}(\partial_z - CG_z)] + F_z^v [\text{abs}(\partial_x - CG_x)] && \text{(Equation 10-42e)} \\ &= (0 \text{ ft-k}) + (0 \text{ k}) [\text{abs}(0 - 4.472 \text{ ft})] + (0.264 \text{ k}) [\text{abs}(4.083 \text{ ft} - 0.040 \text{ ft})] \\ &= 1.067 \text{ ft-k} \end{aligned}$$



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$$\begin{aligned} M_z &= M_z^v + F_x^v [\text{abs}(\partial_y - CG_y)] + F_y^v [\text{abs}(\partial_x - CG_x)] \\ &= (0 \text{ ft-k}) + (0 \text{ k}) [\text{abs}(0 - 0.017 \text{ ft})] + (0.264 \text{ k}) [\text{abs}(4.083 \text{ ft} - 0.040 \text{ ft})] \\ &= 1.067 \text{ ft-k} \end{aligned} \quad (\text{Equation 10-42f})$$

Primary Vertical Translation: (ref: Attachment 10:H3, H4, H5, H6)

$$\begin{aligned} a_1 &= (K_{tz}) - (\text{mass})(MS)^2 \\ &= (66,391 \text{ k/ft}) - (3.435 \text{ k-sec}^2/\text{ft})(373 \text{ rad/sec})^2 \\ &= -411,517 \text{ k/ft} \end{aligned} \quad (\text{Equation 10-43})$$

$$a_2 = (C_{tz})(MS) = (236.0 \text{ k-sec/ft})(373 \text{ rad/sec}) = 88,028 \text{ k/ft} \quad (\text{Equation 10-44})$$

$$\Delta_z = F_z / \sqrt{(a_1)^2 + (a_2)^2} = (0.264 \text{ k}) / \sqrt{(-411,517 \text{ k/ft})^2 + (88,028 \text{ k/ft})^2} = 6.27 \text{ E-7 ft} \quad (\text{Equation 10-45})$$

by trial and error, $NF_1 = 126.4 \text{ rad/sec}$ (1,207 rpm)

$$FR_1 = MS / NF_1 = (373 \text{ rad/sec}) / (126.4 \text{ rad/sec}) = 3.0 > 1.2, \text{ OK} \quad (\text{Equation 10-47})$$

Primary Torsional Rocking: (ref: Attachment 10:H7)

$$\begin{aligned} a_1 &= (K_{tz}) - (I_z)(MS)^2 \\ &= (2,729,193 \text{ ft-k/rad}) - (71.148 \text{ ft-k-sec}^2)(373 \text{ rad/sec})^2 \\ &= -7,169,557 \text{ ft-k/rad} \end{aligned} \quad (\text{Equation 10-43})$$

$$a_2 = (C_{tz})(MS) = (10,899 \text{ ft-k-sec/rad})(373 \text{ rad/sec}) = 4,065,327 \text{ ft-k/rad} \quad (\text{Equation 10-44})$$

$$\begin{aligned} \phi_z &= F_z / \sqrt{(a_1)^2 + (a_2)^2} \\ &= (1.067 \text{ ft-k}) / \sqrt{(-7,169,557 \text{ ft-k/rad})^2 + (4,065,327 \text{ ft-k/rad})^2} \\ &= 1.29 \text{ e-7 rad} \end{aligned} \quad (\text{Equation 10-45})$$

by trial and error, $NF_2 = 192.7 \text{ rad/sec}$ (1,840 rpm)

$$FR_2 = MS / NF_2 = (373 \text{ rad/sec}) / (192.7 \text{ rad/sec}) = 1.9 > 1.2, \text{ OK} \quad (\text{Equation 10-47})$$

Primary Transverse Translation & Rocking: (ref: Attachment 10:I3, I4, I5, I6)

$$\begin{aligned} a_1 &= [(K_{rx}) - (I_x)(MS)^2](F_y) - (K_{cx})(M_x) \\ &= [(964,650 \text{ ft-k/rad}) - (29.784 \text{ ft-k-sec}^2)(373 \text{ rad/sec})^2](0.264 \text{ k}) - (-139,133 \text{ k/rad})(1.185 \text{ ft-k}) \\ &= -674,428 \text{ ft-k}^2 \end{aligned} \quad (\text{Equation 10-48})$$

$$\begin{aligned} a_2 &= (C_{rx})(MS)(F_y) - (C_{cx})(MS)(M_x) \\ &= (4,605 \text{ ft-k-sec/rad})(373 \text{ rad/sec})(0.264 \text{ k}) - (-822 \text{ k-sec/rad})(373 \text{ rad/sec})(1.185 \text{ ft-k}) \\ &= 816,792 \text{ ft-k}^2 \end{aligned} \quad (\text{Equation 10-49})$$



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$$\begin{aligned} b_1 &= [(K_{ty}) - (\text{mass})(MS)^2](M_x) - (K_{cx})(F_y) & \text{(Equation 10-50)} \\ &= [(32,349 \text{ k/ft}) - (3.435 \text{ k-sec}^2/\text{ft})(373 \text{ rad/sec})^2](1.185 \text{ ft-k}) - (-139,133 \text{ k/rad})(0.264 \text{ k}) \\ &= -491,256 \text{ k}^2 \end{aligned}$$

$$\begin{aligned} b_2 &= (C_{ty})(MS)(M_x) - (C_{cx})(MS)(F_y) & \text{(Equation 10-51)} \\ &= (274.6 \text{ k-sec/ft})(373 \text{ rad/sec})(1.185 \text{ ft-k}) - (-822 \text{ k-sec/rad})(373 \text{ rad/sec})(0.264 \text{ k}) \\ &= 202,319 \text{ k}^2 \end{aligned}$$

$$\begin{aligned} c_1 &= (\text{mass})(I_x)(MS)^4 + (K_{ty})(K_{rx}) - (K_{cx})^2 - [(\text{mass})(K_{rx}) + (I_x)(K_{ty}) + (C_{ty})(C_{rx}) - (C_{cx})^2](MS)^2 & \text{(Equation 10-52)} \\ &= (3.435 \text{ k-sec}^2/\text{ft})(29.784 \text{ ft-k-sec}^2)(373 \text{ rad/sec})^4 \\ &\quad + (32,349 \text{ k/ft})(964,650 \text{ ft-k/rad}) - (-139,133 \text{ k/rad})^2 \\ &\quad - [(3.435 \text{ k-sec}^2/\text{ft})(964,650 \text{ ft-k/rad}) + (29.784 \text{ ft-k-sec}^2)(32,349 \text{ k/ft}) \\ &\quad + (274.6 \text{ k-sec/ft})(4,605 \text{ ft-k-sec/rad}) - (-822 \text{ k-sec/rad})^2] * (373 \text{ rad/sec})^2 \\ &= 1.3152 \text{ E12 k}^2 \end{aligned}$$

$$\begin{aligned} c_2 &= [(C_{ty})(K_{rx}) + (C_{rx})(K_{ty}) - 2(C_{cx})(K_{cx})](MS) - [(\text{mass})(C_{rx}) + (I_x)(C_{ty})](MS)^3 & \text{(Equation 10-53)} \\ &= [(274.6 \text{ k-sec/ft})(964,650 \text{ ft-k/rad}) + (4,605 \text{ ft-k-sec/rad})(32,349 \text{ k/ft}) \\ &\quad - 2(-822 \text{ k-sec/rad})(-139,133 \text{ k/rad})](373 \text{ rad/sec}) \\ &\quad - [(3.435 \text{ k-sec}^2/\text{ft})(4,605 \text{ ft-k-sec/rad}) + (29.784 \text{ ft-k-sec}^2)(274.6 \text{ k-sec/ft})] * (373 \text{ rad/sec})^3 \\ &= -1.1763 \text{ E12 k}^2 \end{aligned}$$

$$\begin{aligned} \Delta_y &= \sqrt{[(a_1)^2 + (a_2)^2] / [(c_1)^2 + (c_2)^2]} & \text{(Equation 10-54)} \\ &= \sqrt{[(-674,428 \text{ ft} - \text{k}^2)^2 + (816,792 \text{ ft} - \text{k}^2)^2] / [(1.3152 \text{ E12 k}^2)^2 + (-1.1763 \text{ E12 k}^2)^2]} \\ &= 6.00 \text{ E-7 ft} \end{aligned}$$

$$\begin{aligned} \phi_x &= \sqrt{[(b_1)^2 + (b_2)^2] / [(c_1)^2 + (c_2)^2]} & \text{(Equation 10-55)} \\ &= \sqrt{[(-491,256 \text{ k}^2)^2 + (202,319 \text{ k}^2)^2] / [(1.3152 \text{ E12 k}^2)^2 + (-1.1763 \text{ E12 k}^2)^2]} \\ &= 3.01 \text{ E-7 rad} \end{aligned}$$

by trial and error, $NF_3 = 185.7 \text{ rad/sec}$ (1,773 rpm) and $NF_4 = 53.0 \text{ rad/sec}$ (506 rpm)

$$FR_3 = MS / FR_3 = (373 \text{ rad/sec}) / (185.7 \text{ rad/sec}) = 2.0 > 1.2, \text{ OK} \quad \text{(Equation 10-60a)}$$

$$FR_4 = MS / FR_4 = (373 \text{ rad/sec}) / (53.0 \text{ rad/sec}) = 7.0 > 1.2, \text{ OK} \quad \text{(Equation 10-60b)}$$

Primary Longitudinal Translation & Rocking: (ref: Attachment 10:I7)

$$\begin{aligned} a_1 &= [(K_{ry}) - (I_y)(MS)^2](F_x) - (K_{cy})(M_y) & \text{(Equation 10-48)} \\ &= [(3,453,742 \text{ ft-k/rad}) - (77.332 \text{ ft-k-sec}^2)(373 \text{ rad/sec})^2](0 \text{ k}) - (-134,700 \text{ k/rad})(1.067 \text{ ft-k}) \\ &= 143,725 \text{ ft-k}^2 \end{aligned}$$

$$\begin{aligned} a_2 &= (C_{ry})(MS)(F_x) - (C_{cy})(MS)(M_y) & \text{(Equation 10-49)} \\ &= (11,712 \text{ ft-k-sec/rad})(373 \text{ rad/sec})(0 \text{ k}) - (-812 \text{ k-sec/rad})(373 \text{ rad/sec})(1.067 \text{ ft-k}) \\ &= 323,169 \text{ ft-k}^2 \end{aligned}$$



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$$\begin{aligned} b_1 &= [(K_{tx}) - (\text{mass})(MS)^2](M_y) - (K_{cy})(F_x) & \text{(Equation 10-50)} \\ &= [(31,030 \text{ k/ft}) - (3.435 \text{ k-sec}^2/\text{ft})(373 \text{ rad/sec})^2](1,067 \text{ ft-k}) - (-134,700 \text{ k/rad})(0 \text{ k}) \\ &= -476,819 \text{ k}^2 \end{aligned}$$

$$\begin{aligned} b_2 &= (C_{tx})(MS)(M_y) - (C_{cy})(MS)(F_x) & \text{(Equation 10-51)} \\ &= (271.7 \text{ k-sec/ft})(373 \text{ rad/sec})(1.067 \text{ ft-k}) - (-812 \text{ k-sec/rad})(373 \text{ rad/sec})(0 \text{ k}) \\ &= 108,134 \text{ k}^2 \end{aligned}$$

$$\begin{aligned} c_1 &= (\text{mass})(I_y)(MS)^4 + (K_{tx})(K_{ry}) - (K_{cy})^2 - [(\text{mass})(K_{ry}) + (I_y)(K_{tx}) + (C_{tx})(C_{ry}) - (C_{cy})^2](MS)^2 & \text{(Equation 10-52)} \\ &= (3.435 \text{ k-sec}^2/\text{ft})(77.332 \text{ ft-k-sec}^2)(373 \text{ rad/sec})^4 \\ &\quad + (31,030 \text{ k/ft})(3,453,742 \text{ ft-k/rad}) - (-134,700 \text{ k/rad})^2 \\ &\quad - [(3.435 \text{ k-sec}^2/\text{ft})(3,453,742 \text{ ft-k/rad}) + (77.332 \text{ ft-k-sec}^2)(31,030 \text{ k/ft}) \\ &\quad + (271.7 \text{ k-sec/ft})(11,712 \text{ ft-k-sec/rad}) - (-812 \text{ k-sec/rad})^2] * (373 \text{ rad/sec})^2 \\ &= 2.8955 \text{ E12 k}^2 \end{aligned}$$

$$\begin{aligned} c_2 &= [(C_{tx})(K_{ry}) + (C_{ry})(K_{tx}) - 2(C_{cy})(K_{cy})](MS) - [(\text{mass})(C_{ry}) + (I_y)(C_{tx})](MS)^3 & \text{(Equation 10-53)} \\ &= [(271.7 \text{ k-sec/ft})(3,453,742 \text{ ft-k/rad}) + (11,712 \text{ ft-k-sec/rad})(31,030 \text{ k/ft}) \\ &\quad - 2(-812 \text{ k-sec/rad})(-134,700 \text{ k/rad})](373 \text{ rad/sec}) \\ &\quad - [(3.435 \text{ k-sec}^2/\text{ft})(11,712 \text{ ft-k-sec/rad}) + (77.332 \text{ ft-k-sec}^2)(271.7 \text{ k-sec/ft})] * (373 \text{ rad/sec})^3 \\ &= -2.7742 \text{ E12 k}^2 \end{aligned}$$

$$\begin{aligned} \Delta_x &= \sqrt{[(a_1)^2 + (a_2)^2] / [(c_1)^2 + (c_2)^2]} & \text{(Equation 10-54)} \\ &= \sqrt{[(143,725 \text{ ft} - \text{k}^2)^2 + (323,169 \text{ ft} - \text{k}^2)^2] / [(2.8955 \text{ E12 k}^2)^2 + (-2.7742 \text{ E12 k}^2)^2]} \\ &= 0.88 \text{ E-7 ft} \end{aligned}$$

$$\begin{aligned} \phi_y &= \sqrt{[(b_1)^2 + (b_2)^2] / [(c_1)^2 + (c_2)^2]} & \text{(Equation 10-55)} \\ &= \sqrt{[(-476,819 \text{ k}^2)^2 + (108,134 \text{ k}^2)^2] / [(2.8955 \text{ E12 k}^2)^2 + (-2.7742 \text{ E12 k}^2)^2]} \\ &= 1.22 \text{ E-7 rad} \end{aligned}$$

by trial and error, $NF_5 = 206.1 \text{ rad/sec}$ (1,968 rpm) and $NF_6 = 76.4 \text{ rad/sec}$ (730 rpm)

$$FR_5 = MS / FR_5 = (373 \text{ rad/sec}) / (206.1 \text{ rad/sec}) = 1.8 > 1.2, \text{ OK} \quad \text{(Equation 10-60a)}$$

$$FR_6 = MS / FR_6 = (373 \text{ rad/sec}) / (76.4 \text{ rad/sec}) = 4.9 > 1.2, \text{ OK} \quad \text{(Equation 10-60b)}$$

Double Amplitudes at Point #3: (ref: Attachment 10:J1)

$$D_x = \text{abs}[PT_x - CG_x] = \text{abs}[(-6.833 \text{ ft}) - (0.040 \text{ ft})] = 6.873 \text{ ft} \quad \text{(Equation 10-61a)}$$

$$D_y = \text{abs}[PT_y - CG_y] = \text{abs}[(2.417 \text{ ft}) - (-0.017 \text{ ft})] = 2.434 \text{ ft} \quad \text{(Equation 10-61b)}$$

$$D_z = \text{abs}[PT_z - CG_z] = \text{abs}[(-2.333 \text{ ft}) - (-4.472 \text{ ft})] = 2.139 \text{ ft} \quad \text{(Equation 10-61c)}$$

$$\begin{aligned} DA_x &= \{ \Delta_x + D_y (\phi_z) + D_z (\phi_y) \} (24000) & \text{(Equation 10-62a)} \\ &= \{ (0.88 \text{ E-7 ft}) + (2.434 \text{ ft})(1.29 \text{ E-7 rad}) + (2.139 \text{ ft})(1.22 \text{ E-7 rad}) \} (24000) \\ &= 0.016 \text{ mils} \end{aligned}$$



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$$\begin{aligned} DA_y &= \{\Delta_y + D_x (\phi_z) + D_z (\phi_x)\} (24000) && \text{(Equation 10-62b)} \\ &= \{(6.00 \text{ E-7 ft}) + (6.873 \text{ ft})(1.29 \text{ E-7 rad}) + (2.139 \text{ ft})(3.01 \text{ E-7 rad})\} (24000) \\ &= 0.051 \text{ mils} \end{aligned}$$

$$\begin{aligned} DA_z &= \{\Delta_z + D_x (\phi_y) + D_y (\phi_x)\} (24000) && \text{(Equation 10-62c)} \\ &= \{(6.27 \text{ E-7 ft}) + (6.873 \text{ ft})(1.22 \text{ E-7 rad}) + (2.434 \text{ ft})(3.01 \text{ E-7 rad})\} (24000) \\ &= 0.053 \text{ mils} \end{aligned}$$